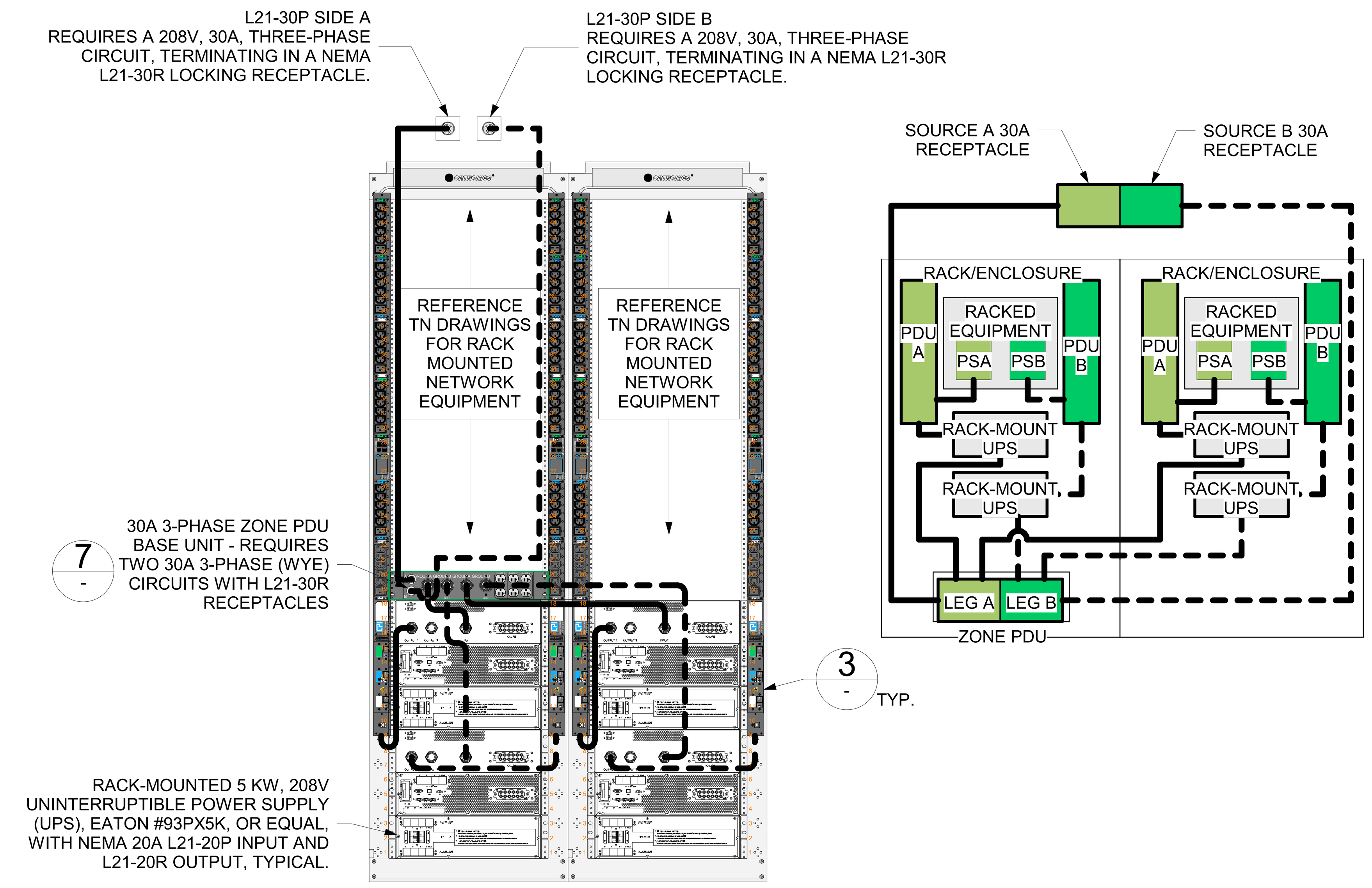
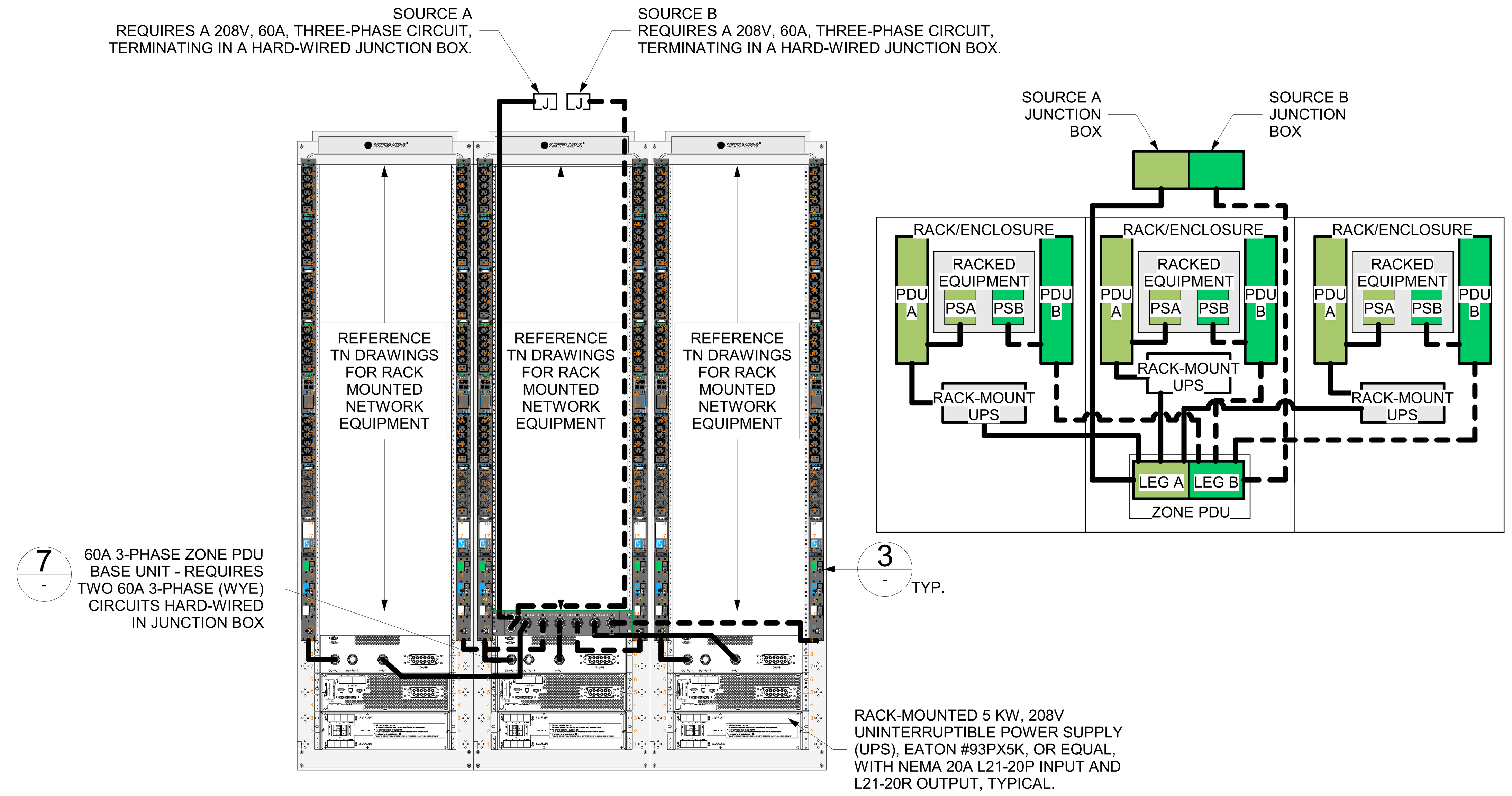
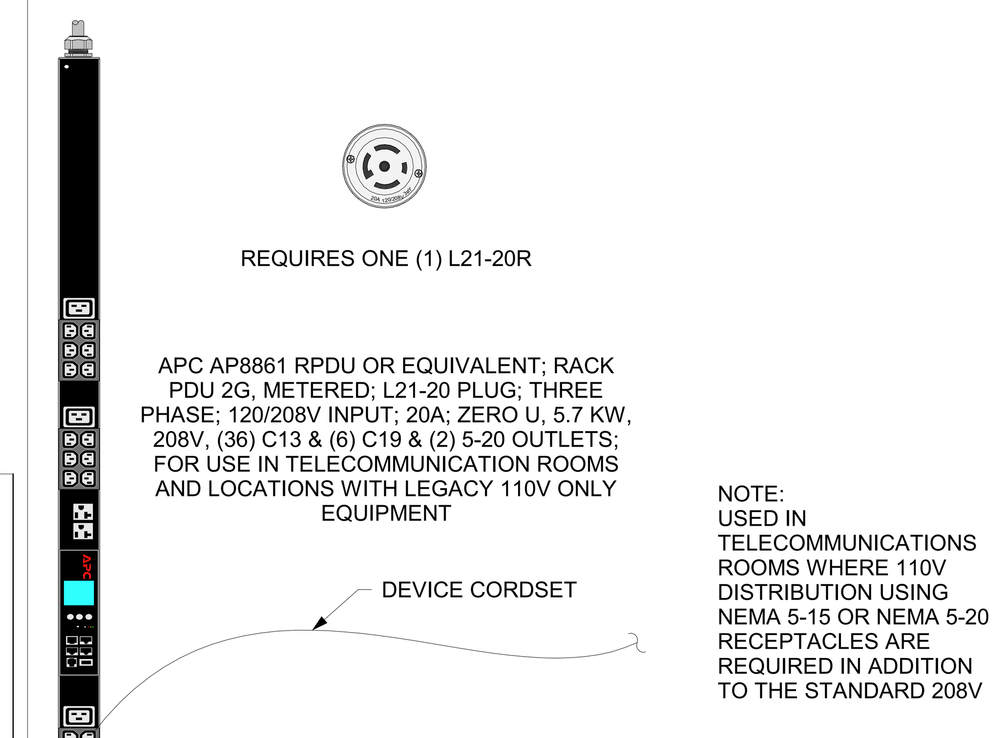
GENERAL NOTES SHALL BE APPLIED TO ALL UPS UNITS:
UPS SHALL SUPPORT BOTH MONITORING AND REMOTE MANAGEMENT TO INCLUDE:
1. FULL COMPLIANCE WITH RFC-1628 - UPS MANAGEMENT INFORMATION BASE.
2. SUPPORT SNMP V3.0, BACNET AND MODBUS PROTOCOLS.
3. PROVIDE CONNECTION TO BMS SYSTEM VIA CAT-6A ETHERNET CONNECTION.
4. UPS SHALL BE MANGED AND REPORT THE FOLLOWING TO THE FACILITY BMS:
   I. OVERLOAD (CRITICAL).
   II. BATTERY TEMP (CRITICAL).
   III. LOW BATTERY.
   IV. BATTERY FAILURE (CRITICAL).
   V. GENERAL CRITICAL ALARM (IE: FAN FAILURE, INTERNAL FAULT, BYPASS, ETC.)



1 POWER ELEVATION AND SCHEMATIC FOR A TWO-RACK ENTRANCE ROOM
01EP400Gb
01EP400Gh
NTS



2 POWER ELEVATION AND SCHEMATIC FOR A THREE-RACK TR
01SEP401
01SEP402
NTS



FOR INFORMATIONAL REFERENCE ONLY:
208V PDU TO BE COMPATIBLE WITH CISCO IEC C14-C15 CABINET JUMPER POWER CORD, DEVICE CORDSET PART NUMBER CAB-C15-CBN.

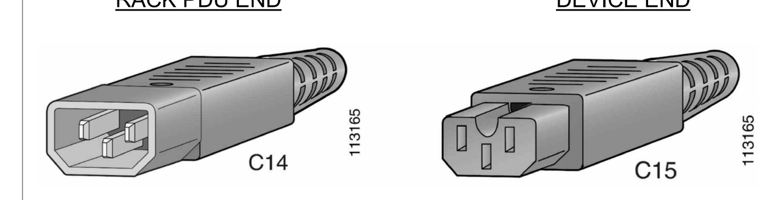
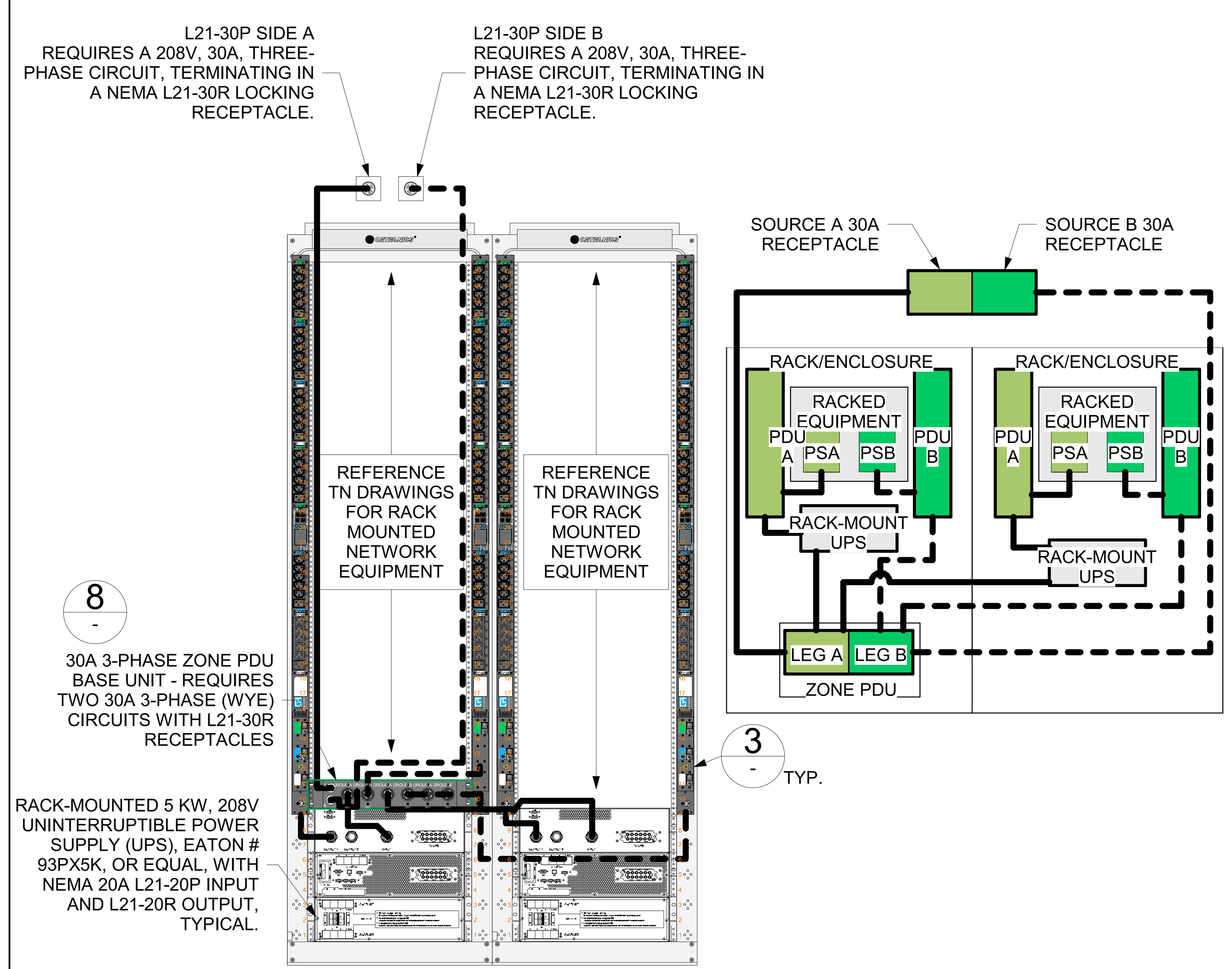


FIGURE 1: IEC 60320 C14 PLUG CONNECTOR.
SOURCE: CISCO SYSTEMS INC.
ILLUSTRATION #113165

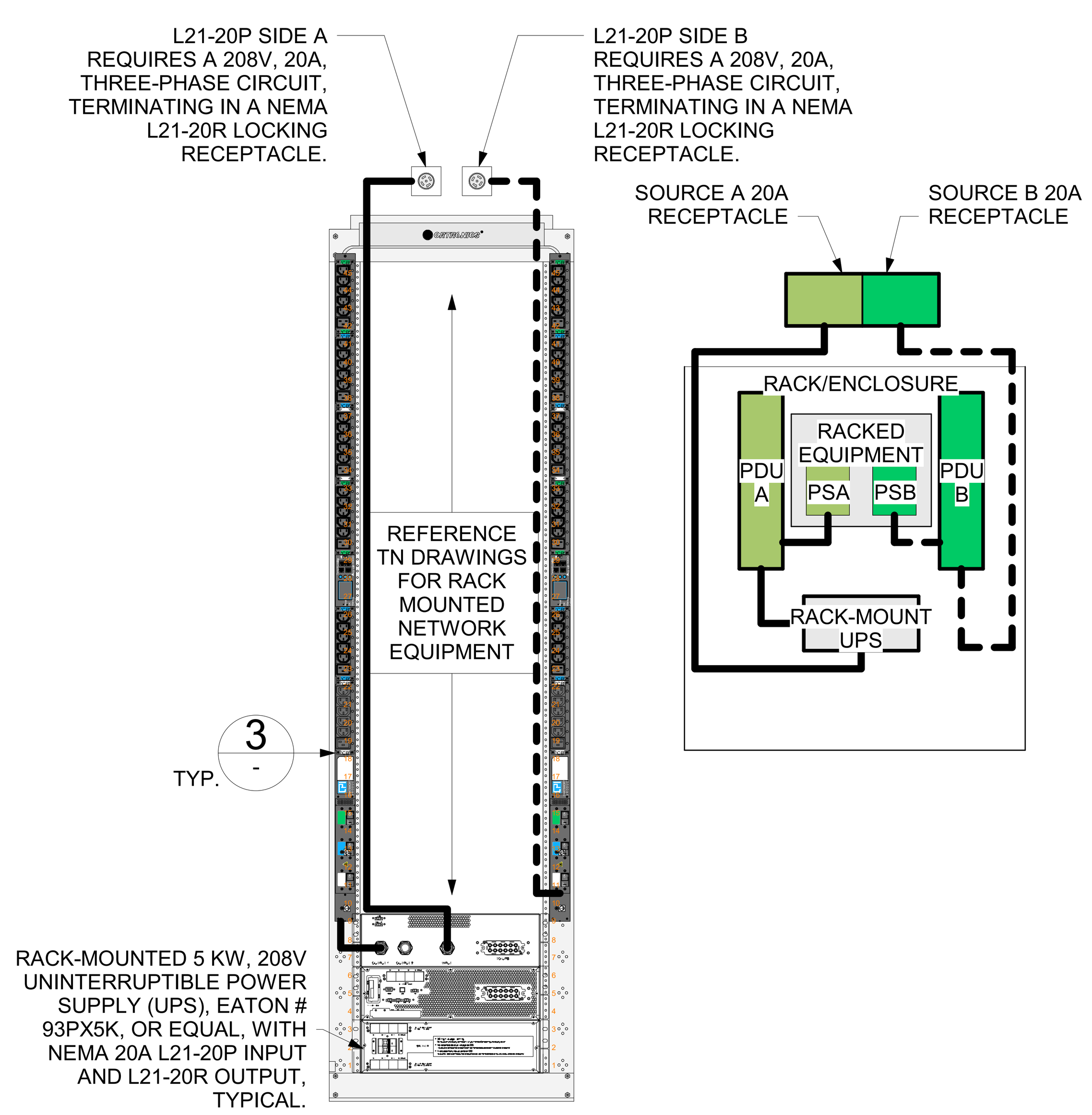
FIGURE 2: IEC 60320 C15 PLUG.
SOURCE: CISCO SYSTEMS INC.
ILLUSTRATION #113165

CAB-C15-CBN DETAILS:
1. CORDSET RATED FOR 250VAC, 13A, 60 HZ, 120° C (248° F) AT COUPLER PINS.
2. MAXIMUM OVERALL LENGTH OF 4 FT.
3. VERIFY CORDSET COMPATIBILITY WITH 208VAC SOURCE POWER BEFORE CONNECTING
4. VERIFY DEVICE COMPATIBILITY WITH 208VAC INPUT POWER BEFORE CONNECTING.

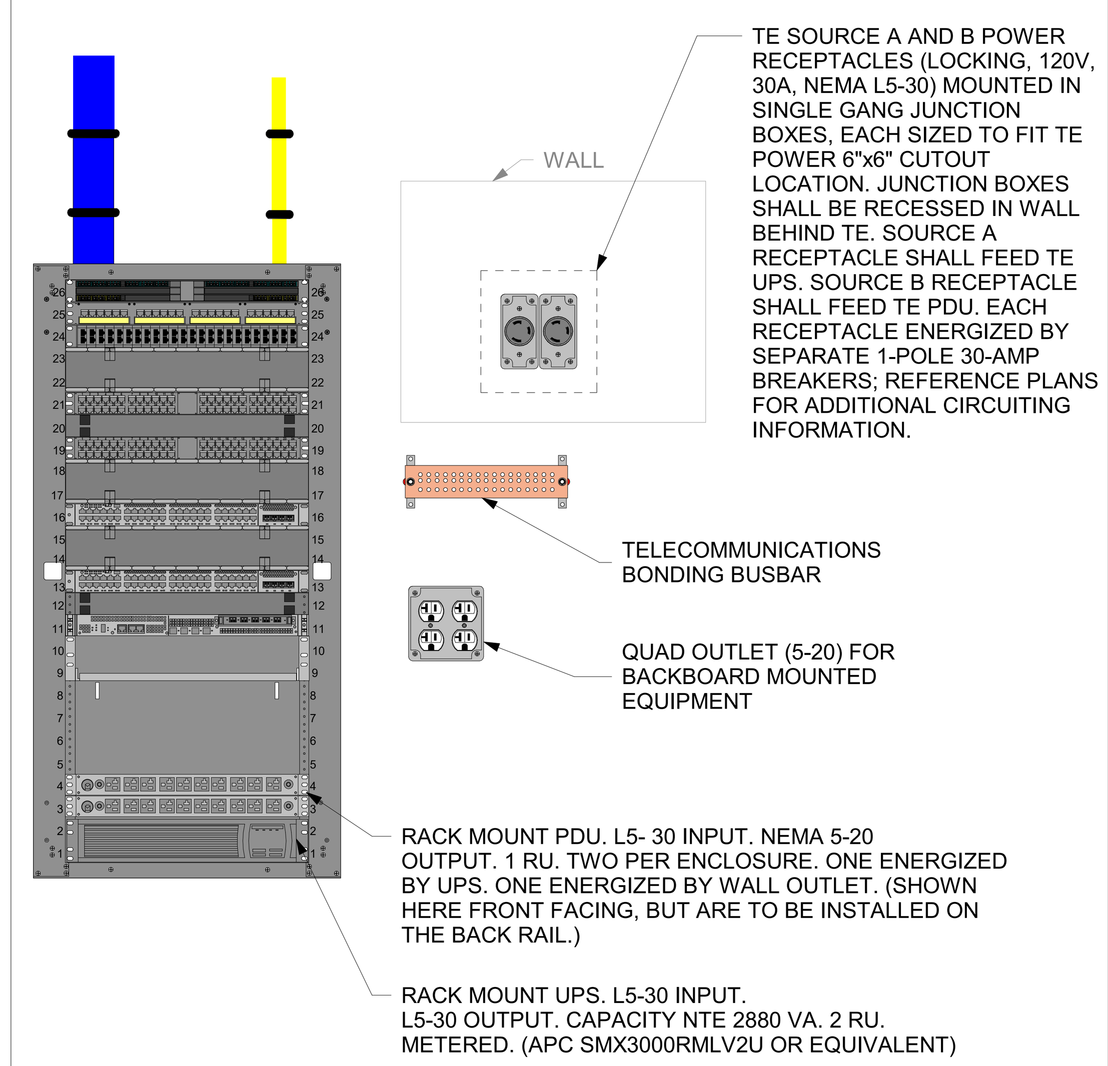
3 208V RACK PDU FOR TRs
NTS



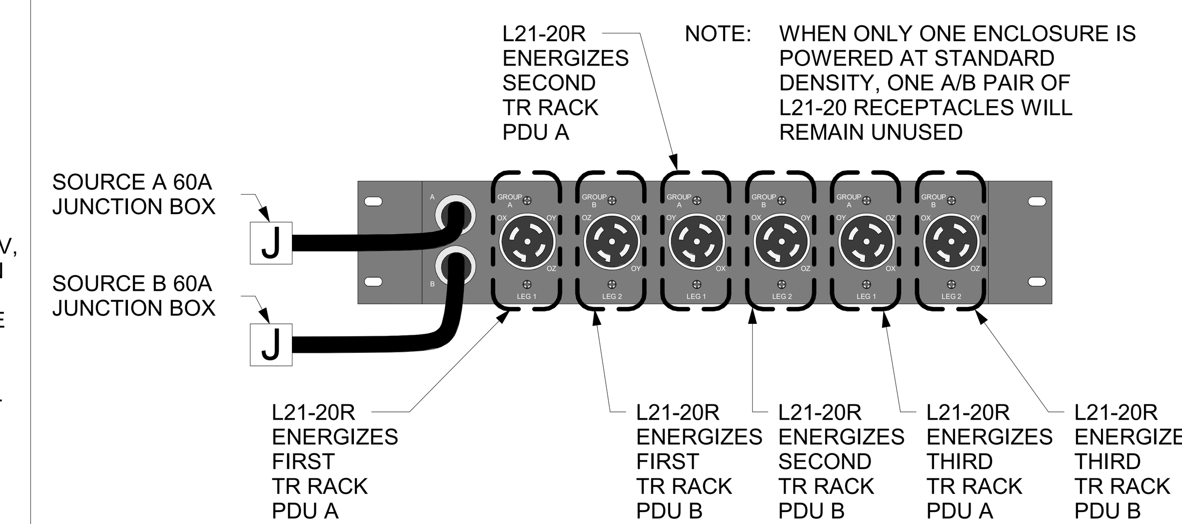
4 POWER ELEVATION AND SCHEMATIC FOR A TWO-RACK TR
NTS



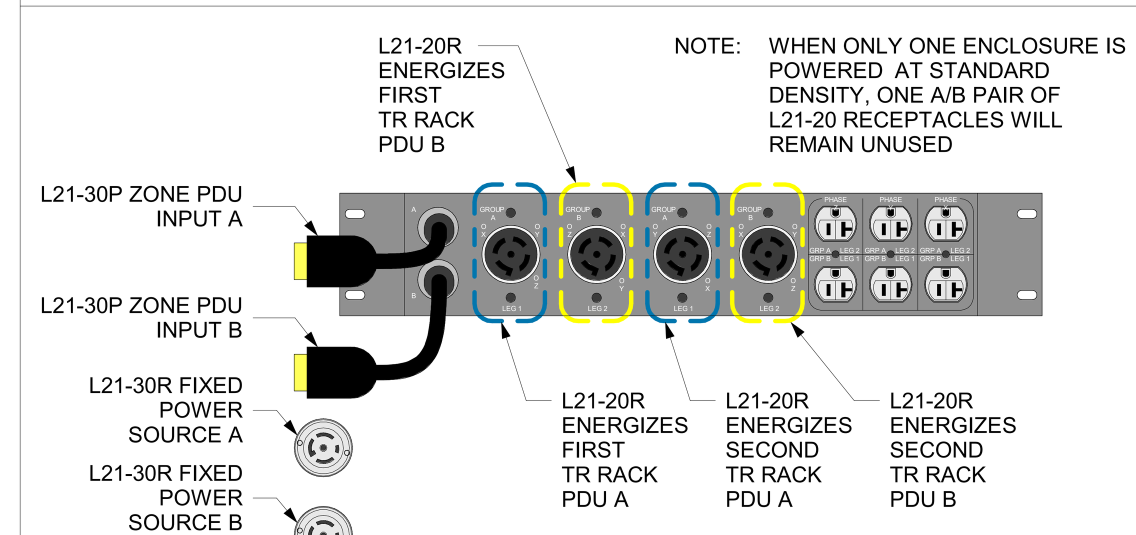
5 POWER ELEVATION AND SCHEMATIC FOR ONE-RACK TR
NTS



6 TELECOMMUNICATIONS ENCLOSURE ELEVATION DETAIL
NTS



7 CONNECTION DETAIL FOR 60A ZONE PDU UNIT SUPPLYING THREE-PHASE
NTS



8 CONNECTION DETAIL FOR 30A ZONE PDU UNIT SUPPLYING THREE-PHASE
NTS

Revisions:	Date:

CONSULTANT
ENGINEERING DISCIPLINE:

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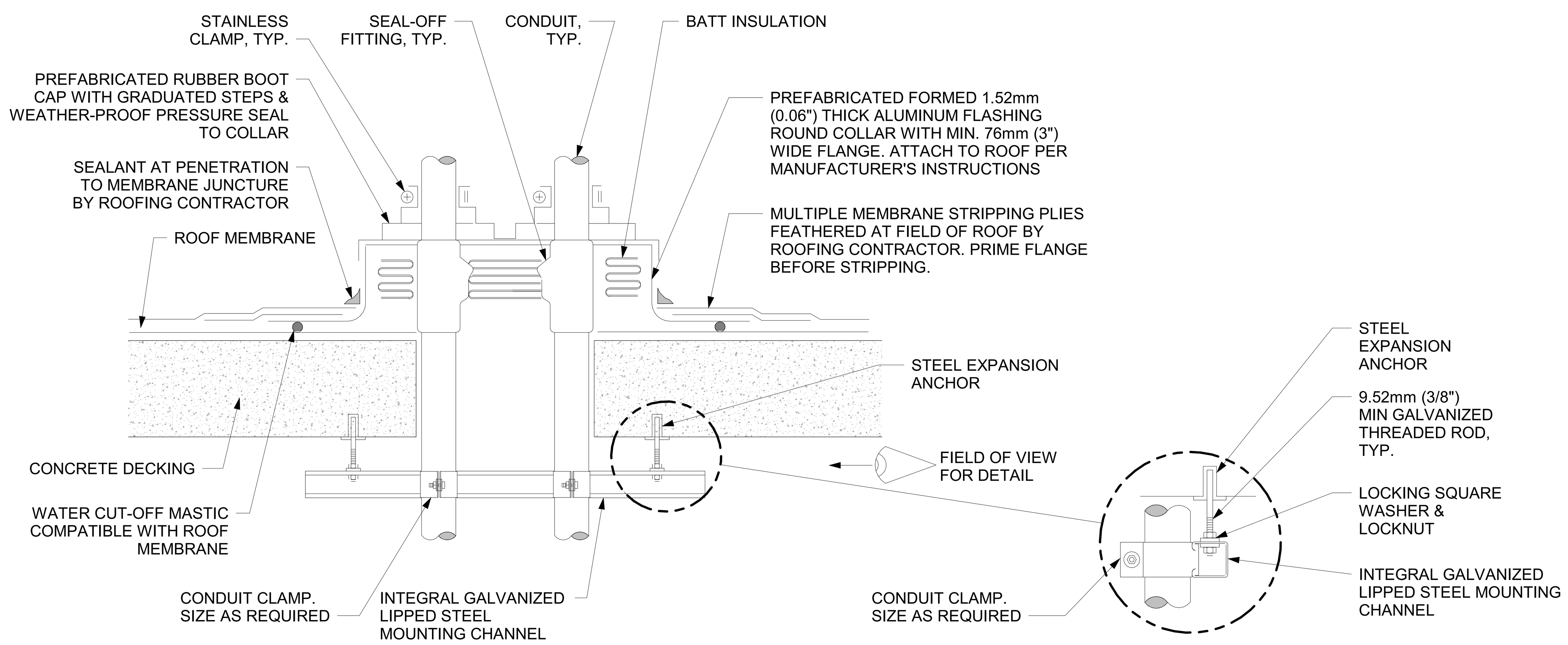
Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title: ELECTRICAL DETAILS
Approved:
Phase: 100% CONSTRUCTION DOCUMENTS
FULLY SPRINKLERED
Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date: 09/03/2025
Checked: DCW/NVP
Drawn: RTS/JNV
Project Number: 581-22-700
Building Number: -
Drawing Number: EP502
440 OF 693

VA FORM 08 - 6231

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:20:57 AM



GENERAL NOTES:

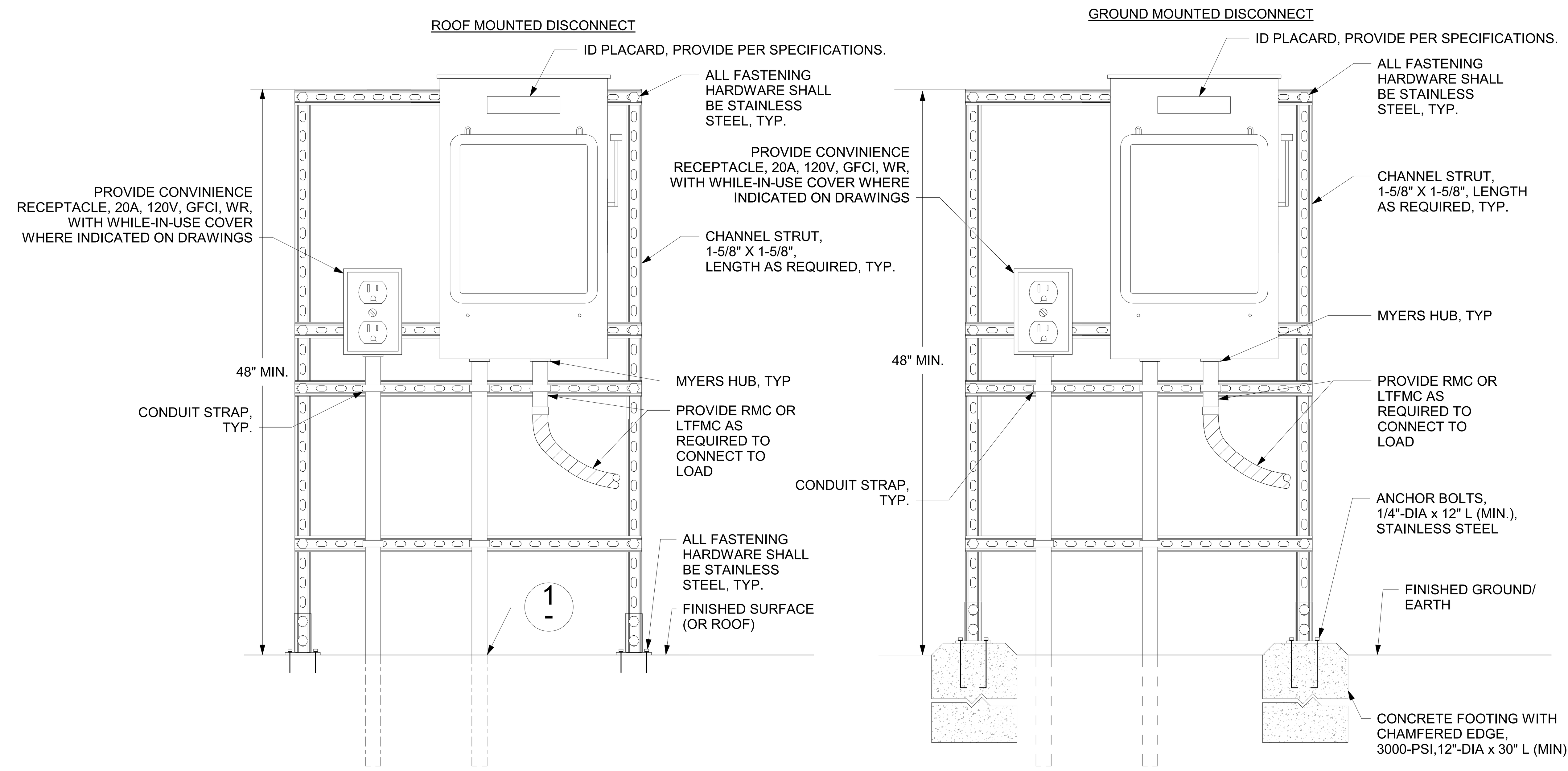
1. MAINTAIN A MINIMUM CLEARANCE OF 305mm (12") ON ALL SIDES OF ROOF PENETRATION FROM WALLS, CURBS, AND OTHER PROJECTIONS TO FACILITATE PROPER FLASHING.
2. FLANGES OF ADJACENT FLASHINGS SHALL NOT BE CUT OR OVERLAPPED.
3. VERIFY ROOF & STRUCTURAL SYSTEM WITH ARCHITECT.
4. COORDINATE FLASHING INSTALLATION WITH ROOFING CONTRACTOR TO ENSURE PROPER METHODS & MATERIALS ARE USED TO MAINTAIN ROOF WARRANTY.

1 EP503 CONDUIT ROOF PENETRATION DETAIL

NTS

DETAIL NOTES:

1. ALL CHANNEL STRUT, POST BASES AND ACCESSORIES SHALL BE HOT-DIPPED GALVANIZED, UNLESS SPECIFICALLY NOTED OTHERWISE.
2. DETAIL INDICATES GENERAL LAYOUT REQUIREMENTS. CONTRACTOR SHALL SIZE RACK ACCORDINGLY TO MOUNT ALL REQUIRED EQUIPMENT AS SHOWN ON PLANS; ADDITIONAL HORIZONTAL AND/OR VERTICAL STRUCTURAL MEMBERS MAY BE REQUIRED.
3. WHERE SHOWN ON PLANS, PROVIDE RACK MOUNTED CONVENIENCE RECEPTACLE, 20A, 120V, GFCI, WP, WITH WHILE-IN-USE COVER. PROVIDE ALL REQUIRED CONDUIT, STRUT, AND FASTENING HARDWARE AS REQUIRED.

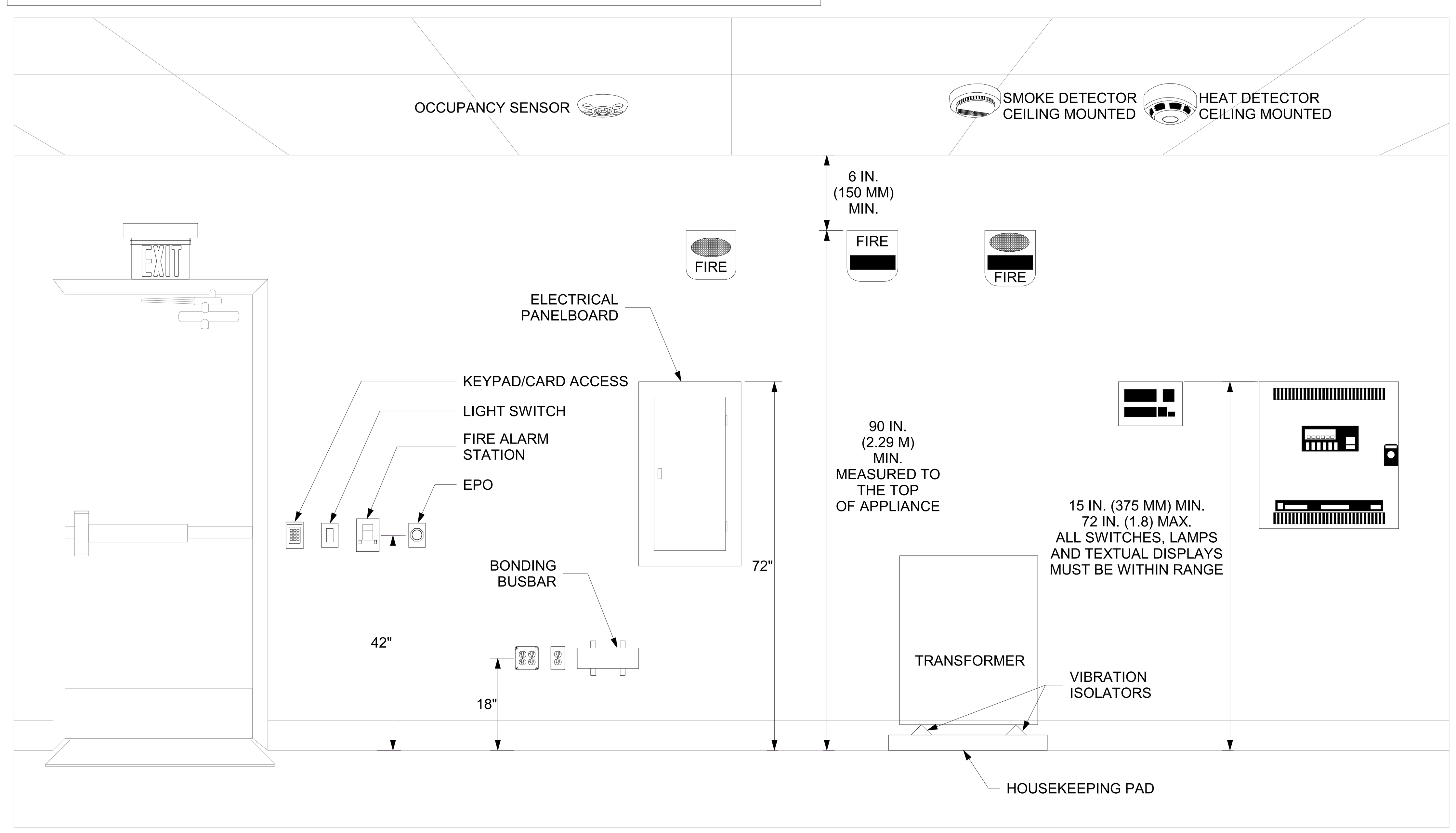


2 - TYPICAL UNISTRUT RACK MOUNTED DISCONNECT SWITCH

NTS

DETAIL NOTE(S):

1. THESE TYPICAL MOUNTING HEIGHTS SHALL BE USED UNLESS OTHERWISE NOTED ON THE PLANS.
2. EQUIPMENT HOUSEKEEPING PADS SHALL BE 5000-PSI CONCRETE, 3-1/2"H, #4 REBAR EVERY 12" ON CENTER. COORDINATE WITH CAST-IN-PLACE CONCRETE SPECIFICATION FOR ADDITIONAL INFORMATION. HOUSEKEEPING PAD SHALL EXTEND 4" BEYOND EQUIPMENT FOOTPRINT.



3 - TYPICAL MOUNTING HEIGHT DETAIL

NTS

4 - NOT USED

Revisions:	Date:

CONSULTANT

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REGISTERED PROFESSIONAL ENGINEER

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: ELECTRICAL DETAILS

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

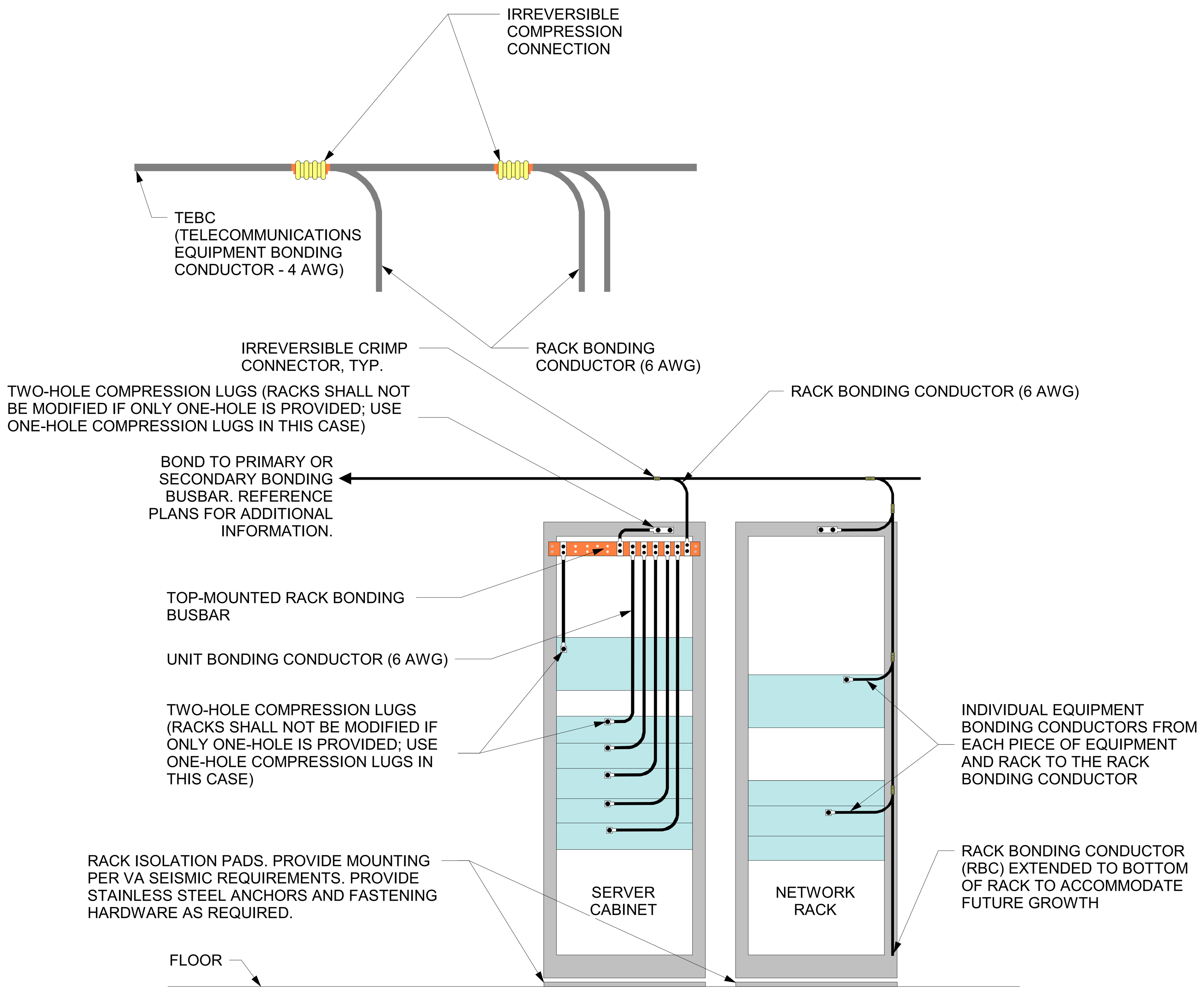
Project Number: 581-22-700

Building Number: -

Drawing Number: EP503

441 OF 693

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:20:01 AM

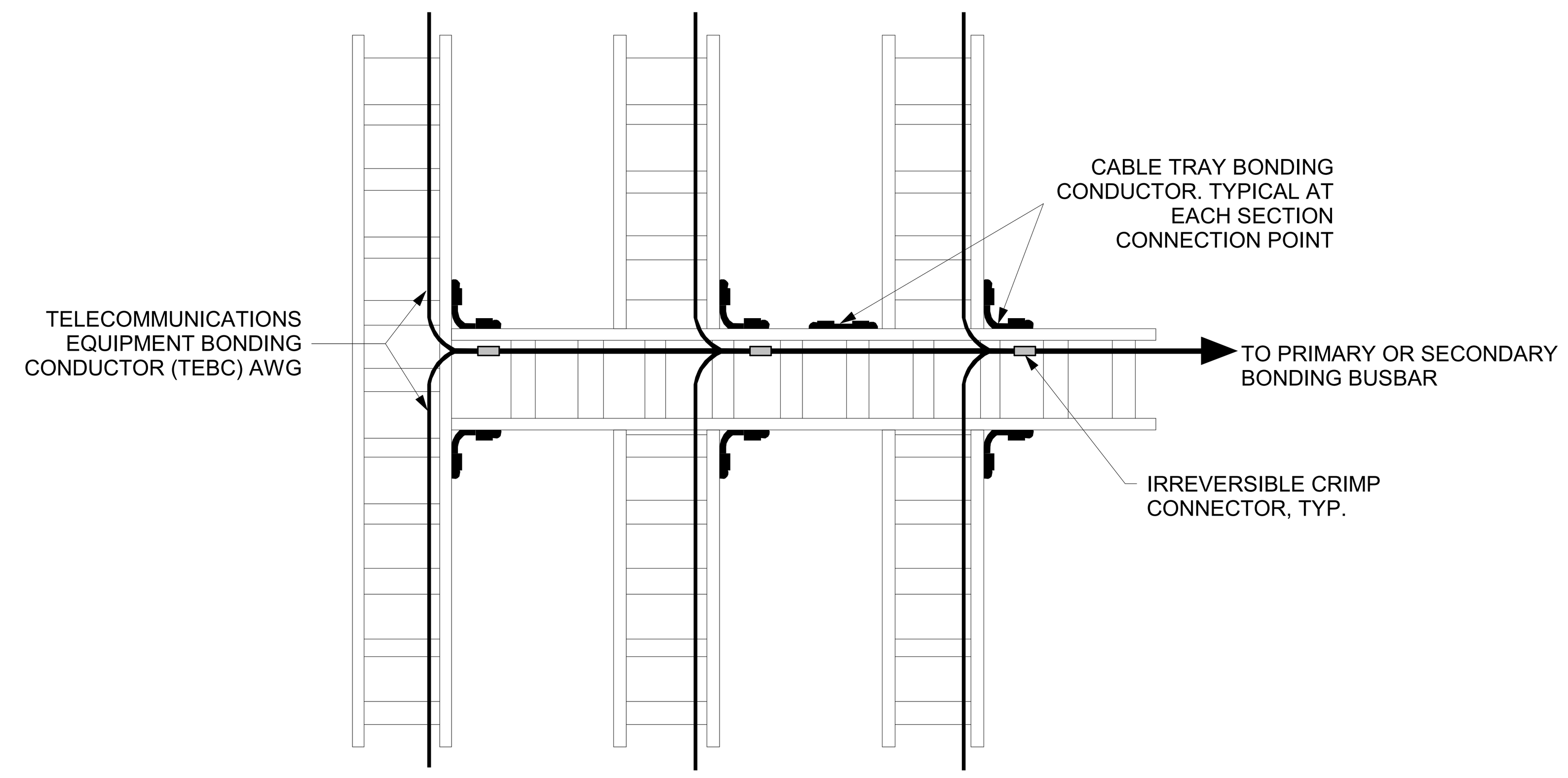


1 - TYPICAL TELECOMMUNICATIONS CABINET & RACK BONDING DETAIL

NTS

DETAIL NOTES:

1. ALL BONDING CONDUCTORS SHALL FOLLOW SIZING REQUIREMENTS SPECIFIED IN ANSI/TIA-607-D.
2. REFERENCE SHEET EG601 AND EG602 FOR ADDITIONAL INFORMATION.

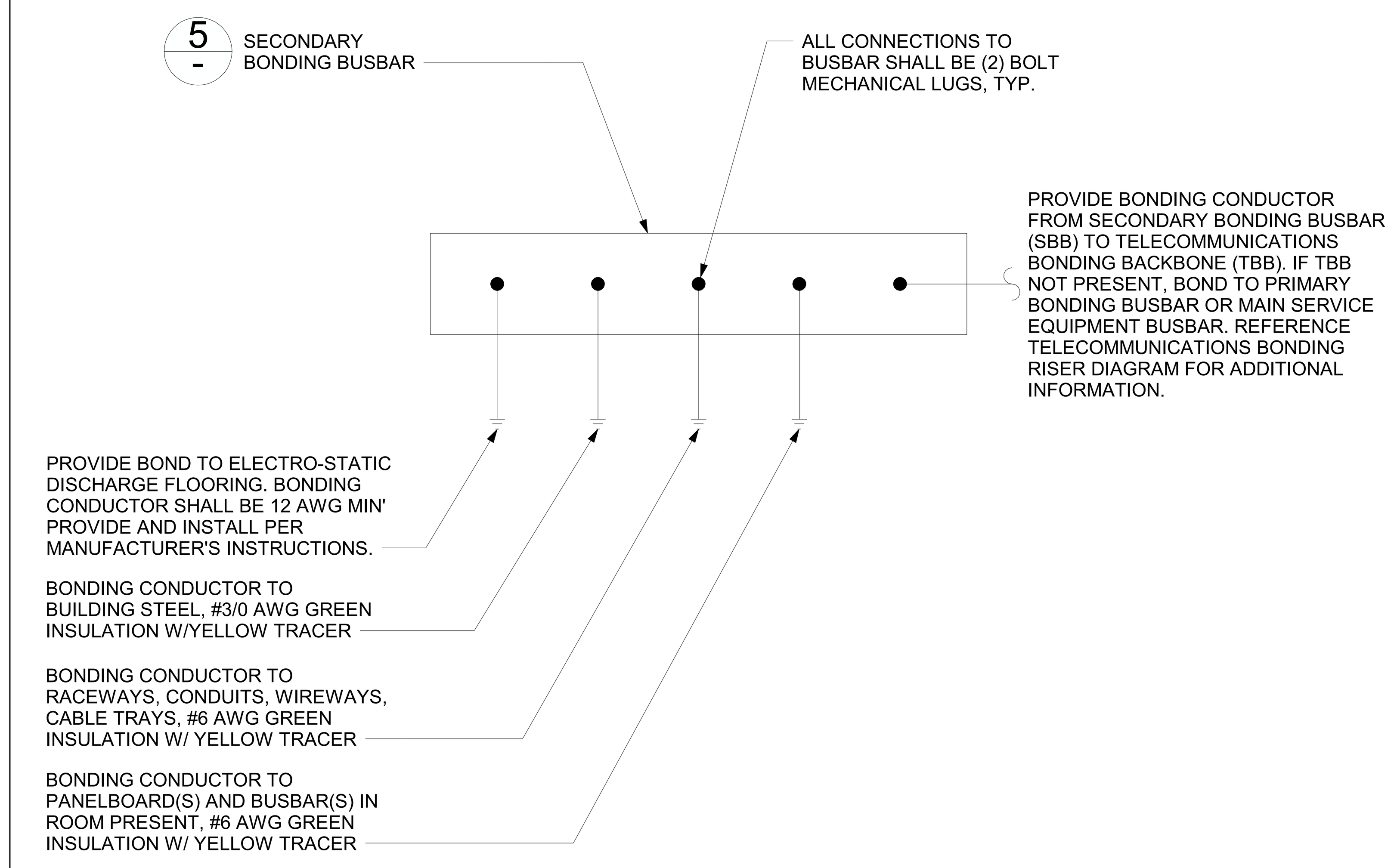


2 - CABLE TRAY GROUNDING/BONDING REQUIREMENTS

NTS

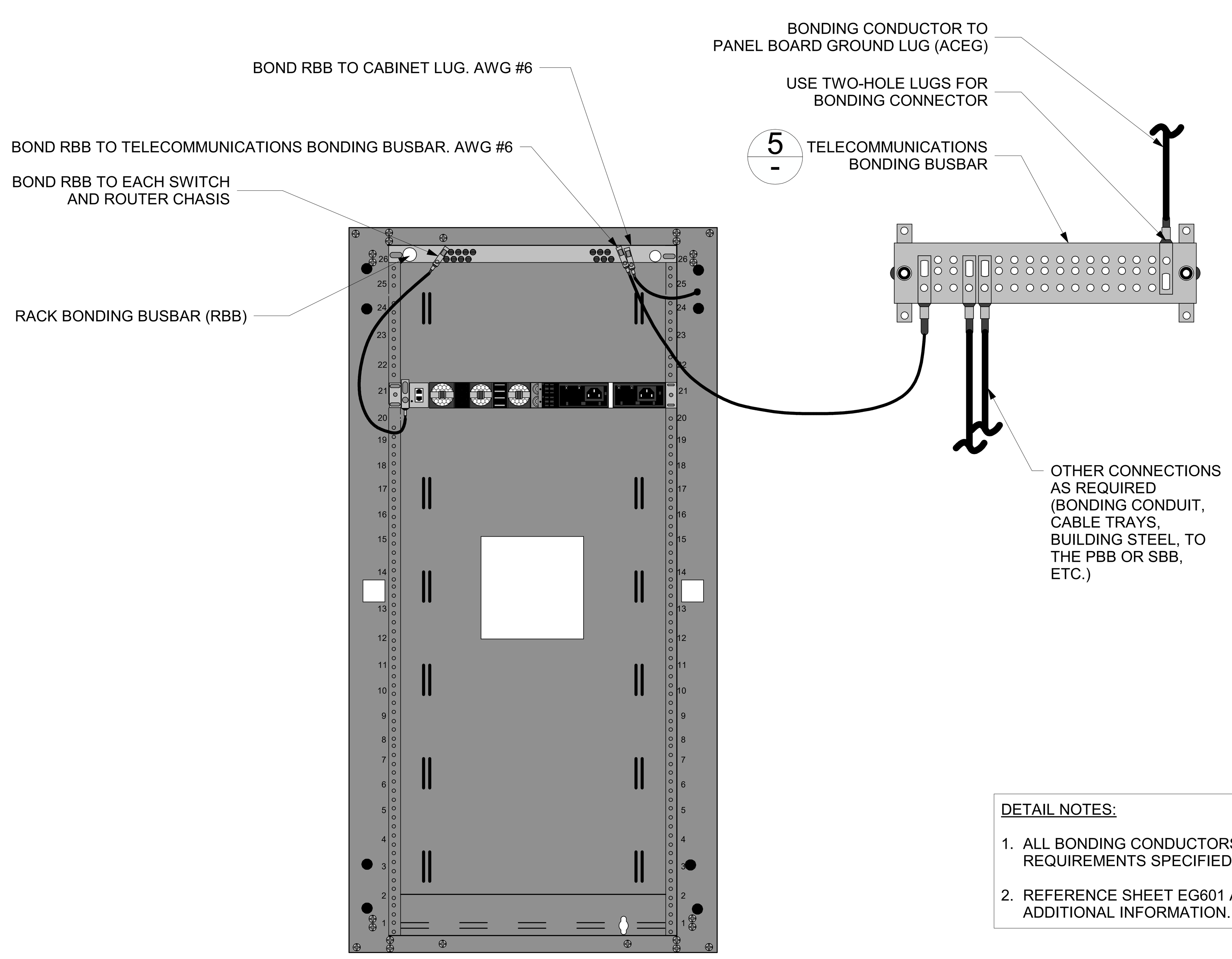
DETAIL NOTES:

1. ALL BONDING CONDUCTORS SHALL FOLLOW SIZING REQUIREMENTS SPECIFIED IN ANSI/TIA-607-D.
2. REFERENCE SHEET EG601 AND EG602 FOR ADDITIONAL INFORMATION.



3 - SBB BONDING DETAIL

NTS



DETAIL NOTES:

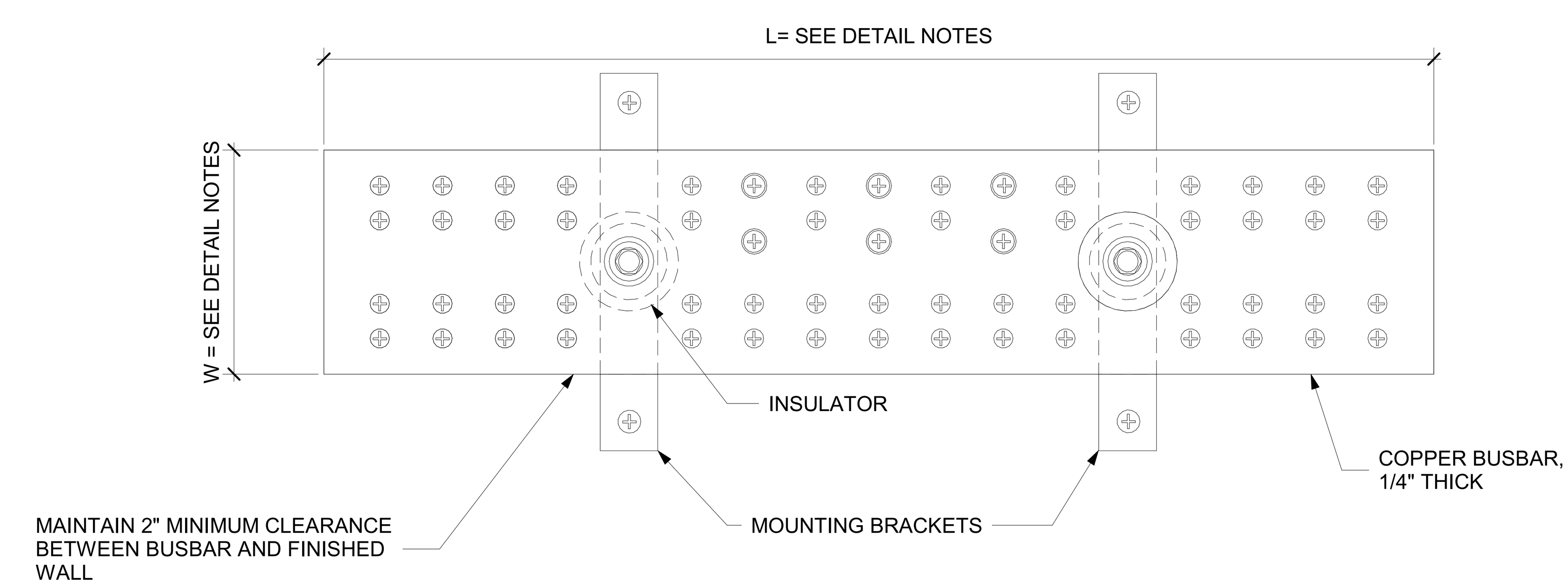
1. ALL BONDING CONDUCTORS SHALL FOLLOW SIZING REQUIREMENTS SPECIFIED IN ANSI/TIA-607-D.
2. REFERENCE SHEET EG601 AND EG602 FOR ADDITIONAL INFORMATION.

4 - TELECOMMUNICATIONS ENCLOSURE GROUNDING DETAIL

NTS

DETAIL NOTES:

1. SECONDARY BONDING BUSBAR (SBB) SHALL BE 16 INCHES LONG, 2 INCHES WIDE.
2. PRIMARY BONDING BUSBAR (PBB) SHALL BE 20 INCHES LONG, 4 INCHES WIDE.



5 - PBB AND SBB BONDING BUSBAR DETAIL

NTS

Revisions:	Date:

CONSULTANT

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ARCHITECT/ENGINEER OF RECORD

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SPEESDESIGNBUILD

STAMP

LOUIS R. LITZKO · STATE OF WASHINGTON · 55681 · REGISTERED PROFESSIONAL ENGINEER · 4-3-25

Office of Construction and Facilities Management

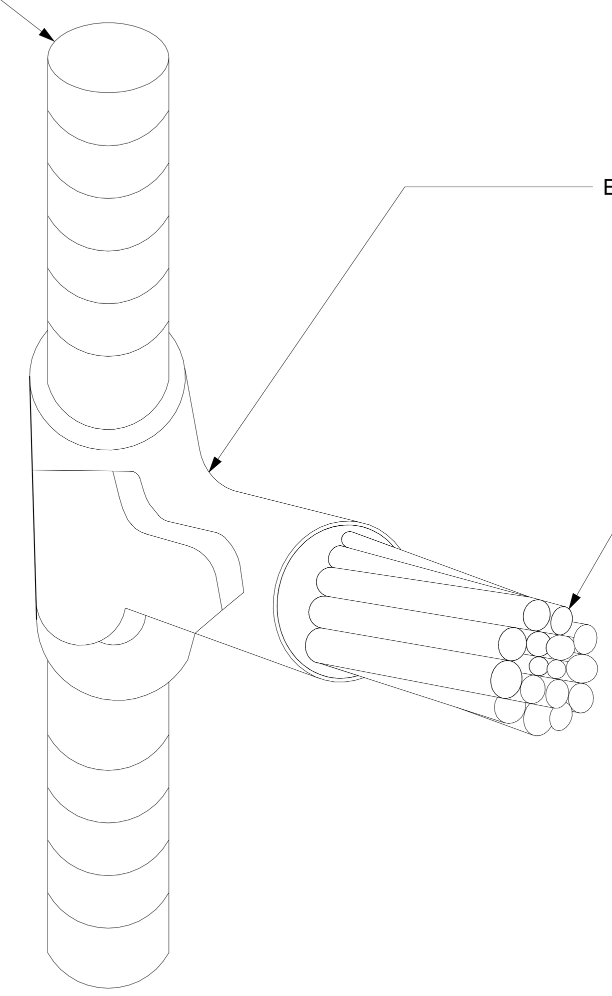
VA U.S. Department of Veterans Affairs

Drawing Title: ELECTRICAL DETAILS	Phase: 100% CONSTRUCTION DOCUMENTS	Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	Project Number: 581-22-700
			Building Number: -
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Drawing Number: EP504
		Issue Date: 09/03/2025 · Checked: DCW/NVP · Drawn: RTS/JNV	442 OF 693

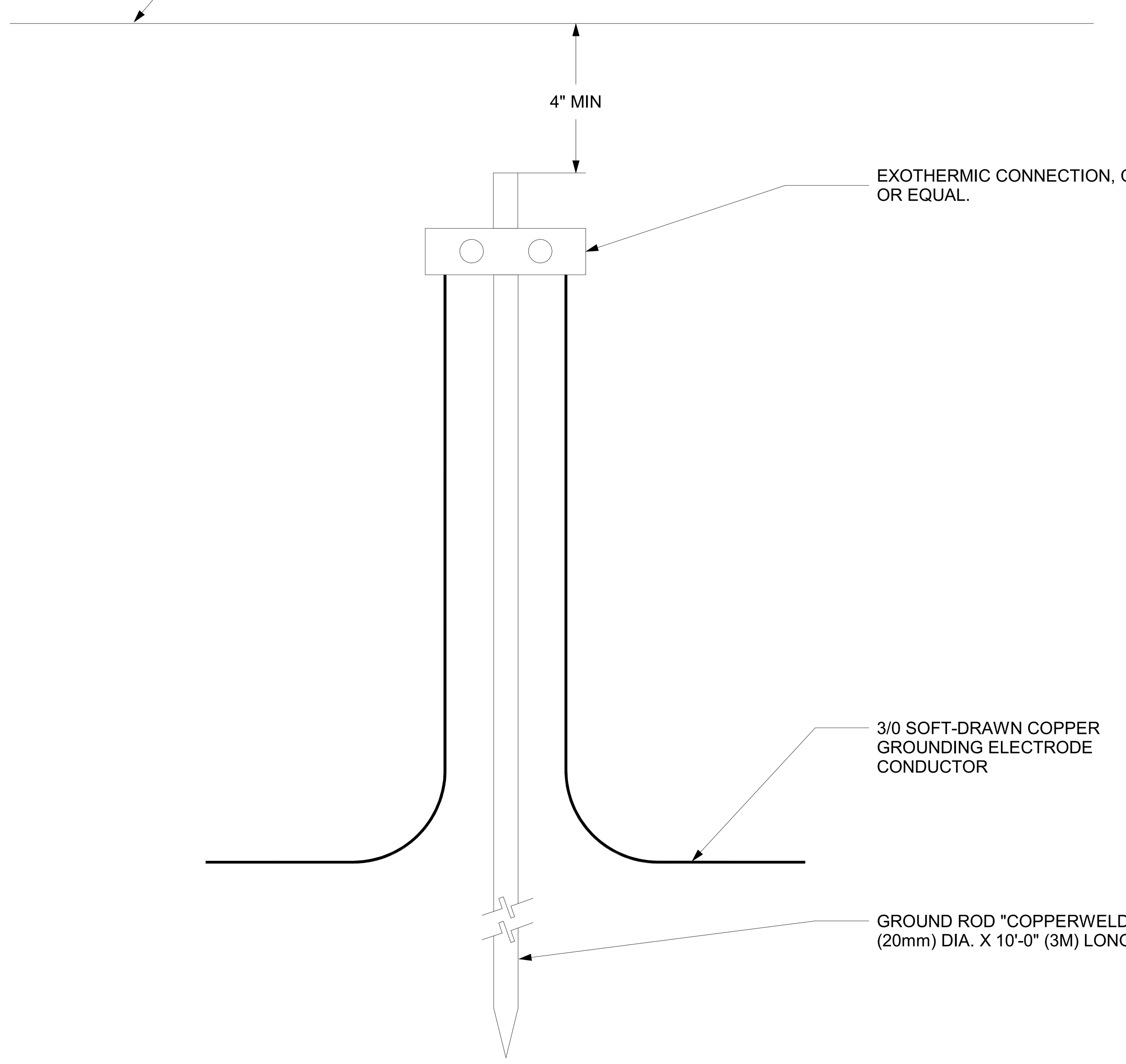
BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:20:08 AM

VA FORM 08 - 6231

DETAIL NOTES:
1. ALL CONDUCTORS SHALL BE SIZED PER PLANS. EXOTHERMIC CONNECTOR SHALL BE SIZED PER CONDUCTOR AND STRUCTURAL STEEL.



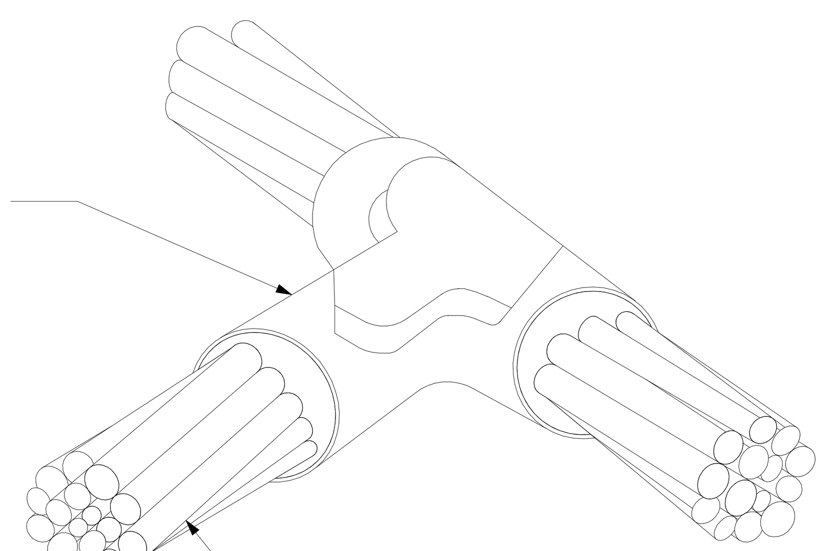
1 TYPICAL BUILDING STRUCTURE EXOTHERMIC CONNECTION DETAIL
-
NTS



2 TYPICAL BURIED GROUND ROD DETAIL
01EG400Ga-1B
NTS

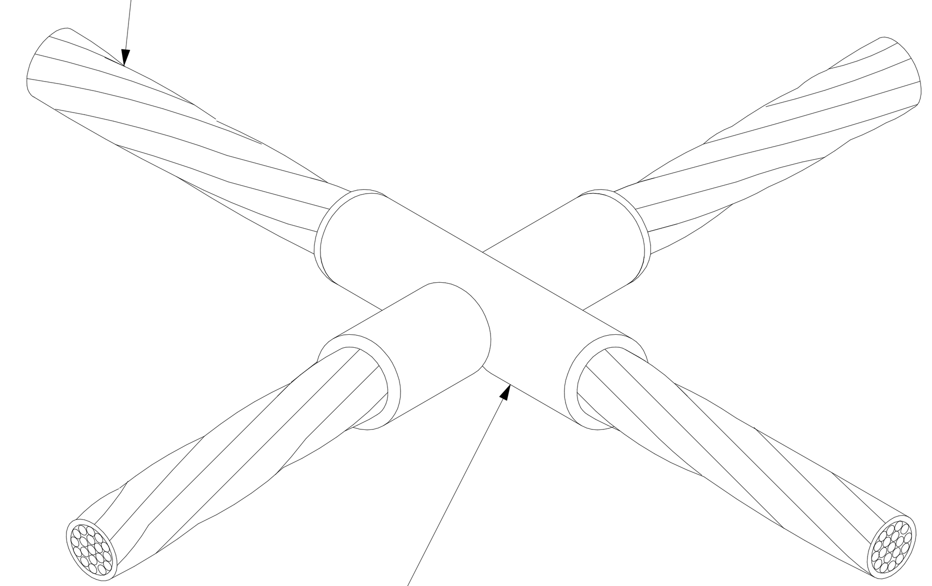
3 NOT USED
-
NTS

DETAIL NOTES:
1. ALL CONDUCTORS SHALL BE SIZED PER PLANS. EXOTHERMIC CONNECTOR SHALL BE SIZED PER CONDUCTORS.

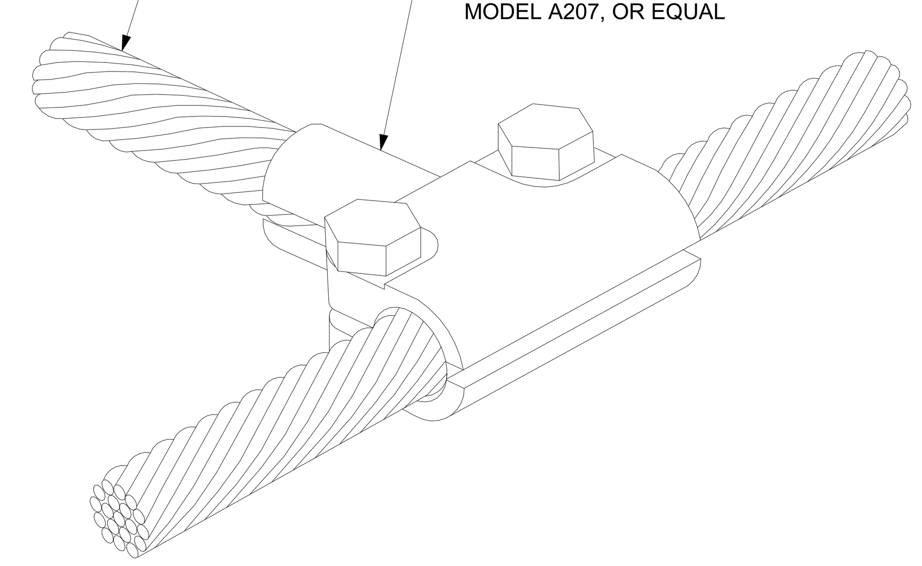


4 TYPICAL EXOTHERMIC "T" CONNECTION DETAIL
01EG400Ga-1B
NTS

DETAIL NOTES:
1. ALL CONDUCTORS SHALL BE SIZED PER PLANS. EXOTHERMIC CONNECTOR SHALL BE SIZED PER CONDUCTORS.

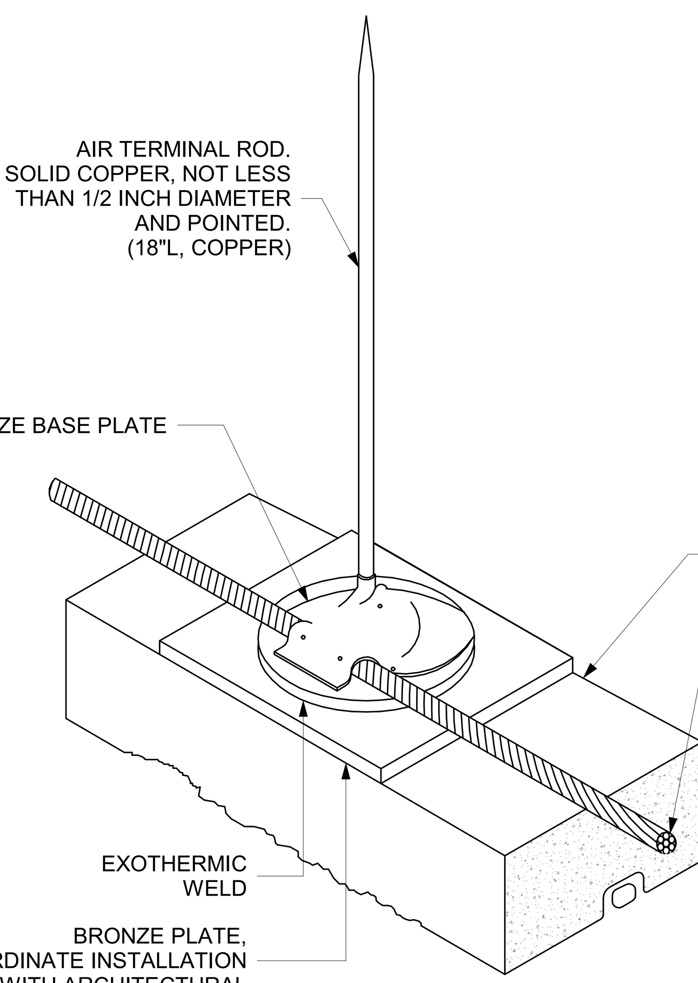


5 TYPICAL EXOTHERMIC CROSS CONNECTION DETAIL
01EG400Ga-1B
NTS



6 LIGHTNING CONDUCTOR TEE CONNECTOR
01EG400Gg-2B
NTS

DETAIL NOTES:
1. BRONZE CLAMP-TYPE CONNECTORS SHALL BE USED FOR ROOF CONDUCTOR SPLICES, AND THE CONNECTION TO THE ROOF CONDUCTOR TO AIR TERMINALS AND BONDING PLATES.
2. CRIMP-TYPE CONNECTORS ARE NOT ALLOWED.



7 NON-PEAKED AIR TERMINAL ROD INSTALLATION DETAIL
01EG400Gg-2B
01SEP407
01WEP403
NTS

Revisions:	Date:

CONSULTANT

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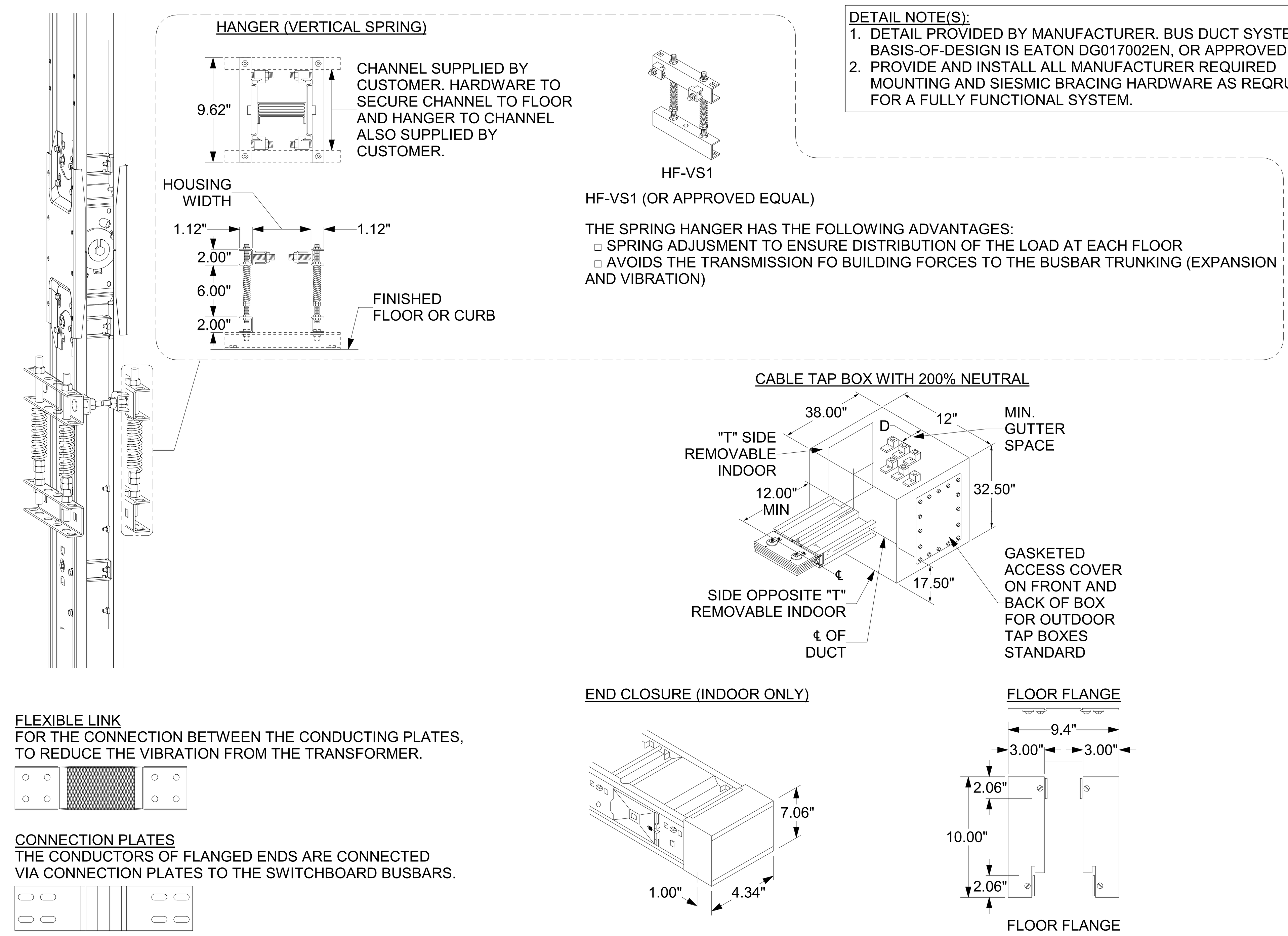
Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

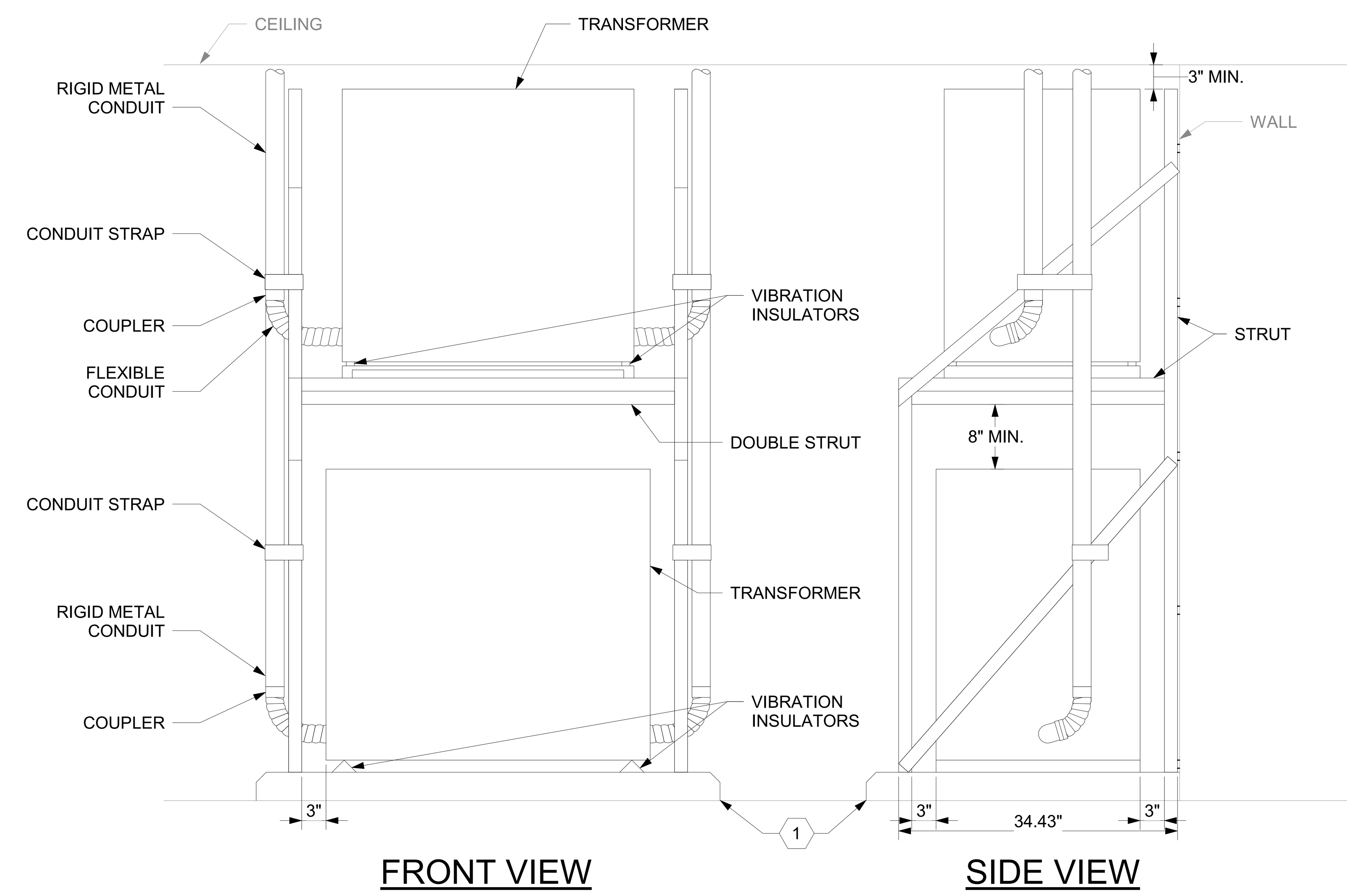
Drawing Title	Phase	Project Title	Project Number
ELECTRICAL DETAILS	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700
			Building Number: -
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Drawing Number: EP505
		Issue Date: 09/03/2025; Checked: DCW/NVP; Drawn: RTS/JNV	443 OF 693

DETAIL NOTE(S):
1. DETAIL PROVIDED BY MANUFACTURER. BUS DUCT SYSTEM BASIS-OF-DESIGN IS EATON DG017002EN, OR APPROVED EQUAL.
2. PROVIDE AND INSTALL ALL MANUFACTURER REQUIRED MOUNTING AND SIESMIC BRACING HARDWARE AS REQRUIED FOR A FULLY FUNCTIONAL SYSTEM.



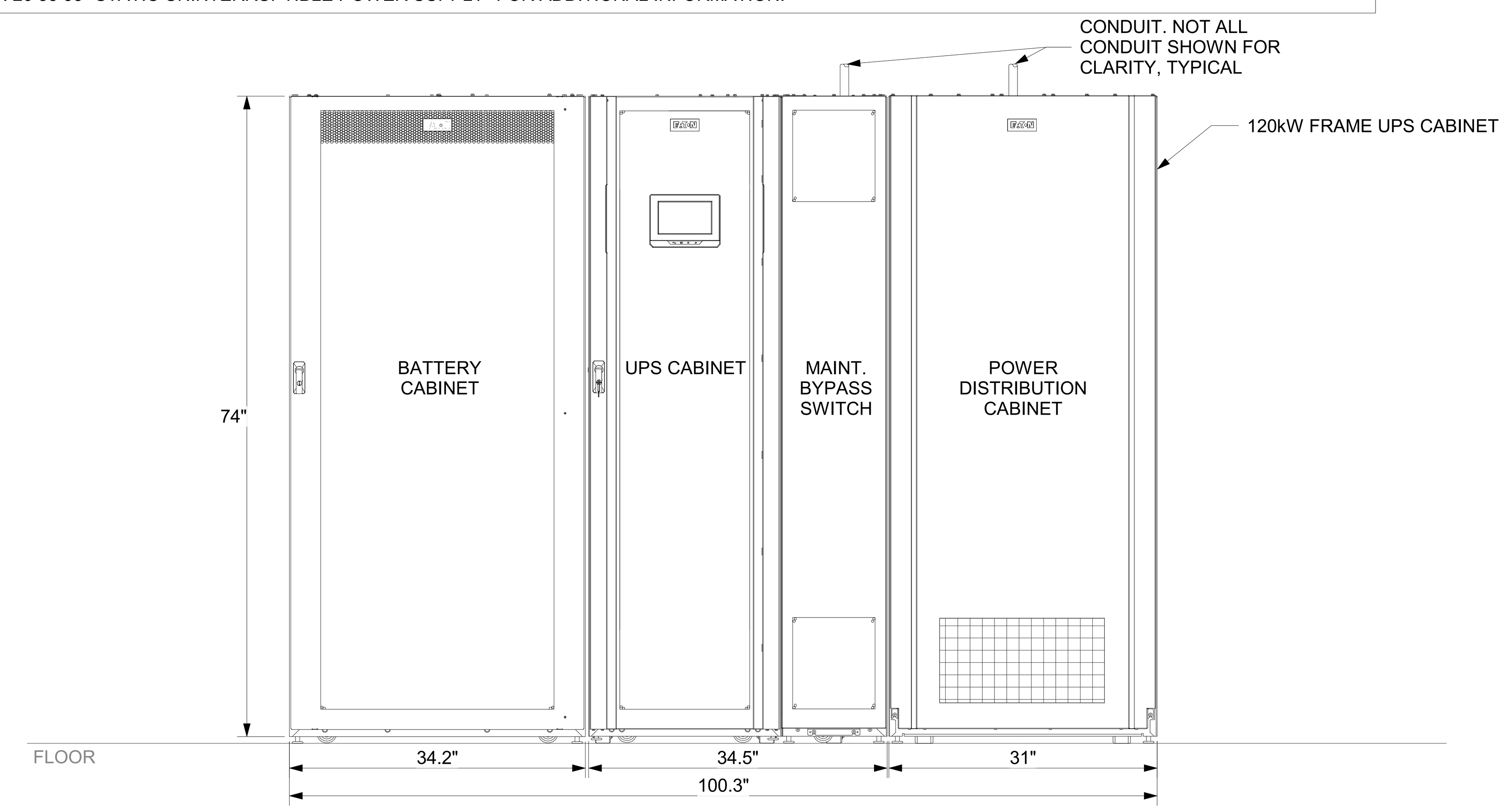
1 - TYPICAL VERTICAL BUS DUCT RISER DETAIL
NTS

DETAIL NOTE(S):
1. TYPICAL DETAIL SHALL BE USED FOR CONSTRUCTABILITY REFERENCE. EXACT DIMENSIONS, MOUNTING SUPPORTS, AND FASTENING HARDWARE MAY DIFFER FROM FIELD INSTALLATION.
2. ALL CONDUIT NOT SHOWN FOR CLARITY.
3. WHERE A NEW TRANSFORMER IS BEING MOUNTED ABOVE AN EXISTING TRANSFORMER, THE FACE OF THE NEW TRANSFORMER SHALL BE EVEN WITH THE FACE OF THE EXISTING TRANSFORMER. UNISTRUT STRUT MOUNTING SUPPORTS SHALL BE ARRANGED AS REQUIRED TO COORDINATE WITH EXISTING CONDITIONS.

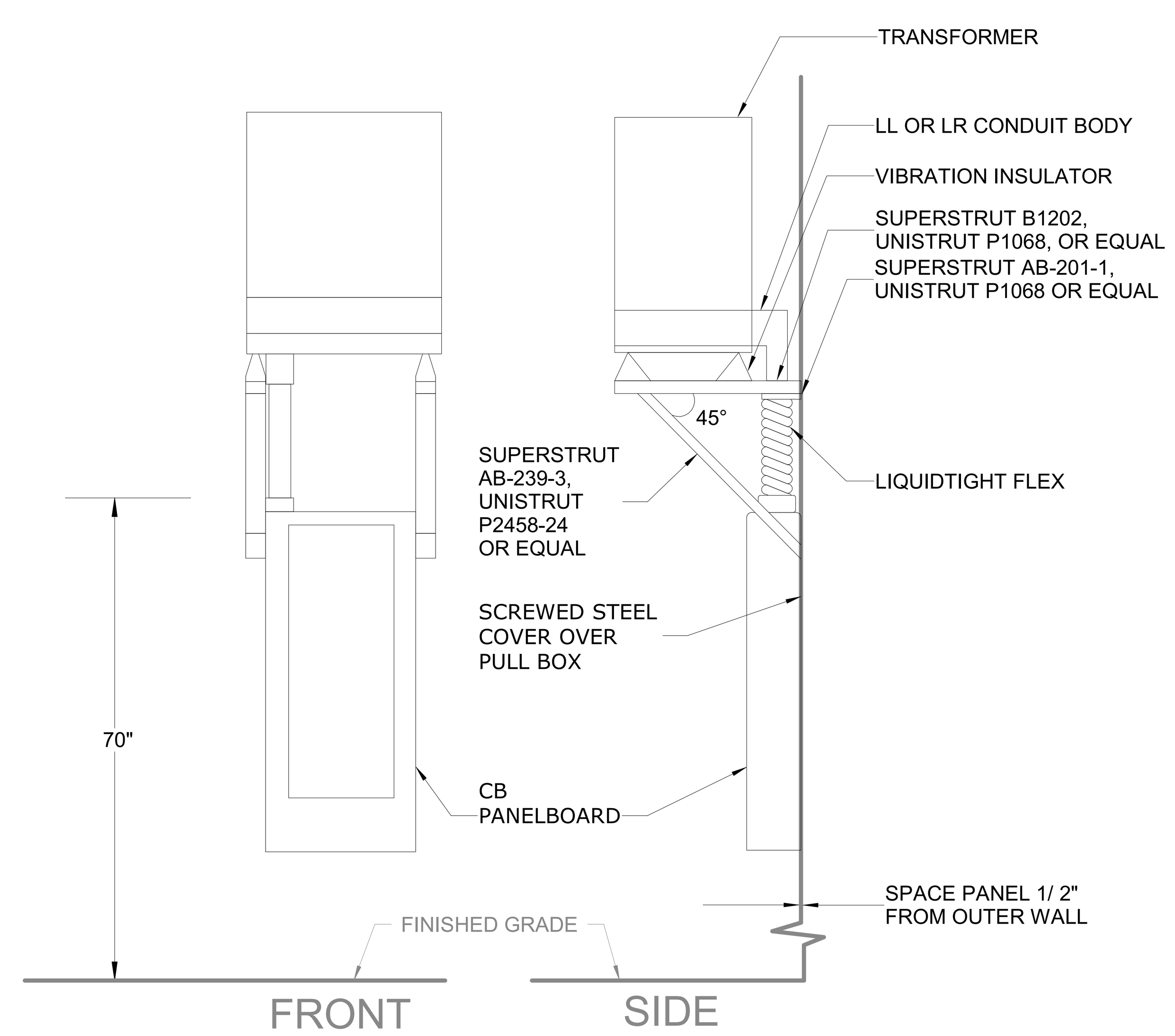


2 - TYPICAL TRANSFORMER STACKING MOUNTING DETAIL
1" = 1'-0"

DETAIL NOTE(S):
1. UPS A AND UPS B SHALL HAVE A MIRRORED LAYOUT WITH POWER DISTRIBUTION CABINETS CLOSEST TO MAIN DISTRIBUTION AREA AND BATTERY CABINET FURTHEST AWAY.
2. MAINTENANCE BYPASS SWITCH CAN BE INSTALLED ON THE RIGHT OR LEFT OF UPS CABINET.
3. EACH CABINET DEPTH IS 42"
4. UPS IS TOP AIR EXHAUSTED.
5. UPS SHALL BE ENERGY SAVER SYSTEM (ESS) PROVIDING 99% EFFECIENCY RATING.
6. UPS SHALL SUPPORT BOTH MONITORING AND REMOTE MANAGEMENT TO INCLUDE:
   A. FULL COMPLIANCE WITH RFC-1628 - UPS MANAGEMENT INFORMATION BASE.
   B. SUPPORT SNMP V3.0, BACNET AND MODBUS PROTOCOLS.
   C. PROVIDE CONNECTION TO BMS SYSTEM VIA CAT-6A ETHERNET CONNECTION.
   D. UPS SHALL BE MANAGED AND REPORT THE FOLLOWING TO THE FACILITY BMS:
      I. OVERLOAD (CRITICAL).
      II. BATTERY TEMP (CRITICAL).
      III. LOW BATTERY.
      IV. BATTERY FAILURE (CRITICAL.)
      V. GENERAL CRITICAL ALARM (IE: FAN FAILURE, INTERNAL FAULT, BYPASS, ETC.)
      VI. LOAD ON BYBASS.
7. UPS SHALL CONTAIN BATTERY MANAGEMENT SYSTEM.
8. REFERENCE SPECIFICATION 26 33 53 "STATIC UNINTERRUPTIBLE POWER SUPPLY" FOR ADDITIONAL INFORMATION.



3 - TYPICAL UPS ELEVATION
01EP400Gb
01EP400Gh
NTS



4 - PANELBOARD AND TRANSFORMER STACKING INSTALLATION DETAIL
NTS

GENERAL NOTES

1. NOT ALL RACEWAYS, BOXES, OR CONNECTIONS MAY BE SHOWN FOR CLARITY. REFERENCE ONE-LINE DIAGRAMS AND FLOOR PLANS FOR ADDITIONAL INFORMATION.
2. CONTRACTOR SHALL ENSURE ALL ELECTRICAL CLEARANCES ARE MET PRIOR TO INSTALLATION.

KEY NOTES

1. HOUSEKEEPING PAD, 5000-PSI CONCRETE, 3-1/2"H, #4 REBAR EVERY 12" ON CENTER. COORDINATE WITH CAST-IN-PLACE CONCRETE SPECIFICATION FOR ADDITIONAL INFORMATION. HOUSEKEEPING PAD SHALL EXTEND 4" BEYOND EQUIPMENT FOOTPRINT.

Revisions:	Date:

CONSULTANT
ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD
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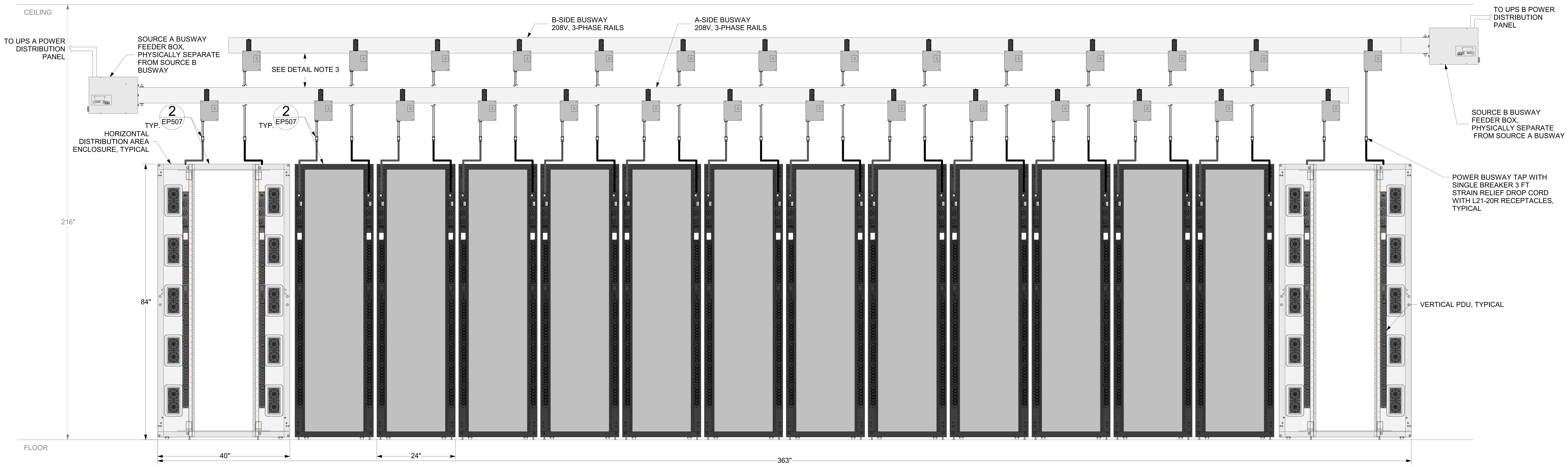
Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
ELECTRICAL DETAILS	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Building Number: -
		Issue Date: 09/03/2025 · Checked: DCW/NVP · Drawn: RTS/JNV	Drawing Number: EP506

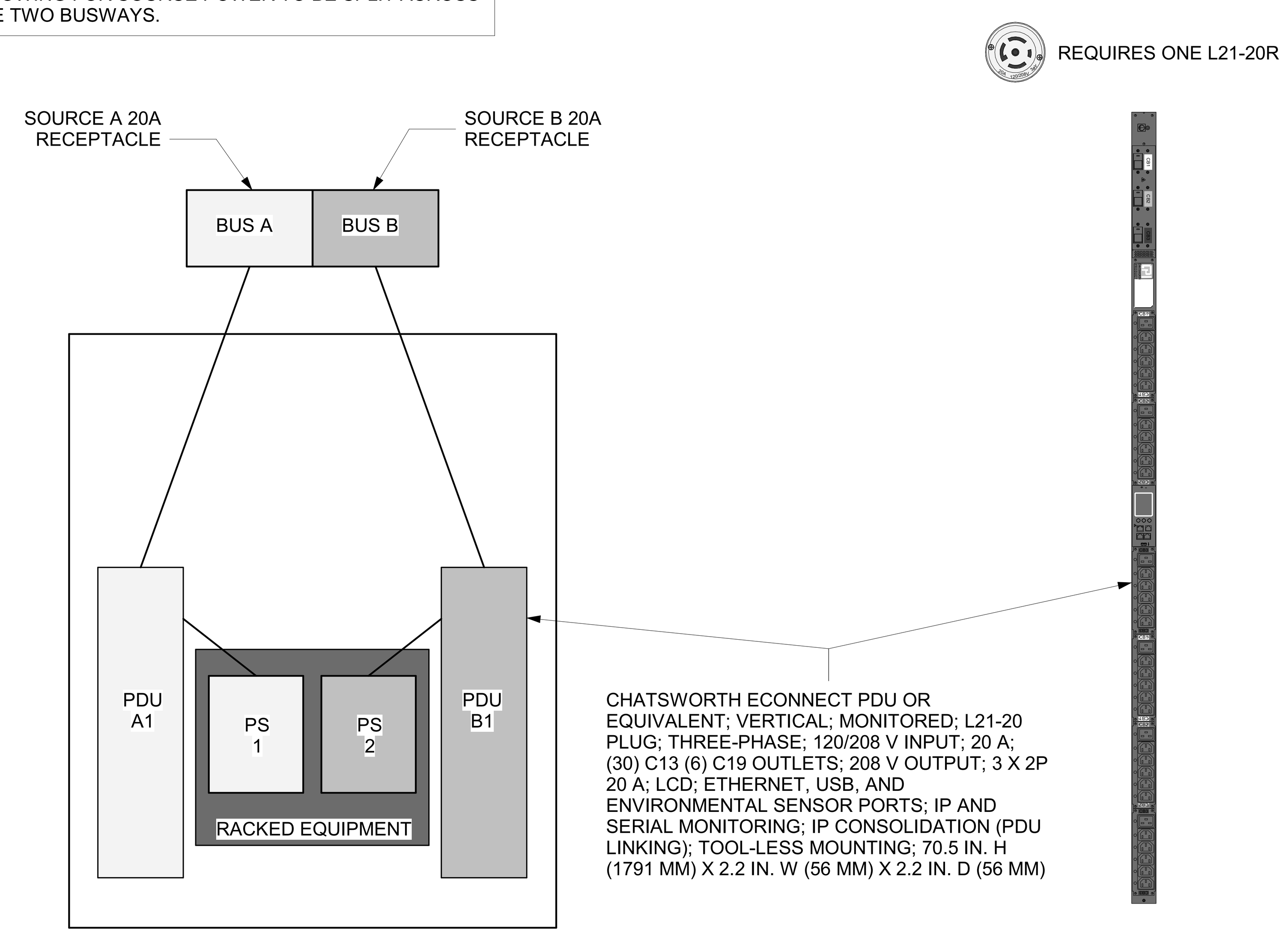


DETAIL NOTES:
1. MDA A AND B WILL HAVE A MIRRORED LAYOUT, WITH THE OUTER MOST CABINETS BEING THE TR DISTRIBUTION CABINETS.
2. EACH CABINET HAS A DEPTH OF 43.3".
3. BUSWAY 'A' AND 'B' ARE PHYSICALLY SEPARATE BY AT LEAST 1-FT HORIZONTALLY AND 1-FT VERTICALLY WITH SOURCE 'B' BUSWAY FURTHEST FROM THE CABINETS.

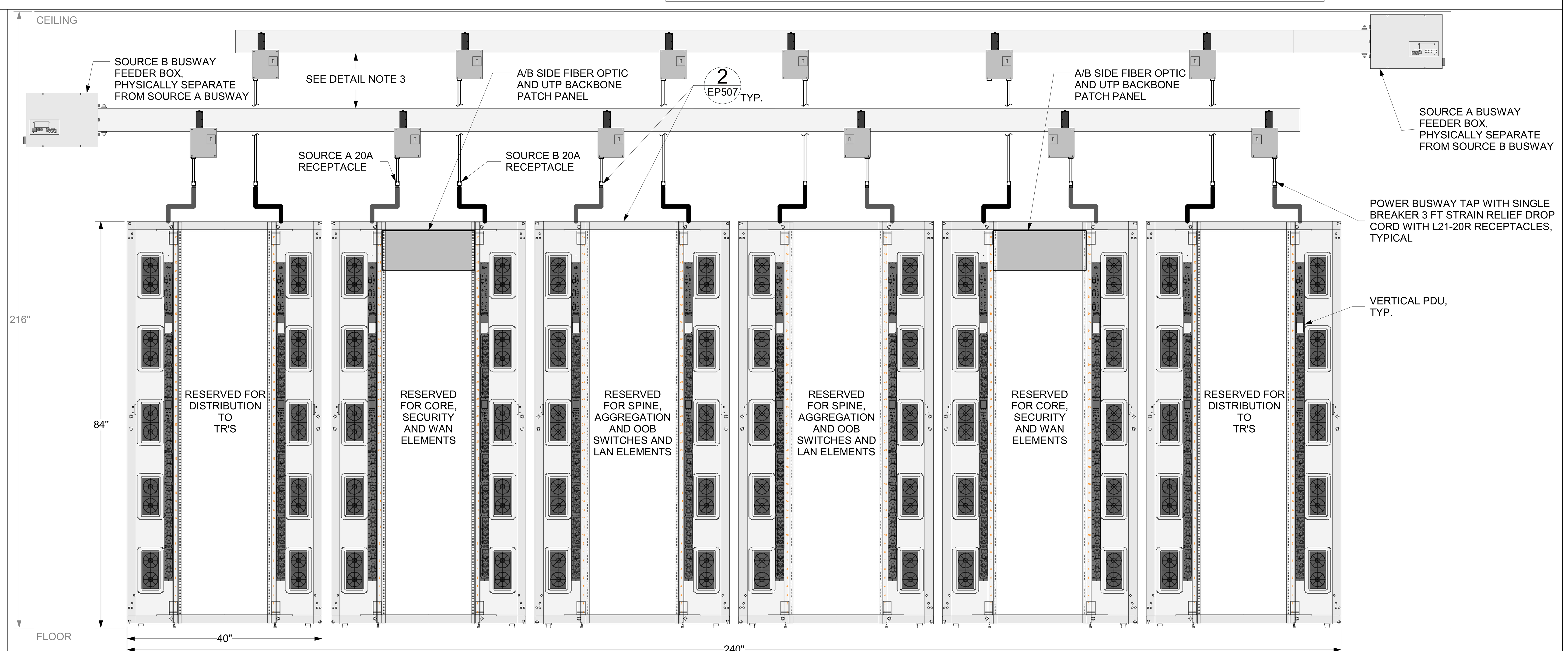
1 SERVER CABINET ROW POWER ELEVATION
01EP400Gh NTS

EACH RACK WILL CONTAIN:

1. ONE L21-20C DROP PER BUSWAY.
2. A MINIMUM OF TWO ZERO U VERTICAL RACK PDUs UTILIZING SEPARATE BUSWAY TAP INPUTS.
3. DIVERSE POWER INPUTS FOR ACTIVE EQUIPMENT ALLOWING FOR SOURCE POWER TO BE SPLIT ACROSS THE TWO BUSWAYS.



2 POWER SCHEMATIC FOR STANDARD DENSITY DATA CENTER ENCLOSURE DETAIL
EP507 NTS



DETAIL NOTES:
1. MDA A AND B WILL HAVE A MIRRORED LAYOUT, WITH THE OUTER MOST CABINETS BEING THE TR DISTRIBUTION CABINETS.
2. EACH CABINET HAS A DEPTH OF 43.3".
3. BUSWAY 'A' AND 'B' ARE PHYSICALLY SEPARATE BY AT LEAST 1-FT HORIZONTALLY AND 1-FT VERTICALLY WITH SOURCE 'B' BUSWAY FURTHEST FROM THE CABINETS.

3 MDA POWER ELEVATION
01EP400Gh NTS

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

A/E:
SPEES DESIGN BUILD
23830 PACIFIC HIGHWAY S. SUITE 300
KENT, WA 98032

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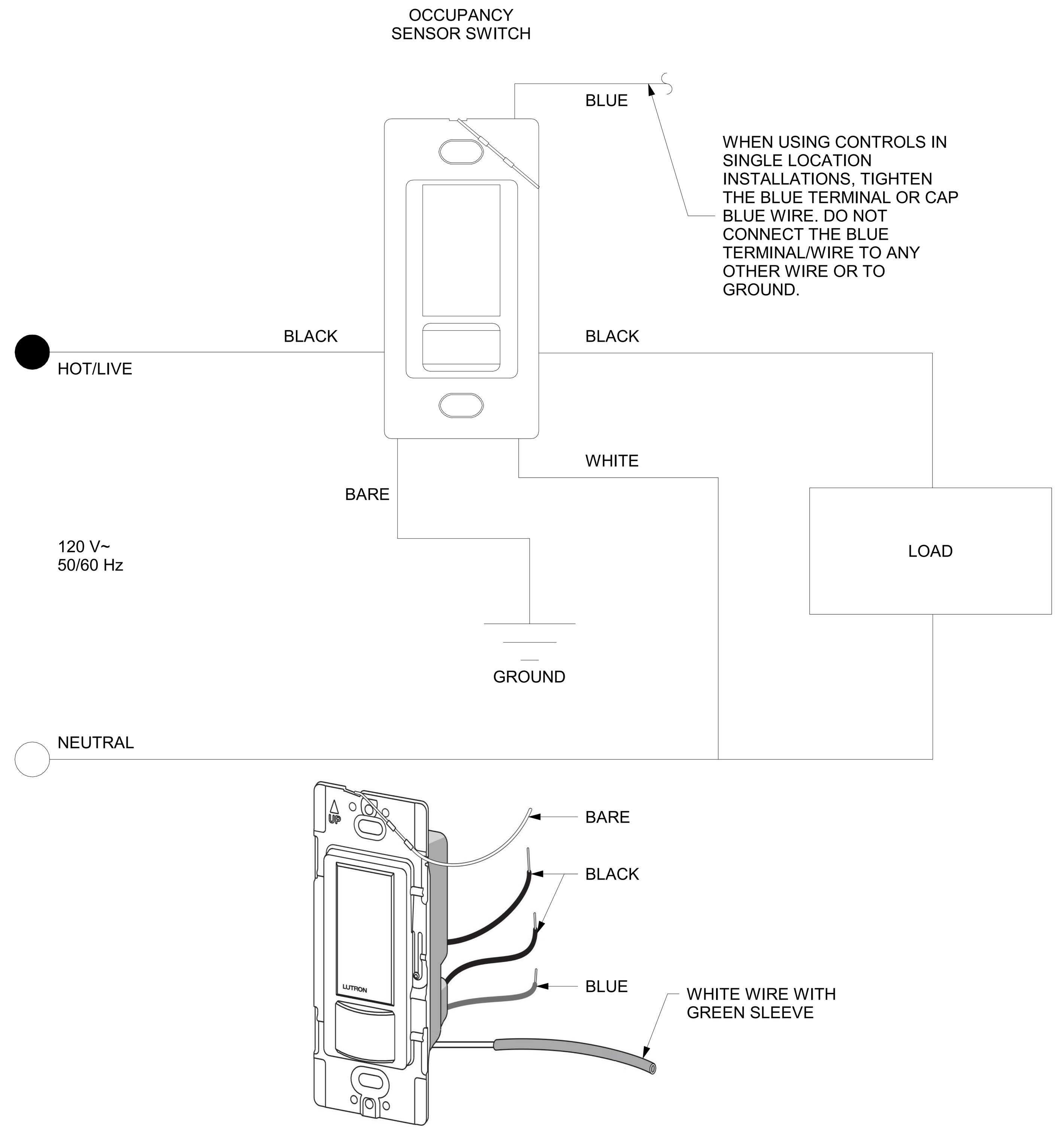
STAMP

LOUIS R. LITZKO — STATE OF WASHINGTON — 55681 — REGISTERED PROFESSIONAL ENGINEER — 4-3-25

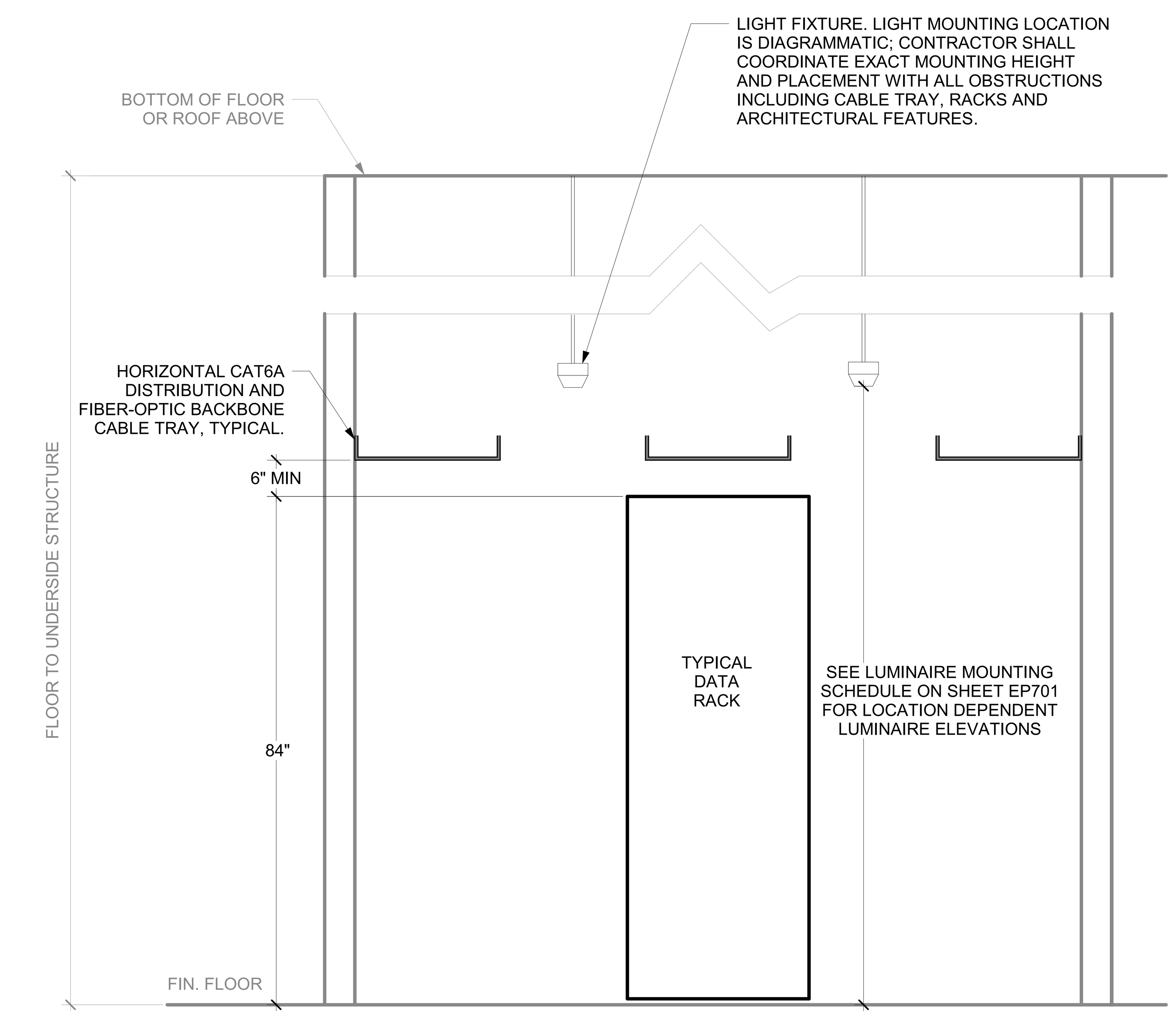
Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
ELECTRICAL DETAILS	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Building Number: -
		Issue Date: 09/03/2025; Checked: DCW/NVP; Drawn: RTS/JNV	Drawing Number: EP507



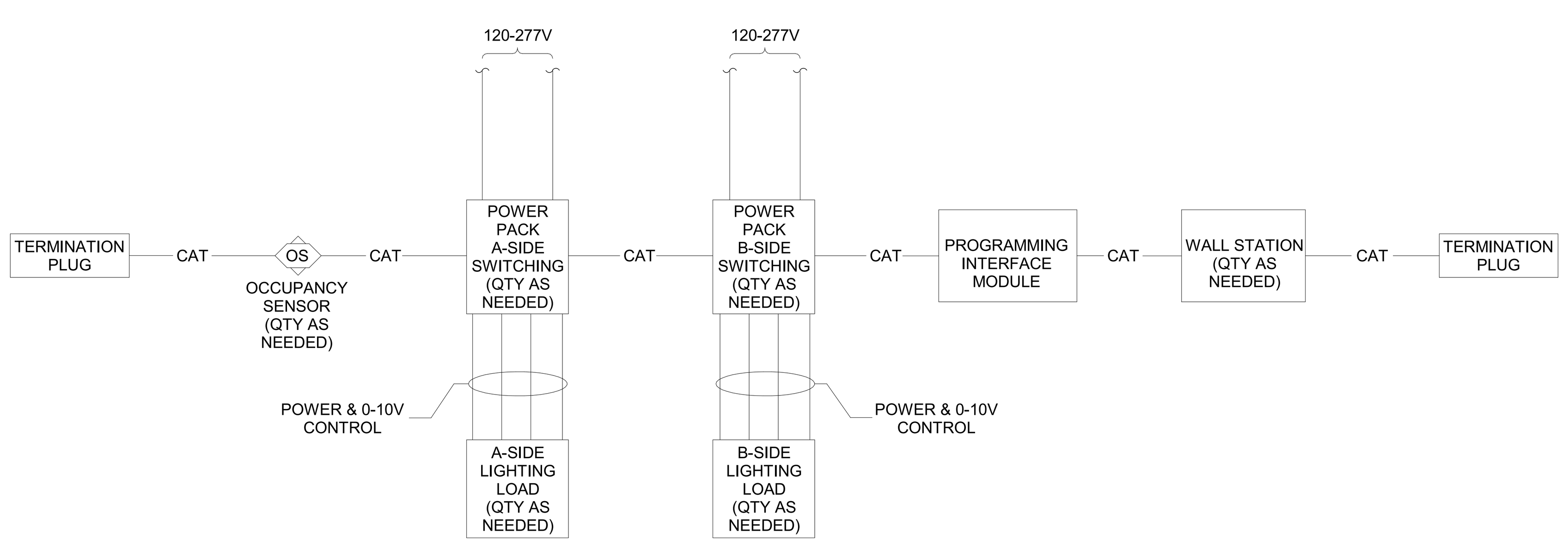
1 - TYPICAL WALL SWITCH OCCUPANCY SENSOR
NTS



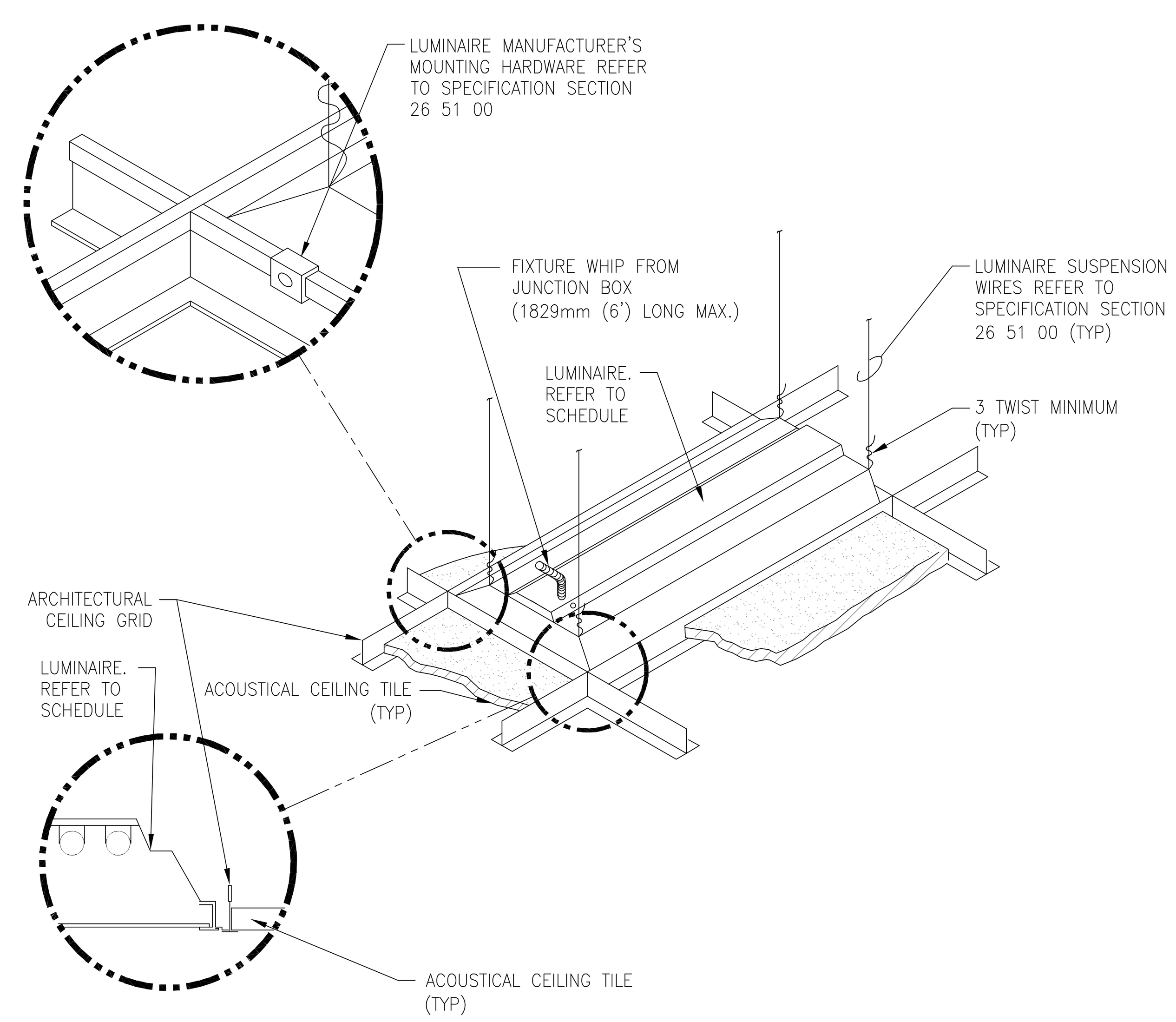
2 - TYPICAL TELECOMMUNICATIONS ROOM ELEVATION
3/4" = 1'-0"

3 - NOT USED
NTS

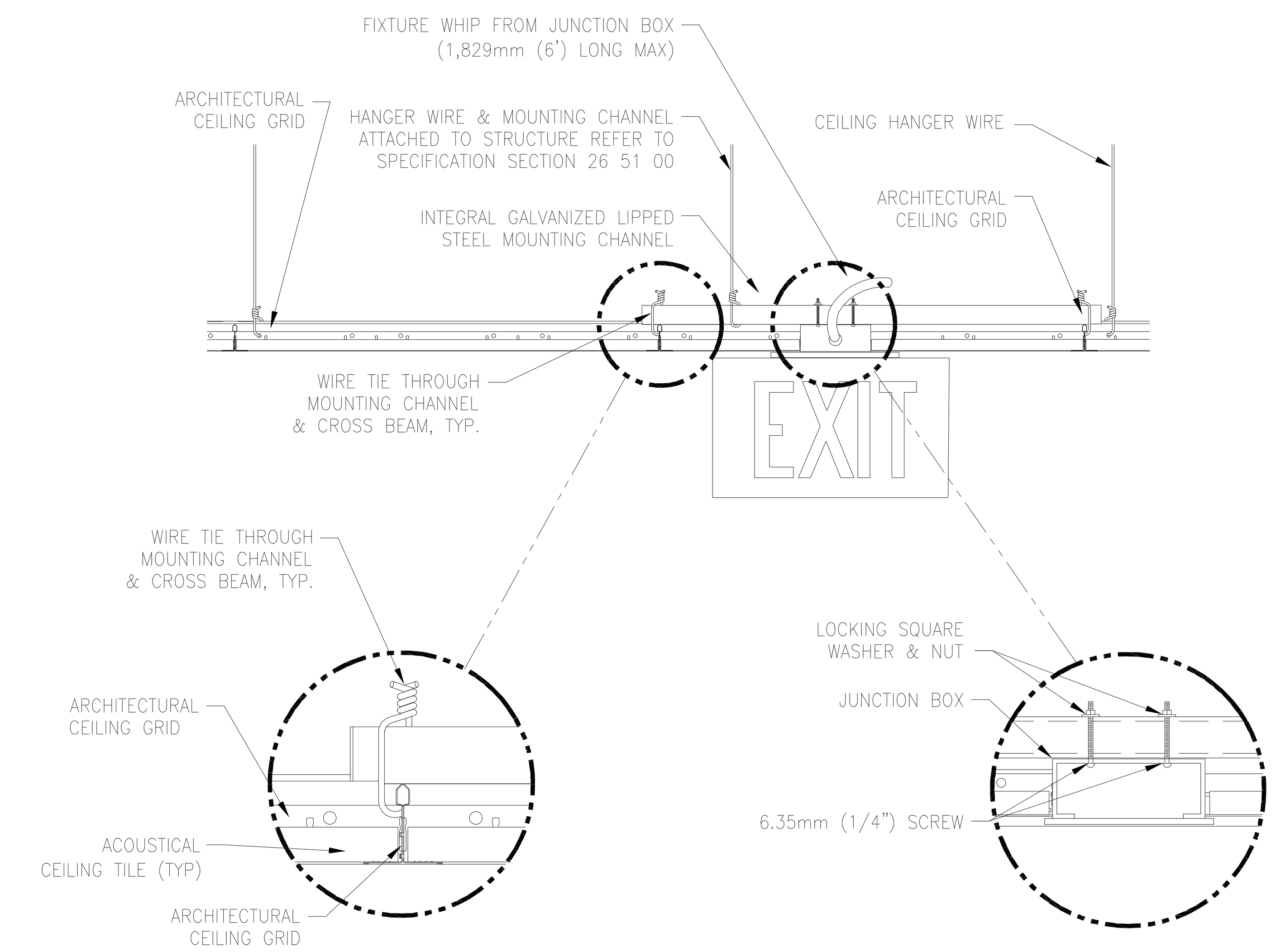
DETAIL NOTE(S):
1. REFERENCE PLANS FOR FIELD DEVICE QUANTITIES. PROVIDE CONTROL CABLING AND WIRING AS REQUIRED.
2. PROVIDE A COMPLETE AND FULLY FUNCTIONAL LIGHTING SYSTEM.



4 - TYPICAL LOW-VOLTAGE SENSOR LIGHTING WIRING DIAGRAM
NTS



5 - TYPICAL RECESSED LIGHTING INSTALLATION
NTS



6 - TYPICAL EXIT SIGN INSTALLATION DETAIL
NTS

Revisions:	Date:

CONSULTANT
ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD
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SPEES DESIGN BUILD
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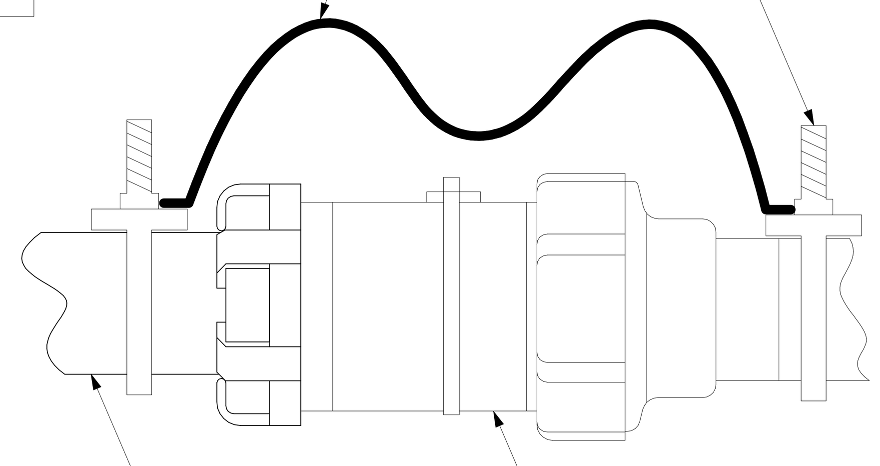
REGISTERED PROFESSIONAL ENGINEER
4-3-25

Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
ELECTRICAL DETAILS	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700
			Building Number: -
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Drawing Number: EP508
		Issue Date: 09/03/2025; Checked: DCW/NVP; Drawn: RTS/JNV	446 OF 693

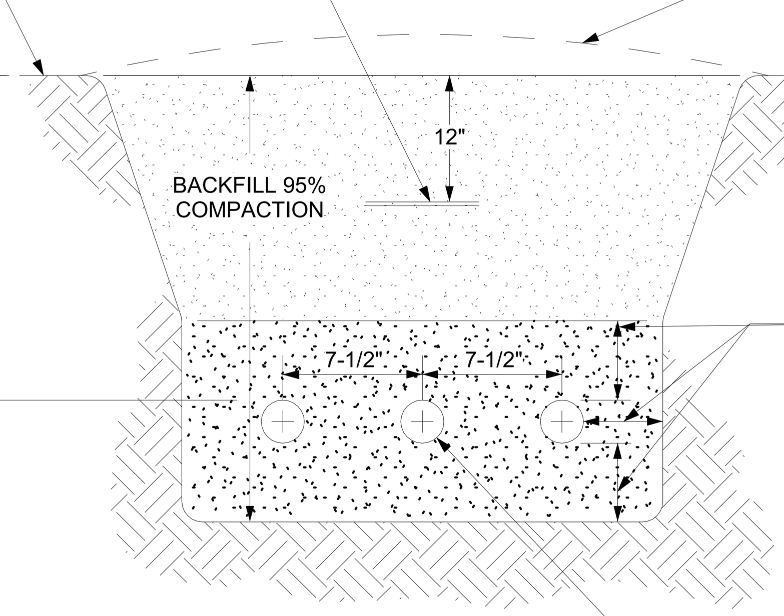
BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:31:44 AM

DETAIL NOTE(S):
1. PROVIDE CONDUIT EXPANSION COUPLINGS WHEN CONDUIT IS ROUTED ACROSS BUILDING EXPANSION JOINTS. REFERENCE ELECTRICAL AND ARCHITECTURAL PLANS AS REQUIRED TO IDENTIFY REQUIRED LOCATIONS.

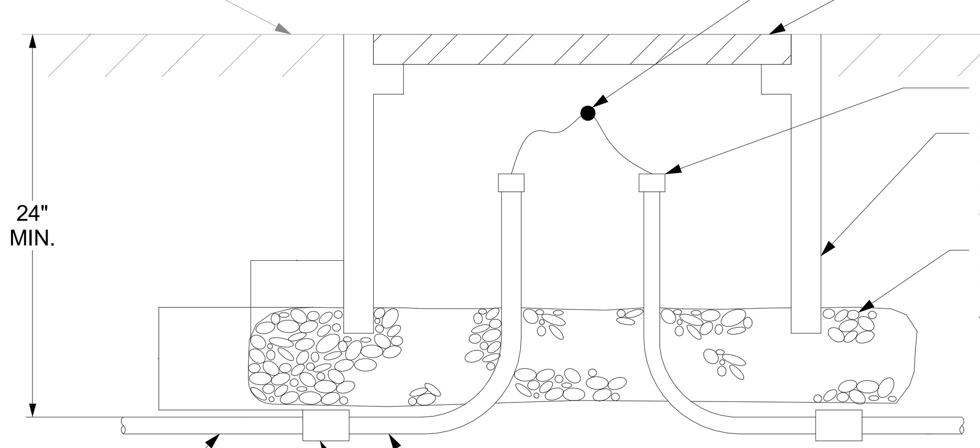


1 CONDUIT EXPANSION JOINT DETAIL
\-
NTS

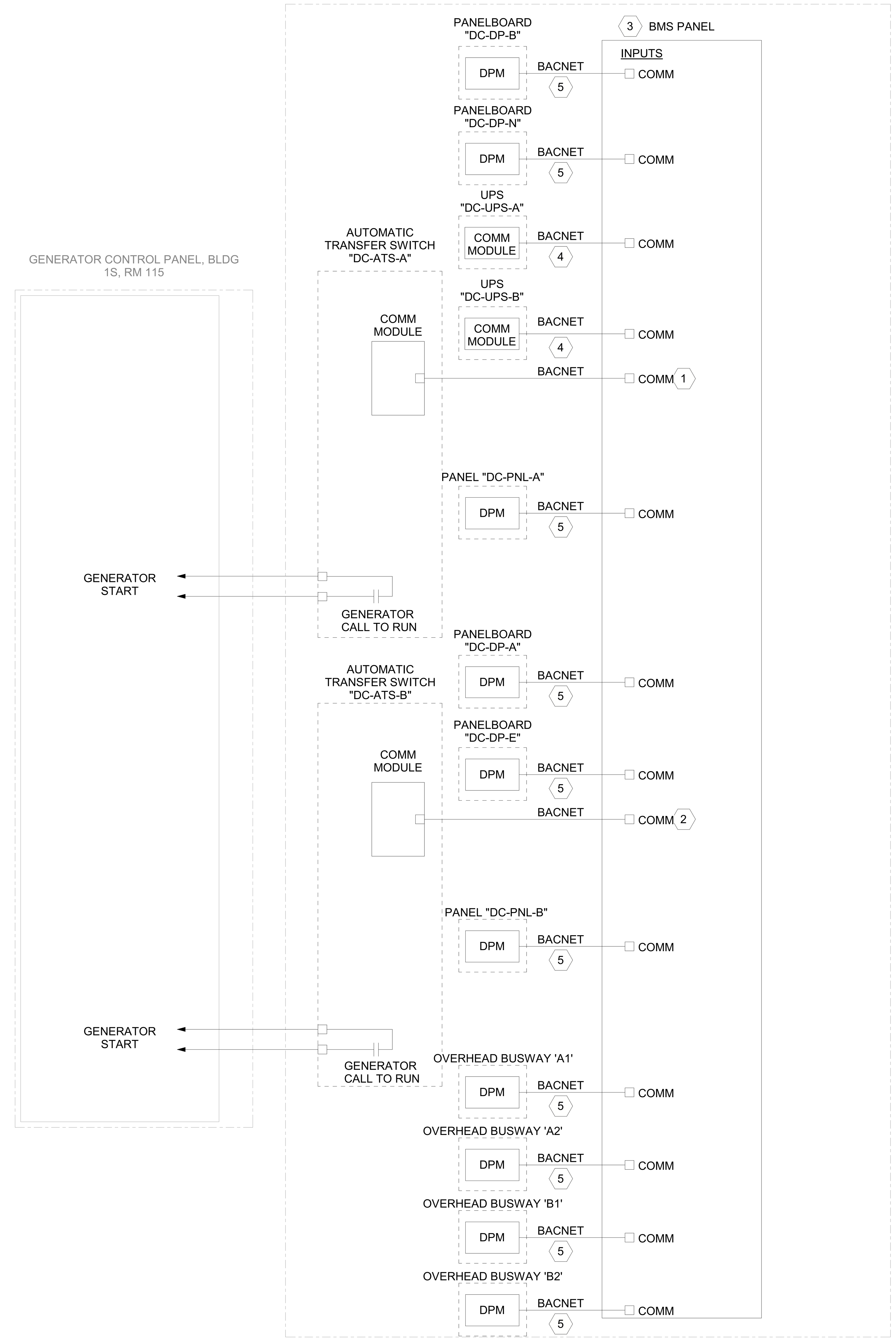
GENERAL NOTE:
REFERENCE CIVIL PLANS FOR TRENCHING, BACKFILL, AND SURFACE REPAIR INFORMATION.



2 TYPICAL TRENCHING SECTION
ES101
TR1EP101
NTS



3 TYPICAL HANDHOLE SECTION
ES101
NTS



4 DATA CENTER EQUIPMENT CONTROL AND STATUS WIRING DIAGRAM
01EP400Gb-1B
NTS

GENERAL NOTES

KEY NOTES

1. ATS "A" POINTS TO BE MONITORED, AT MINIMUM, SHALL BE AS FOLLOWS:
   - ATS IN NORMAL POSITION "DC-ATS-A-NPS"
   - ATS IN EMERGENCY POSITION "DC-ATS-A-EPS"
   - ATS NORMAL POWER AVAILABLE "DC-ATS-A-NPA"
   - ATS EMERGENCY POWER AVAILABLE "DC-ATS-A-EPA"
   - ATS FAIL "DC-ATS-A-FAL"
2. ATS "B" POINTS TO BE MONITORED, AT MINIMUM, SHALL BE AS FOLLOWS:
   - ATS IN NORMAL POSITION "DC-ATS-B-NPS"
   - ATS IN EMERGENCY POSITION "DC-ATS-B-EPS"
   - ATS NORMAL POWER AVAILABLE "DC-ATS-B-NPA"
   - ATS EMERGENCY POWER AVAILABLE "DC-ATS-B-EPA"
   - ATS FAIL "DC-ATS-B-FAL"
3. BMS PANEL SHALL BE ACCOMPANIED BY I/O EXPANSION MODULES AS REQUIRED FOR ALL ELECTRICAL AND MECHANICAL EQUIPMENT CONTROL AND STATUS. PROVIDE ALL TERMINATIONS, CABLING, RACEWAYS, PROGRAMMING AND APPURTENANCES AS REQUIRED TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM. REFERENCE MECHANICAL DRAWINGS AND SPECIFICATIONS FOR ADDITIONAL INFORMATION.
4. UPS BATTERY MANAGEMENT SYSTEM SHALL BE MONITORED WITH THE FOLLOWING MINIMUM DATA POINTS:
   - OVERLOAD (CRITICAL).
   - BATTERY TEMP (CRITICAL).
   - LOW BATTERY.
   - BATTERY FAILURE (CRITICAL.)
   - GENERAL CRITICAL ALARM (IE: FAN FAILURE, INTERNAL FAULT, BYPASS, ETC.)
5. DATA POINT SHALL RECORD kVA, kW, AND AMPERAGE AT A MINIMUM.

Revisions:	Date:

CONSULTANT
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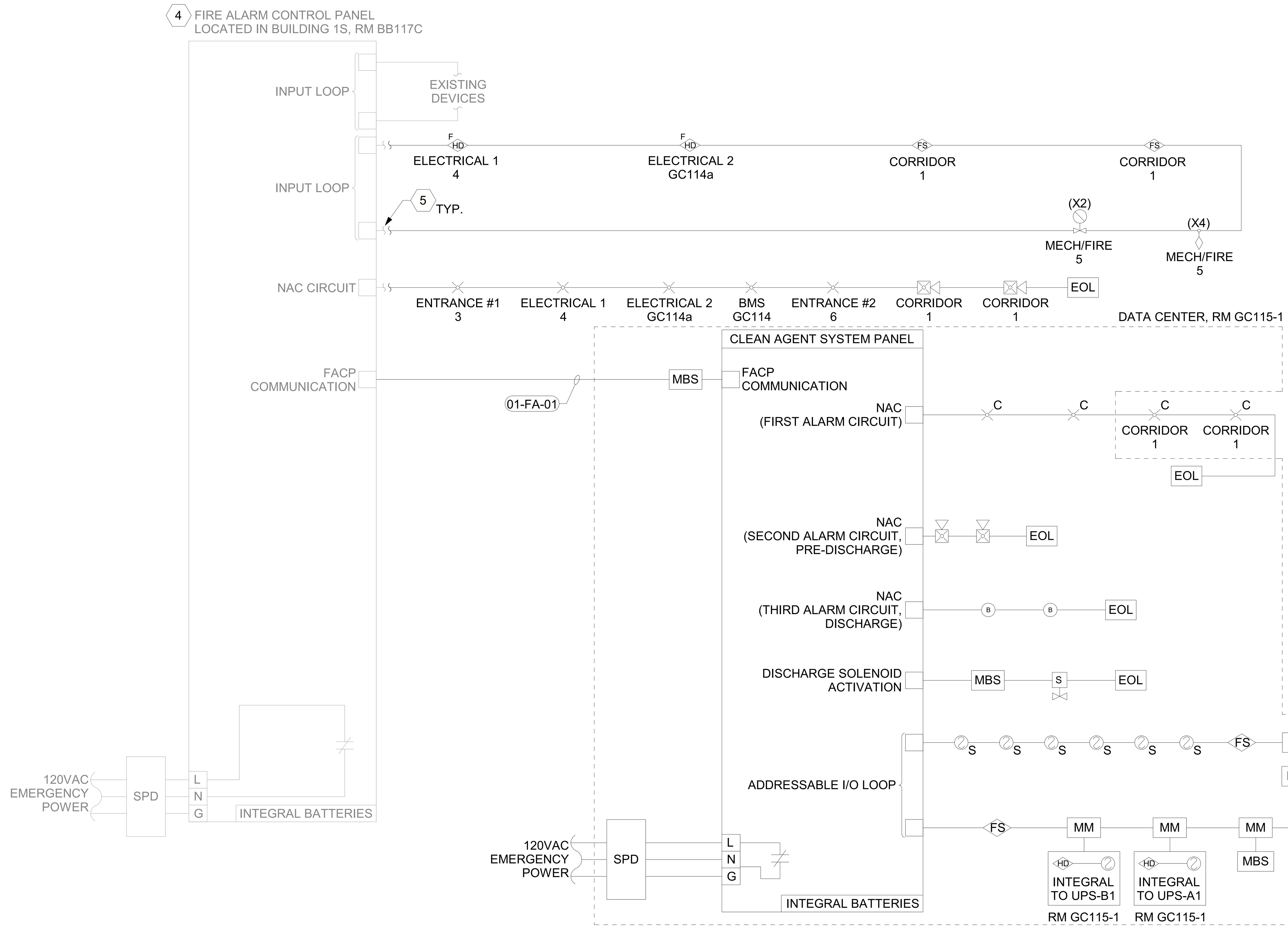
Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

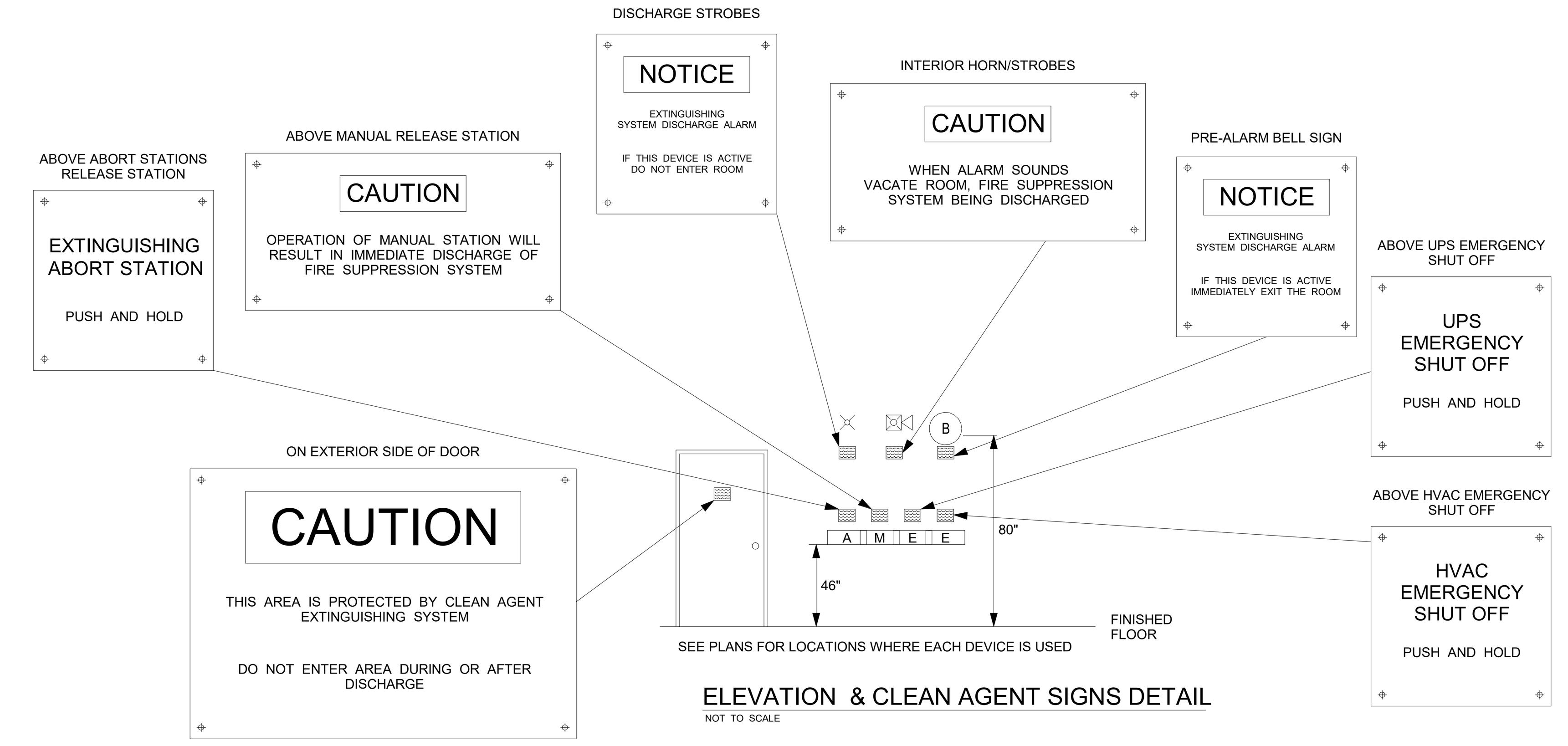
Drawing Title	Phase	Project Title	Project Number
ELECTRICAL DETAILS	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700
			Building Number: -
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Drawing Number: EP509
		Issue Date: 09/03/2025 / Checked: DCW/NVP / Drawn: RTS/JNV	447 OF 693

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:29:27 AM



1 DATA CENTER FIRE ALARM RISER DIAGRAM AND I/O MATRIX

NTS



ELEVATION & CLEAN AGENT SIGNS DETAIL
NOT TO SCALE

CLEAN AGENT SIGNAGE NOTES:
1. ALL SIGNS HAVE RED BACKGROUNDS WITH WHITE LETTERS.
2. SIGNS INCLUDE FIXED ADHESIVE ON THE REVERSE SIDES FOR EASY MOUNTING.
3. SIGNS TO BE WALL MOUNTED AS INDICATED OR AT EYE LEVEL.

2 CLEAN AGENT FIELD DEVICE SIGNAGE DETAIL

01EF400Gb-1B NTS
01EF400Gf-2B

GENERAL NOTES

1. FIRE ALARM DETAILS ARE PROVIDED FOR GENERAL FUNCTIONAL REQUIREMENTS ONLY; CONTRACTOR SHALL PROVIDE ALL REQUIRED EQUIPMENT, RACEWAYS, CABLING, CONNECTIONS, PROGRAMMING, AND COMMISIONING. COORDINATE WITH MECHANICAL, FIRE ALARM SYSTEM VENDOR, AND CLEAN AGENT SYSTEM VENDOR AS REQUIRED. CONTRACTOR SHALL PROVIDE FIRE ALARM SHOP DRAWINGS REVIEWED AND STAMPED BY A LICENSED FIRE PROTECTION ENGINEER. REFERENCE SPECIFICATIONS 21 22 00, 28 31 00, AND DIVISION 26 FOR ALL REQUIREMENTS TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
2. REFERENCE SHEET 01EF401 FOR DEVICE QUANTITIES AND LOCATIONS. DEVICE LOCATIONS AND QUANTITIES MAY CHANGE DEPENDING ON SYSTEM SELECTED. CONTRACTOR SHALL PROVIDE A COMPLETE SUBMITTAL INCLUDING, BUT NOT LIMITED TO: FUNCTIONAL NARRATIVE, WIRING DIAGRAMS, SHOP DRAWINGS, DETAILS, AND RED-LINED FLOOR PLANS STAMPED BY A LICENSED FIRE PROTECTION ENGINEER INDICATING INSTALLATION LOCATIONS FOR REVIEW PRIOR TO COMMENCING WORK.
3. COORDINATE WITH VA, CAMPUS FIRE ALARM CONTRACTOR, AND MECHANICAL CONTRACTOR AS REQUIRED.
4. DATA CENTER FIRE PROTECTION SHALL ADHERE TO NFPA 75. AUTOMATIC DETECTION SHALL BE PROVIDED BY SMOKE DETECTORS (LOCATE AND PROVIDE QUANTITY AS REQUIRED); AUTOMATIC ACTUATION AND NOTIFICATION SHALL BE PROVIDED BY FIRE ALARM SYSTEM. REFERENCE SPECIFICATIONS 21 22 00 AND 28 31 00 FOR ADDITIONAL INFORMATION.
5. COORDINATE WITH EXISTING FACILITY INFRASTRUCTURE, EQUIPMENT, NATIONAL/LOCAL CODES AND MANUFACTURER TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM AS REQUIRED.
6. ALL WIRING SHALL BE ROUTED IN 3/4" CONDUIT (MINIMUM SIZE). ALL RACEWAYS AND BOXES SHALL BE COLORED RED.
7. COORDINATE WITH CIVIL AND TELECOMMUNICAITONS DRAWINGS FOR FIBER-OPTIC CABLING.
8. COORDINATE WITH MECHANICAL DRAWINGS FOR CLEAN AGENT SYSTEM PANEL CONNECTIONS AND REQUIREMENTS.

KEY NOTES

1. PROVIDE RACEWAYS AND CONDUCTORS TO ACTIVATE SHUNT-TRIP CIRCUIT BREAKERS SUPPLYING POWER TO ALL HVAC EQUIPMENT SUPPORTING MCR. REFERENCE ONE-LINE DIAGRAM ON SHEET 01EP609-2B FOR ADDITIONAL INFORMATION.
2. PROVIDE RACEWAYS AND CONDUCTORS TO DEACTIVATE MCR UPSs. REFERENCE ONE-LINE DIAGRAM ON SHEET 01EP609-2B FOR ADDITIONAL INFORMATION.
3. PROVIDE RACEWAYS AND CONDUCTORS TO ACTIVATE SHUNT-TRIP CIRCUIT BREAKERS SUPPLYING POWER TO ALL INFORMATION TECHNOLOGY EQUIPMENT (ITE) SUPPORTING MCR, INCLUDING BUT NOT LIMITED TO, MCR UPSs. REFERENCE ONE-LINE DIAGRAM ON SHEET 01EP609-2B FOR ADDITIONAL INFORMATION.
4. EXISTING FIRE ALARM CONTROL PANEL SHALL BE MODIFIED TO INTEGRATE NEW FIELD DEVICES AS REQUIRED.
5. EXTEND EXISTING CABLING AND RACEWAYS AS REQUIRED TO INTEGRATE NEW FIELD DEVICES.

CLEAN AGENT FIRE SUPPRESSION SYSTEM (CAFSS) OPERATION MATRIX

		OUTPUTS																		
		ALARM LED (PANEL)	SUPERVISORY LED (PANEL)	TROUBLE LED (PANEL)	ACTIVATED LED (DETECTION DEVICE)	ALARM SIGNAL (MAIN FACP)	SUPERVISORY SIGNAL (MAIN FACP)	TROUBLE SIGNAL (MAIN FACP)	ITE SHUTDOWN	HVAC SHUTDOWN	FIRE/SMOKE DAMPER CLOSE	PRE-DISCHARGE BELL	PRE-DISCHARGE STROBE	INITIATE 20 SECOND DELAY	INITIATE 30 SECOND DELAY	CEASE DELAY	RESUME DELAY	DISCHARGE STROBES	CLEAN AGENT DISCHARGE	SYSTEM RESET	NOTES:
INPUTS	LOW PRESSURE SENSOR		X				X														
	AREA SMOKE DETECTOR (1ST TO DETECT SMOKE)	X			X	X					X										
	AREA SMOKE DETECTOR (2ND TO DETECT SMOKE)				X				X	X	X	X	X		X			X	X		(1)
	DUCT SMOKE DETECTOR IN CRACS		X								X										
	MANUAL ACTIVATION STATION	X				X			X	X	X	X	X	X				X	X		(2)
	UPS EPO PUSHBUTTON			X				X	X												
	HVAC EPO PUSHBUTTON			X				X		X											
	ABORT (DELAY) SWITCH ACTIVATED															X					(3)
	ABORT (DELAY) SWITCH RELEASED																X	X	X		
	MAINTENANCE LOCK-OUT SWITCH		X				X														
	CLEAN AGENT PANEL - FAULT CONDITION			X				X													
	CLEAN AGENT PANEL - BYPASS FUNCTION			X				X													
	VALVE ACTUATOR TAMPER SWITCH			X				X													
	ALARM TRANSMISSION BYPASS SWITCH (CAFSS PNL)			X				DISABLED													
	ALARM OFF SWITCH (CAFSS PNL)			X		X															(5)
	TROUBLE SILENCE SWITCH (CAFSS PNL)			DISABLED																	(6)
	RESET SWITCH (CAFS PANEL)																			X	
	LAMP TEST SWITCH (CAFSS PNL)																				(7)
	AHU BYPASS SWITCH (CAFSS PNL)			X						DISABLED											
	RESET2 SWITCH (CAFSS PNL)																			X	(8)

NOTES:
(1) DISCHARGE STROBE ACTIVED AND CLEAN AGENT DISCHARGE OCCURS AFTER 30 SECOND COUNTDOWN
(2) DISCHARGE STROBE ACTIVED AND CLEAN AGENT DISCHARGE OCCURS AFTER 20 SECOND COUNTDOWN
(3) COUNTDOWN THAT WAS INITIATED RESUMES WITHOUT RESETTING, FINISHES, AND DISCHARGE STROBE ACTIVATED WITH RELEASE OF CLEAN AGENT
(4) FAULT CONDITIONS INCLUDE BATTERY/CHARGER FAULT, OPEN/SHORT/GROUND FAULT IN INITIATING, SIGNAL, AND NOTIFICATION DEVICE WIRING
(5) NOTIFICATION APPLIANCE CIRCUIT DISABLED
(6) SILENCE OF TROUBLE ALARM WITHOUT RESET OF TROUBLE ALARM
(7) LAMP TEST ACTIVATES PANEL LAMPS WHILE PRESSED
(8) CAFSS PANEL MUST BE RESET BEFORE MAIN FACP
(9) MATRIX PROVIDED FOR INFORMATIONAL PURPOSES ONLY. ALL SYSTEM FUNCTIONAL REQUIREMENTS SHALL ADHERE TO SPECIFICATION SECTION 21 22 00.

Revisions: Date:

CONSULTANT
ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD
A/E:
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STATE OF WASHINGTON
55681
REGISTERED PROFESSIONAL ENGINEER
9-3-25

Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title: FIRE ALARM RISER DIAGRAM
Approved:

Phase: 100% CONSTRUCTION DOCUMENTS
FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date: 09/03/2025 | Checked: DCW/NVP | Drawn: RTS/JNV

Project Number: 581-22-700
Building Number: -
Drawing Number: EF601
448 OF 693

VA FORM 08 - 6231

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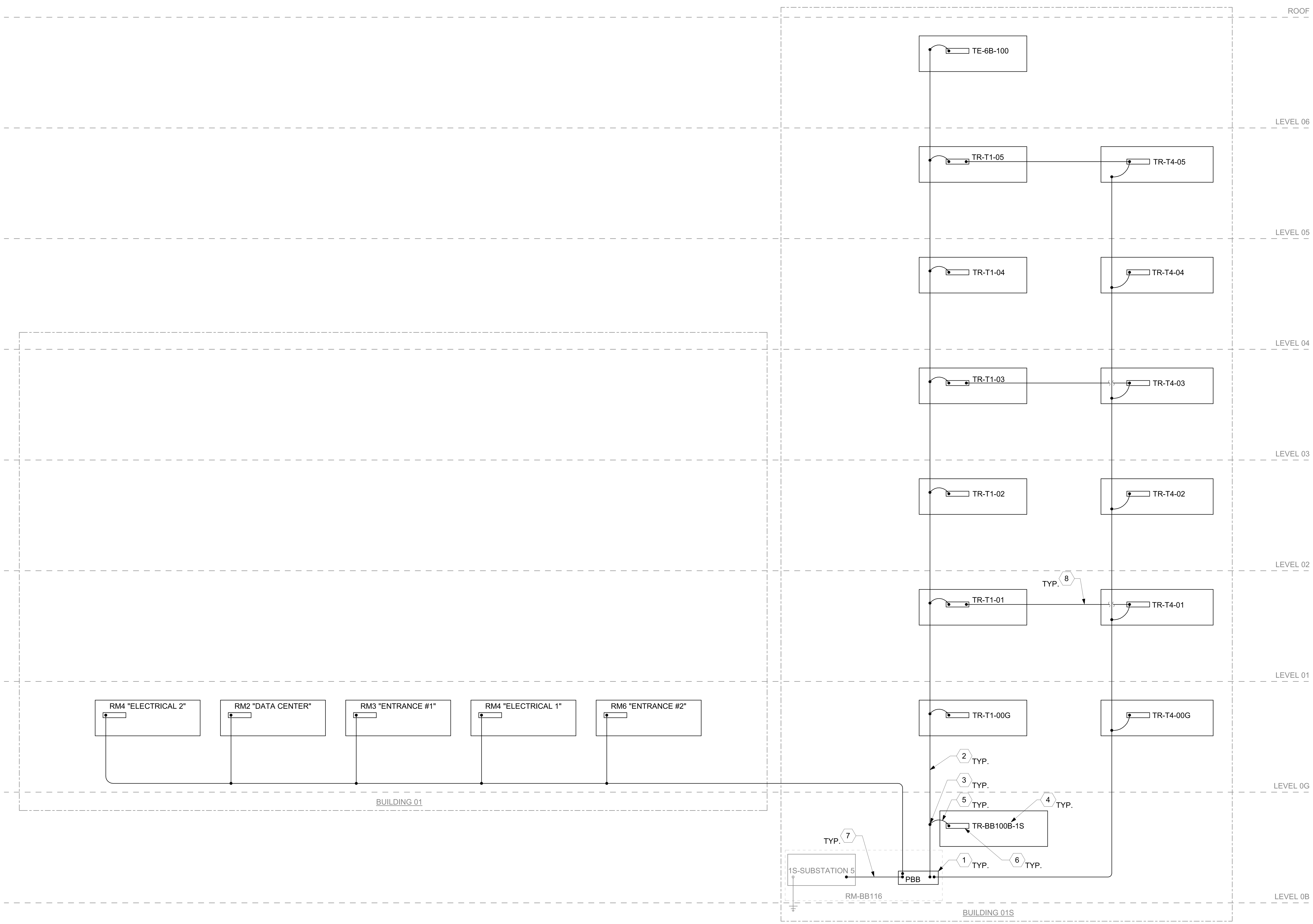

1 TELECOMMUNICATIONS BONDING RISER DIAGRAMS

NTS

GENERAL NOTES

1. ALL BONDING CONDUCTORS, INCLUDING TBB, SHALL FOLLOW SIZING REQUIREMENTS SPECIFIED IN ANSI/TIA-607-E.
2. CONTRACTOR SHALL PROVIDE ALL REQUIRED BONDING, CONDUCTORS, AND PATHWAYS AS REQUIRED FOR THE TELECOMMUNICATIONS SYSTEM BONDING. ALL EXISTING TELECOMMUNICATIONS BONDING FOR EQUIPMENT BEING REMOVED SHALL BE DEMOLISHED. PROVIDE ALL PATCHING, REPAIR AND FIREPROOFING FOR REMOVAL OF ALL EXISTING EQUIPMENT AND CONDUCTORS AND NEW INSTALLATIONS AS REQUIRED.
3. ALL BONDING CONDUCTORS SHALL HAVE BEEN GREEN INSULATION AND SHALL BE ROUTED THROUGH CONDUIT FOR PROTECTION.
4. ALL BONDING CONDUCTORS SHALL HAVE LABELS AT EACH END (TERMINATION POINT); LABEL SHALL READ AS FOLLOWS: "IF CONNECTION OR CONDUCTOR IS LOOSE OR MUST BE REMOVED, NOTIFY OPERATIONS STAFF".

KEY NOTES

1. PROVIDE PRIMARY BONDING BUSBAR (PBB). WHERE EXISTING PBB EXISTS, CONTRACTOR SHALL REPLACE; CONTRACTOR TO FIELD VERIFY EXISTING LOCATION. PROVIDE ALL BONDING CONNECTIONS AS REQUIRED.
2. PROVIDE TELECOMMUNICATIONS BONDING BACKBONE (TBB) CONDUCTOR AS INDICATED. CONTRACTOR SHALL ROUTE AND PROVIDE ALL REQUIRED PATHWAYS FOR TBB AS REQUIRED. EXISTING TBB SHALL BE DEMOLISHED AND REMOVED; PROVIDE ALL PATCHING, REPAIR AND FIREPROOF SEALING AS REQUIRED.
3. PROVIDE IRREVERSIBLE CRIMP CONNECTOR, UL LISTED.
4. TELECOMMUNICATIONS ROOM. REFER TO ENLARGED FLOOR PLANS FOR ADDITIONAL INFORMATION.
5. PROVIDE SECONDARY BONDING CONDUCTOR (SBC) TO CONNECT SBB TO TBB. MINIMUM SIZE SHALL BE 6 AWG.
6. PROVIDE SECONDARY BONDING BUSBAR (SBB). PROVIDE TELECOMMUNICATIONS EQUIPMENT BONDING CONDUCTOR TO BOND SBB TO THE FOLLOWING EQUIPMENT: DATA RACK, CABLE TRAY, TR CONDUITS, ELECTRO-STATIC FLOORING, ELECTRICAL PANEL ETC. REFERENCE SHEET EP504 FOR ADDITIONAL BONDING INFORMATION.
7. PROVIDE TELECOMMUNICATIONS BONDING CONDUCTOR (TBC) TO BOND PBB TO BUILDING ELECTRICAL SERVICE EQUIPMENT GROUND BUSBAR. MINIMUM CONDUCTOR SIZE SHALL MATCH LARGEST TBB.
8. PROVIDE BACKBONE BONDING CONDUCTOR (BBC) TO BOND SSBs WHEN MULTIPLE TRs ARE PRESENT ON A FLOOR. MINIMUM CONDUCTOR SIZE SHALL BE EQUAL TO THE TBB.

TBB/BBC SIZING SCHEDULE (ANSI/TIA-607-E)

TBB/BBC LINEAR LENGTH IN FEET	CONDUCTOR SIZE IN AWG
LESS THAN 13	6
14 - 20	4
21 - 26	3
27 - 33	2
34 - 41	1
42 - 52	1/0
53 - 66	2/0
67 - 84	3/0
85 - 105	4/0
106 - 125	250 kcmil
126 - 150	300 kcmil
151 - 175	350 kcmil
176 - 250	500 kcmil
251 - 300	600 kcmil
GREATER THAN 301	750 kcmil

Revisions: Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: TELECOMMUNICATIONS BONDING RISER DIAGRAMS

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700

Building Number: -

Drawing Number: EG601

VA FORM 08 - 6231

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt

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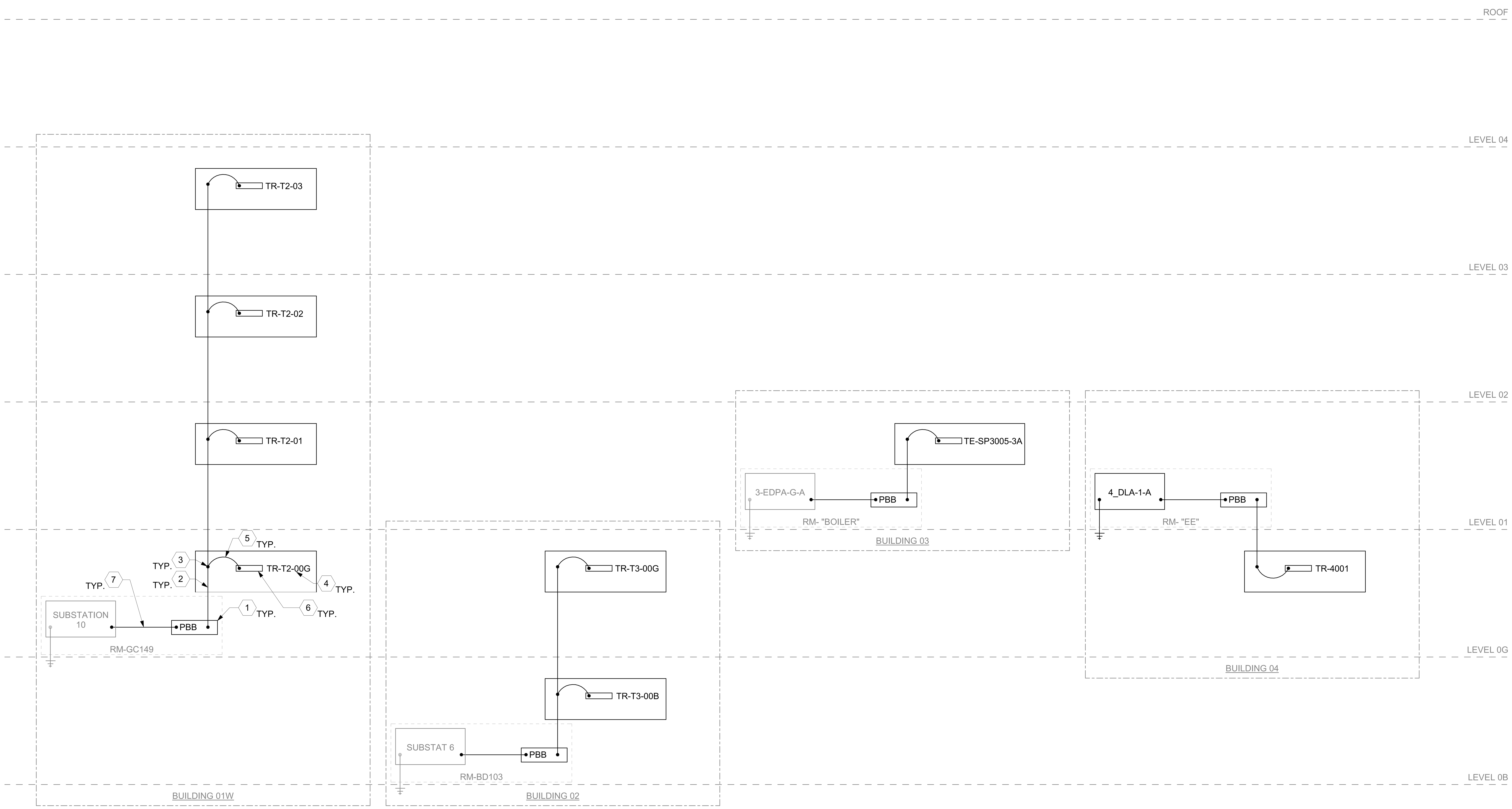

GENERAL NOTES

1. ALL BONDING CONDUCTORS, INCLUDING TBB, SHALL FOLLOW SIZING REQUIREMENTS SPECIFIED IN ANSI/TIA-607-E.
2. CONTRACTOR SHALL PROVIDE ALL REQUIRED BONDING, CONDUCTORS, AND PATHWAYS AS REQUIRED FOR THE TELECOMMUNICATIONS SYSTEM BONDING. ALL EXISTING TELECOMMUNICATIONS BONDING FOR EQUIPMENT BEING REMOVED SHALL BE DEMOLISHED. PROVIDE ALL PATCHING, REPAIR AND FIREPROOFING FOR REMOVAL OF ALL EXISTING EQUIPMENT AND CONDUCTORS AND NEW INSTALLATIONS AS REQUIRED.
3. ALL BONDING CONDUCTORS SHALL HAVE BEEN GREEN INSULATION AND SHALL BE ROUTED THROUGH CONDUIT FOR PROTECTION.
4. ALL BONDING CONDUCTORS SHALL HAVE LABELS AT EACH END (TERMINATION POINT); LABEL SHALL READ AS FOLLOWS: "IF CONNECTION OR CONDUCTOR IS LOOSE OR MUST BE REMOVED, NOTIFY OPERATIONS STAFF".

KEY NOTES

1. PROVIDE PRIMARY BONDING BUSBAR (PBB). WHERE EXISTING PBB EXISTS, CONTRACTOR SHALL REPLACE; CONTRACTOR TO FIELD VERIFY EXISTING LOCATION. PROVIDE ALL BONDING CONNECTIONS AS REQUIRED.
2. PROVIDE TELECOMMUNICATIONS BONDING BACKBONE (TBB) CONDUCTOR AS INDICATED. CONTRACTOR SHALL ROUTE AND PROVIDE ALL REQUIRED PATHWAYS FOR TBB AS REQUIRED. EXISTING TBB SHALL BE DEMOLISHED AND REMOVED; PROVIDE ALL PATCHING, REPAIR AND FIREPROOF SEALING AS REQUIRED.
3. PROVIDE IRREVERSIBLE CRIMP CONNECTOR, UL LISTED.
4. TELECOMMUNICATIONS ROOM. REFER TO ENLARGED FLOOR PLANS FOR ADDITIONAL INFORMATION.
5. PROVIDE SECONDARY BONDING CONDUCTOR (SBC) TO CONNECT SBB TO TBB. MINIMUM SIZE SHALL BE 6 AWG.
6. PROVIDE SECONDARY BONDING BUSBAR (SBB). PROVIDE TELECOMMUNICATIONS EQUIPMENT BONDING CONDUCTOR TO BOND SBB TO THE FOLLOWING EQUIPMENT: DATA RACK, CABLE TRAY, TR CONDUITS, ELECTRO-STATIC FLOORING, ELECTRICAL PANEL ETC. REFERENCE SHEET EP504 FOR ADDITIONAL BONDING INFORMATION.
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126 - 150	300 kcmil
151 - 175	350 kcmil
176 - 250	500 kcmil
251 - 300	600 kcmil
GREATER THAN 301	750 kcmil

1 - TELECOMMUNICATIONS BONDING RISER DIAGRAMS

NTS

Revisions: Date:

CONSULTANT

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: TELECOMMUNICATIONS BONDING RISER DIAGRAMS

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

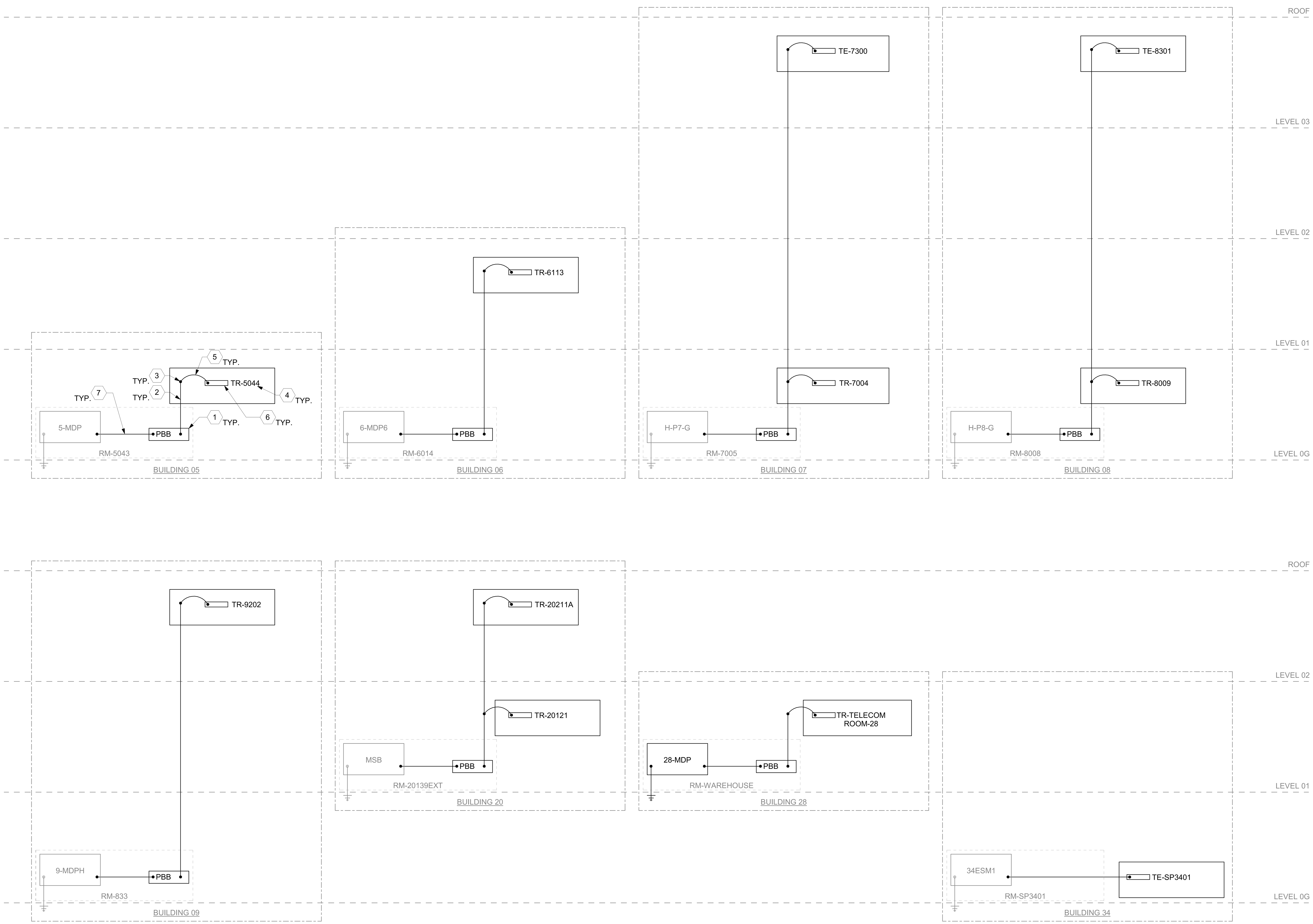
Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date: 09/03/2025
Checked: DCW/NVP
Drawn: RTS/JNV

Project Number: 581-22-700

Building Number: -

Drawing Number: EG602



1 TELECOMMUNICATIONS BONDING RISER DIAGRAMS

NTS

GENERAL NOTES

1. ALL BONDING CONDUCTORS, INCLUDING TBB, SHALL FOLLOW SIZING REQUIREMENTS SPECIFIED IN ANSI/TIA-607-E.
2. CONTRACTOR SHALL PROVIDE ALL REQUIRED BONDING, CONDUCTORS, AND PATHWAYS AS REQUIRED FOR THE TELECOMMUNICATIONS SYSTEM BONDING. ALL EXISTING TELECOMMUNICATIONS BONDING FOR EQUIPMENT BEING REMOVED SHALL BE DEMOLISHED. PROVIDE ALL PATCHING, REPAIR AND FIREPROOFING FOR REMOVAL OF ALL EXISTING EQUIPMENT AND CONDUCTORS AND NEW INSTALLATIONS AS REQUIRED.
3. ALL BONDING CONDUCTORS SHALL HAVE BEEN GREEN INSULATION AND SHALL BE ROUTED THROUGH CONDUIT FOR PROTECTION.
4. ALL BONDING CONDUCTORS SHALL HAVE LABELS AT EACH END (TERMINATION POINT); LABEL SHALL READ AS FOLLOWS: "IF CONNECTION OR CONDUCTOR IS LOOSE OR MUST BE REMOVED, NOTIFY OPERATIONS STAFF".

KEY NOTES

1. PROVIDE PRIMARY BONDING BUSBAR (PBB). WHERE EXISTING PBB EXISTS, CONTRACTOR SHALL REPLACE; CONTRACTOR TO FIELD VERIFY EXISTING LOCATION. PROVIDE ALL BONDING CONNECTIONS AS REQUIRED.
2. PROVIDE TELECOMMUNICATIONS BONDING BACKBONE (TBB) CONDUCTOR AS INDICATED. CONTRACTOR SHALL ROUTE AND PROVIDE ALL REQUIRED PATHWAYS FOR TBB AS REQUIRED. EXISTING TBB SHALL BE DEMOLISHED AND REMOVED; PROVIDE ALL PATCHING, REPAIR AND FIREPROOF SEALING AS REQUIRED.
3. PROVIDE IRREVERSIBLE CRIMP CONNECTOR, UL LISTED.
4. TELECOMMUNICATIONS ROOM. REFER TO ENLARGED FLOOR PLANS FOR ADDITIONAL INFORMATION.
5. PROVIDE SECONDARY BONDING CONDUCTOR (SBC) TO CONNECT SBB TO TBB. MINIMUM SIZE SHALL BE 6 AWG.
6. PROVIDE SECONDARY BONDING BUSBAR (SBB). PROVIDE TELECOMMUNICATIONS EQUIPMENT BONDING CONDUCTOR TO BOND SBB TO THE FOLLOWING EQUIPMENT: DATA RACK, CABLE TRAY, TR CONDUITS, ELECTRO-STATIC FLOORING, ELECTRICAL PANEL ETC. REFERENCE SHEET EP504 FOR ADDITIONAL BONDING INFORMATION.
7. PROVIDE TELECOMMUNICATIONS BONDING CONDUCTOR (TBC) TO BOND PBB TO BUILDING ELECTRICAL SERVICE EQUIPMENT GROUND BUSBAR. MINIMUM CONDUCTOR SIZE SHALL MATCH LARGEST TBB.

TBB/BBC SIZING SCHEDULE (ANSI/TIA-607-E)

TBB/BBC LINEAR LENGTH IN FEET	CONDUCTOR SIZE IN AWG
LESS THAN 13	6
14 - 20	4
21 - 26	3
27 - 33	2
34 - 41	1
42 - 52	1/0
53 - 66	2/0
67 - 84	3/0
85 - 105	4/0
106 - 125	250 kcmil
126 - 150	300 kcmil
151 - 175	350 kcmil
176 - 250	500 kcmil
251 - 300	600 kcmil
GREATER THAN 301	750 kcmil

Revisions: Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

A/E:
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23830 PACIFIC HIGHWAY S. SUITE 300
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SPEESDESIGNBUILD

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LOUIS R. LITZKO STATE OF WASHINGTON 55681 9-3-25 REGISTERED PROFESSIONAL ENGINEER

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: TELECOMMUNICATIONS BONDING RISER DIAGRAMS

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date: 09/03/2025 | Checked: DCW/NVP | Drawn: RTS/JNV

Project Number: 581-22-700

Building Number: -

Drawing Number: EG603

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:29:42 AM

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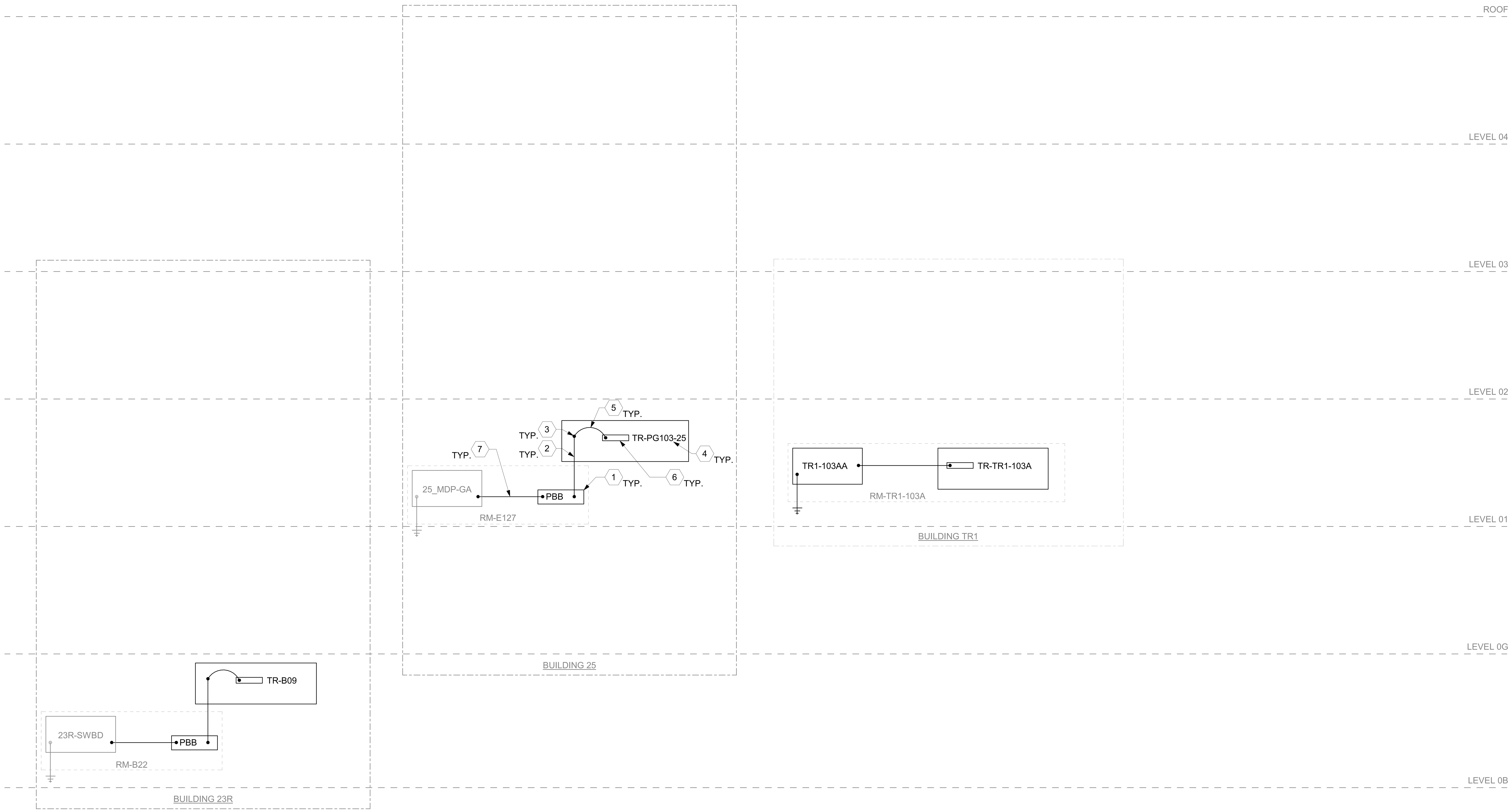

1 - TELECOMMUNICATIONS BONDING RISER DIAGRAMS

NTS

GENERAL NOTES

1. ALL BONDING CONDUCTORS, INCLUDING TBB, SHALL FOLLOW SIZING REQUIREMENTS SPECIFIED IN ANSI/TIA-607-E.
2. CONTRACTOR SHALL PROVIDE ALL REQUIRED BONDING, CONDUCTORS, AND PATHWAYS AS REQUIRED FOR THE TELECOMMUNICATIONS SYSTEM BONDING. ALL EXISTING TELECOMMUNICATIONS BONDING FOR EQUIPMENT BEING REMOVED SHALL BE DEMOLISHED. PROVIDE ALL PATCHING, REPAIR AND FIREPROOFING FOR REMOVAL OF ALL EXISTING EQUIPMENT AND CONDUCTORS AND NEW INSTALLATIONS AS REQUIRED.
3. ALL BONDING CONDUCTORS SHALL HAVE BEEN GREEN INSULATION AND SHALL BE ROUTED THROUGH CONDUIT FOR PROTECTION.
4. ALL BONDING CONDUCTORS SHALL HAVE LABELS AT EACH END (TERMINATION POINT); LABEL SHALL READ AS FOLLOWS: "IF CONNECTION OR CONDUCTOR IS LOOSE OR MUST BE REMOVED, NOTIFY OPERATIONS STAFF".

KEY NOTES

1. PROVIDE PRIMARY BONDING BUSBAR (PBB). WHERE EXISTING PBB EXISTS, CONTRACTOR SHALL REPLACE; CONTRACTOR TO FIELD VERIFY EXISTING LOCATION. PROVIDE ALL BONDING CONNECTIONS AS REQUIRED.
2. PROVIDE TELECOMMUNICATIONS BONDING BACKBONE (TBB) CONDUCTOR AS INDICATED. CONTRACTOR SHALL ROUTE AND PROVIDE ALL REQUIRED PATHWAYS FOR TBB AS REQUIRED. EXISTING TBB SHALL BE DEMOLISHED AND REMOVED; PROVIDE ALL PATCHING, REPAIR AND FIREPROOF SEALING AS REQUIRED.
3. PROVIDE IRREVERSIBLE CRIMP CONNECTOR, UL LISTED.
4. TELECOMMUNICATIONS ROOM. REFER TO ENLARGED FLOOR PLANS FOR ADDITIONAL INFORMATION.
5. PROVIDE SECONDARY BONDING CONDUCTOR (SBC) TO CONNECT SBB TO TBB. MINIMUM SIZE SHALL BE 6 AWG.
6. PROVIDE SECONDARY BONDING BUSBAR (SBB). PROVIDE TELECOMMUNICATIONS EQUIPMENT BONDING CONDUCTOR TO BOND SBB TO THE FOLLOWING EQUIPMENT: DATA RACK, CABLE TRAY, TR CONDUITS, ELECTRO-STATIC FLOORING, ELECTRICAL PANEL ETC. REFERENCE SHEET EP504 FOR ADDITIONAL BONDING INFORMATION.
7. PROVIDE TELECOMMUNICATIONS BONDING CONDUCTOR (TBC) TO BOND PBB TO BUILDING ELECTRICAL SERVICE EQUIPMENT GROUND BUSBAR. MINIMUM CONDUCTOR SIZE SHALL MATCH LARGEST TBB.

TBB/BBC SIZING SCHEDULE (ANSI/TIA-607-E)

TBB/BBC LINEAR LENGTH IN FEET	CONDUCTOR SIZE IN AWG
LESS THAN 13	6
14 - 20	4
21 - 26	3
27 - 33	2
34 - 41	1
42 - 52	1/0
53 - 66	2/0
67 - 84	3/0
85 - 105	4/0
106 - 125	250 kcmil
126 - 150	300 kcmil
151 - 175	350 kcmil
176 - 250	500 kcmil
251 - 300	600 kcmil
GREATER THAN 301	750 kcmil

Revisions: | Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

A/E:
SPEES DESIGN BUILD
23830 PACIFIC HIGHWAY S. SUITE 300
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LOUIS R. LITZKO
STATE OF WASHINGTON
55681
REGISTERED PROFESSIONAL ENGINEER
9-3-25

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: TELECOMMUNICATIONS BONDING RISER DIAGRAMS

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

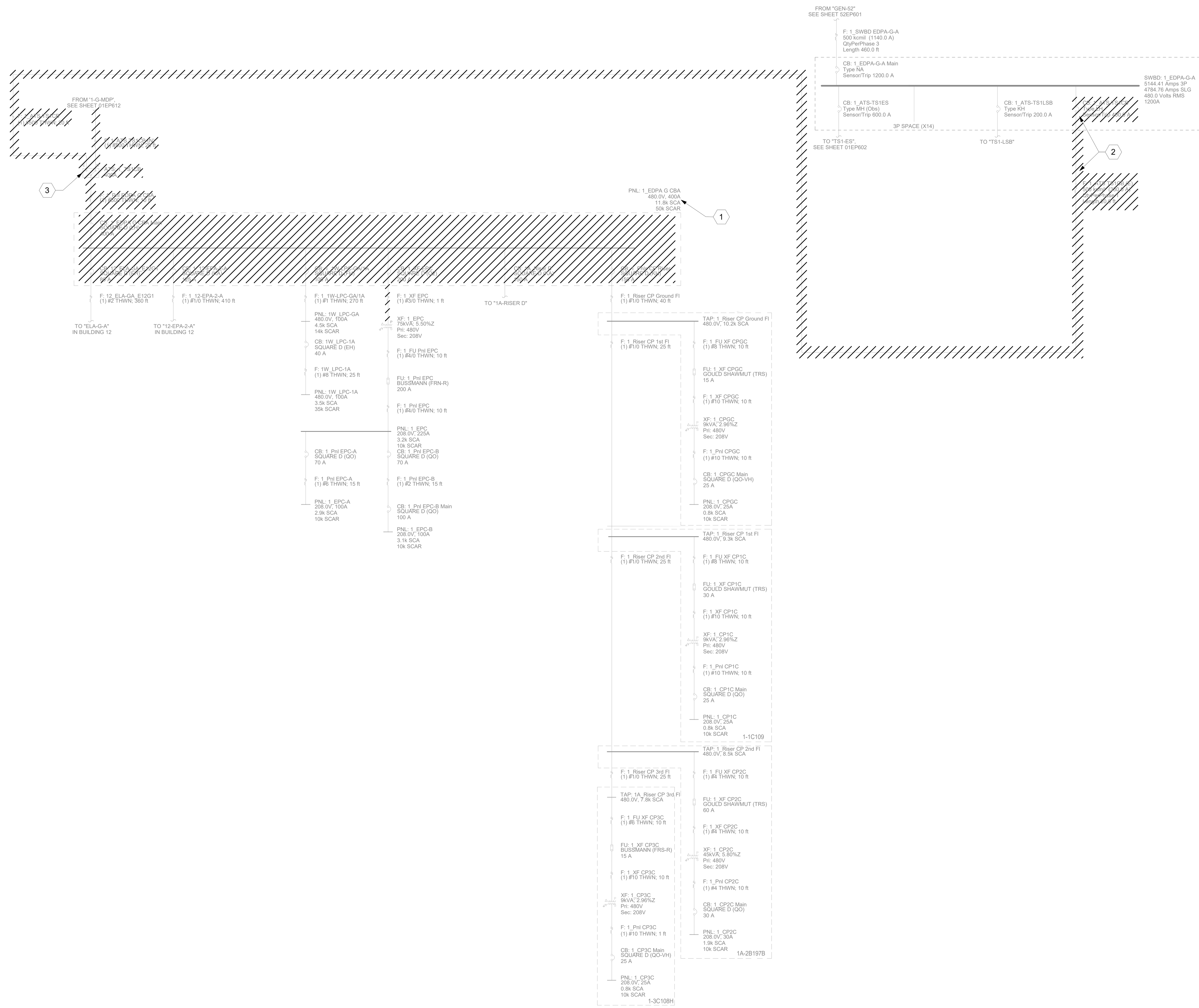
Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date: 09/03/2025 | Checked: DCW/NVP | Drawn: RTS/JNV

Project Number: 581-22-700

Building Number: -

Drawing Number: EG604



BLDG 01 - PARTIAL ONE-LINE DIAGRAM - EXISTING (1 OF 2)

NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. REFERENCE SHEET 01EP611 FOR REVISED ONE-LINE DIAGRAM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTES

1. EXISTING EQUIPMENT SHALL BE DEMOLISHED AND REPLACED. PRESERVE EXISTING CIRCUITS FOR RETERMINATION. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. EXISTING FEEDER AND CIRCUIT BREAKER SHALL BE DEMOLISHED AS INDICATED.
3. EXISTING ATS AND ASSOCIATED FEEDERS AND CONTROL WIRING SHALL BE DEMOLISHED. EXISTING CONTROL WIRING RACEWAY(S) SHALL BE PRESERVED FOR REUSE.

Revisions: Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01 - PARTIAL ONE-LINE DIAGRAM - EXISTING (1 OF 2)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700

Building Number: 01

Drawing Number: 01EP601

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:21:55 AM

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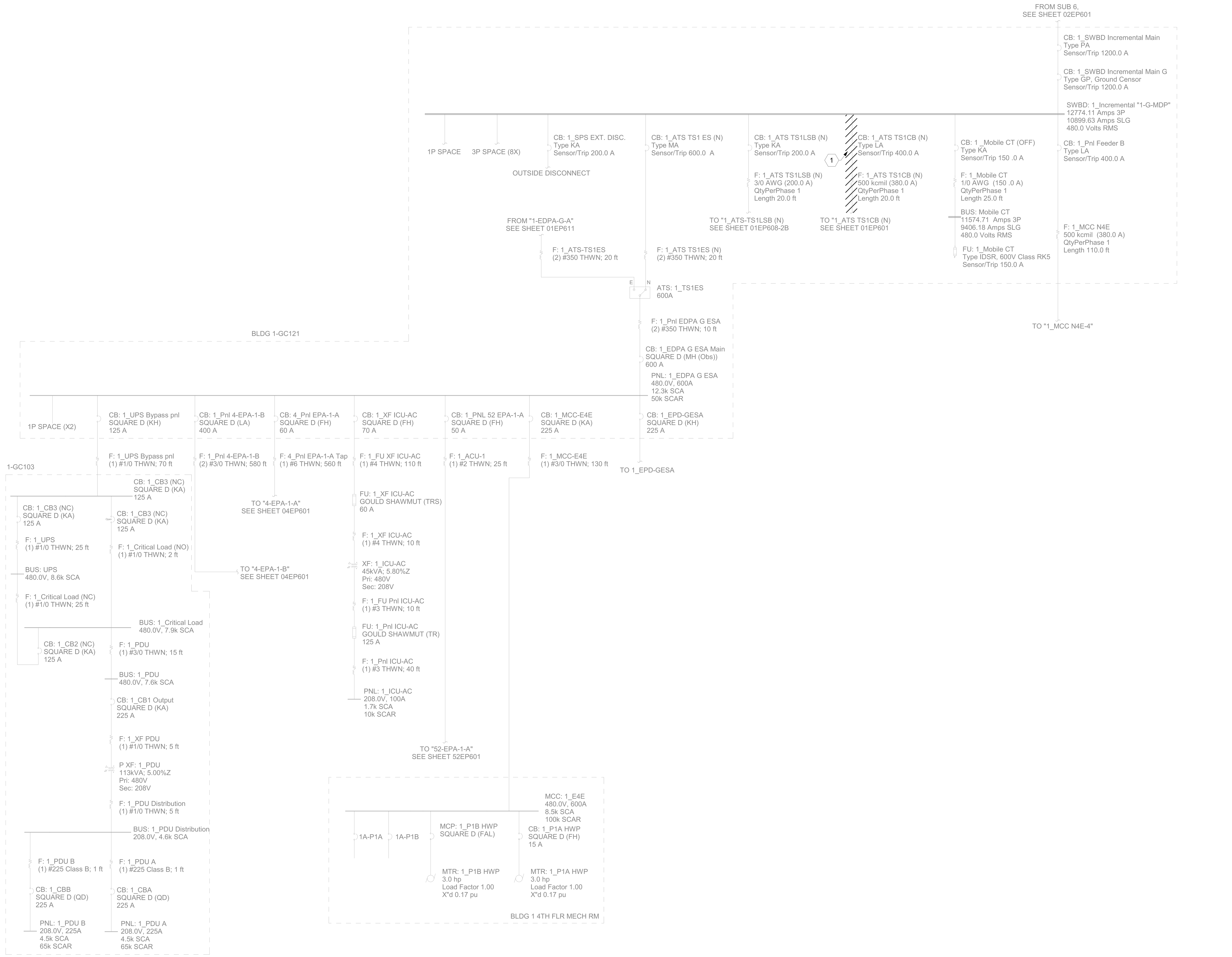

1 / - BLDG 01 - PARTIAL ONE-LINE DIAGRAM - EXISTING (2 OF 2)

1/4" = 1'-0"

GENERAL NOTE

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30-DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL INTERIOR LOCATED EQUIPMENT SHALL BE NEMA 1 RATED AND ALL EXTERIOR LOCATED EQUIPMENT SHALL BE NEMA 3R RATED, UNLESS OTHERWISE NOTED.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTE

1. DEMOLISH EXISTING CIRCUIT BREAKER AND FEEDER SERVING AUTOMATIC TRANSFER SWITCH "1_ATS TS1CB".

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

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LOUIS R. LITZKO — STATE OF WASHINGTON — 55681 — REGISTERED PROFESSIONAL ENGINEER — 4-3-25

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01 - PARTIAL ONE-LINE DIAGRAM - EXISTING (2 OF 2)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Project Number: 581-22-700

Building Number: 01

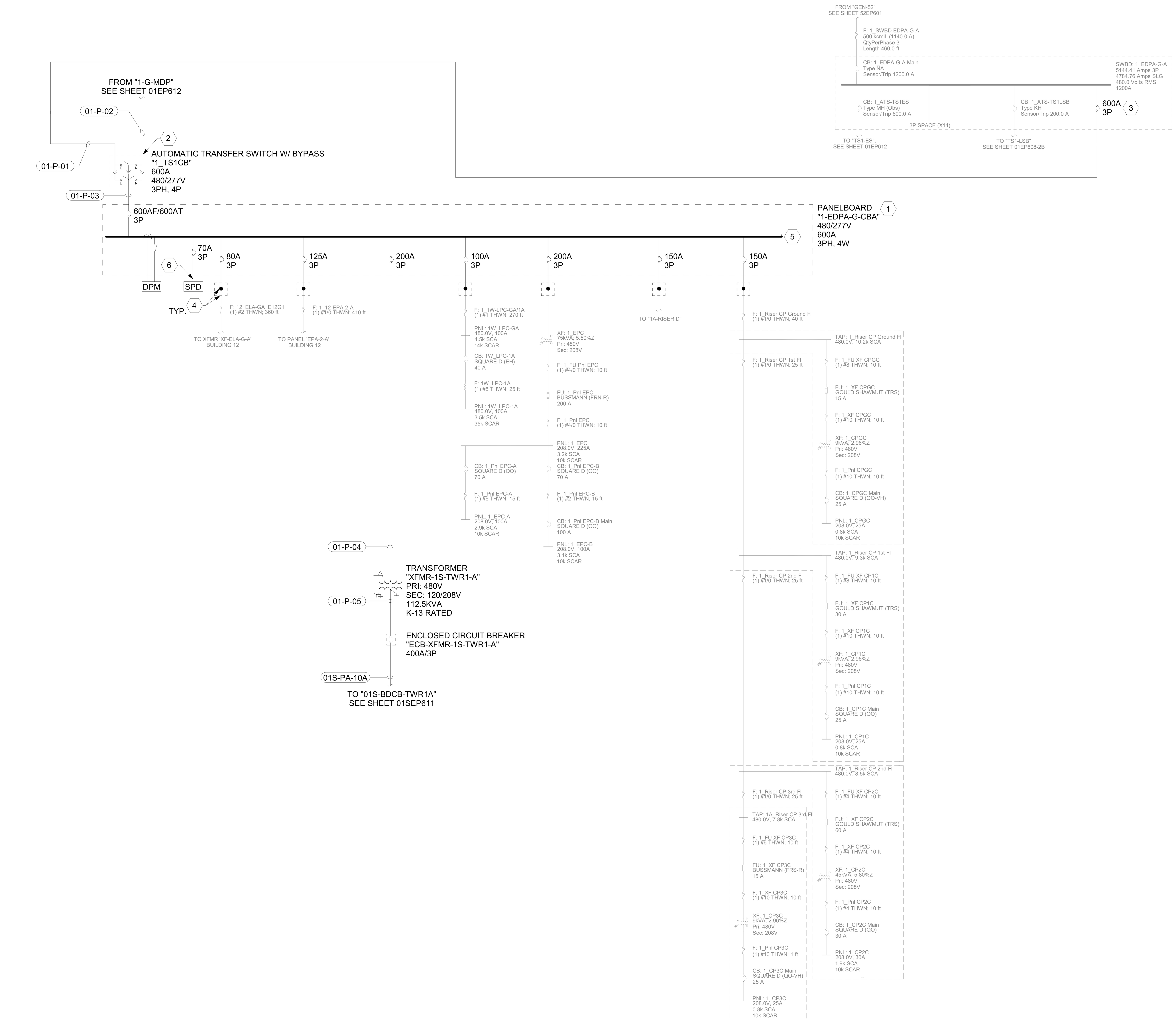
Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Drawing Number: 01EP602

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:32:41 AM

VA FORM 08 - 6231



1 - BLDG 01 - PARTIAL ONE-LINE DIAGRAM - REVISED (1 OF 2)

NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
7. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1. PROVIDE REPLACEMENT PANELBOARD "1-EDPA-G-CBA" AS INDICATED TO SERVE NEW AND EXISTING LOADS. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. PROVIDE REPLACEMENT AUTOMATIC TRANSFER SWITCH "1-TS1CB" AND ASSOCIATED FEEDERS AS INDICATED. EXISTING CONTROL WIRING SHALL BE REPLACED IN ITS ENTIRETY FROM ATS TO GENERATOR "GEN-52"; REUSE AND EXTEND EXISTING RACEWAY AS REQUIRED.
3. UTILIZE EXISTING SPACE TO PROVIDE CIRCUIT BREAKER AS INDICATED. REFERENCE ASSOCIATED REVISED PANEL SCHEDULES FOR CIRCUITING INFORMATION.
4. PROVIDE JUNCTION BOX AND INSULATED TAPS TO EXTEND EXISTING CONDUIT AND WIRE TO PANELBOARD AS REQUIRED. SIZE JUNCTION BOX PER NEC. CONTRACTOR SHALL FIELD COORDINATE JUNCTION BOX INSTALLATION LOCATION AS REQUIRED.
5. REFERENCE ASSOCIATED PANEL SCHEDULE FOR ADDITIONAL INFORMATION.
6. PROVIDE SURGE PROTECTION DEVICE AS INDICATED. PROVIDE 1" CONDUIT WITH (4) 10 AWG CONDUCTORS AND (1) 10 AWG GROUND; CONDUCTOR SIZES ARE MINIMUM. PROVIDE AND INSTALL SURGE PROTECTION EQUIPMENT, BREAKER, RACEWAY, AND CONDUCTORS PER MANUFACTURERS REQUIREMENTS. SURGE PROTECTION DEVICE CONDUCTORS SHALL BE LIMITED TO 12" IN LENGTH UNLESS OTHERWISE NOTED BY MANUFACTURER.

Revisions: | Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

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KENT, WA 98032

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
BLDG 01 - PARTIAL ONE-LINE DIAGRAM - REVISED (1 OF 2)	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700

Building Number: 01

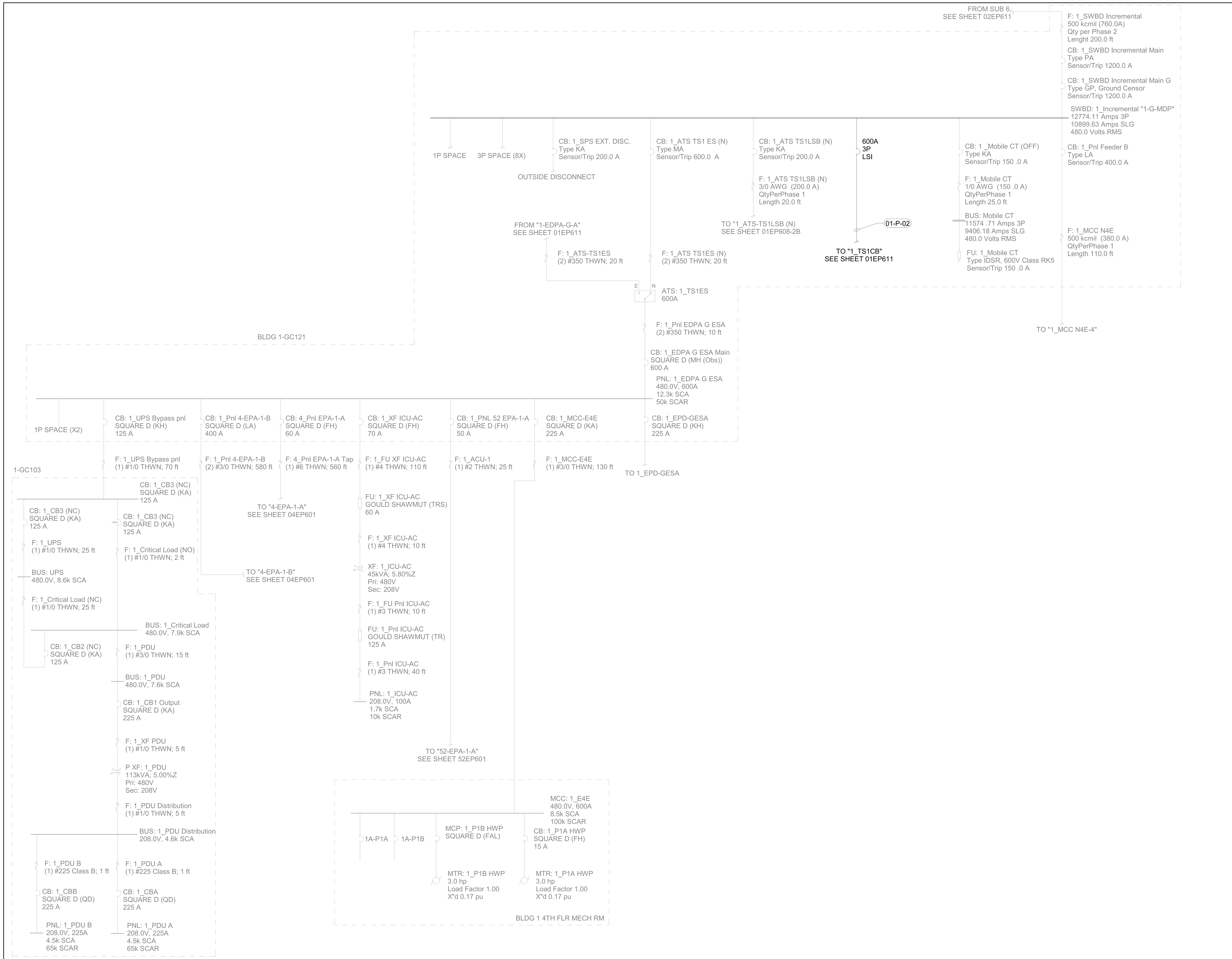
Approved:

FULLY SPRINKLERED

Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Drawing Number: 01EP611



1
-
BLDG 01 - PARTIAL ONE-LINE DIAGRAM - REVISED (2 OF 2)
1/4" = 1'-0"

GENERAL NOTE

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30-DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL INTERIOR LOCATED EQUIPMENT SHALL BE NEMA 1 RATED AND ALL EXTERIOR LOCATED EQUIPMENT SHALL BE NEMA 3R RATED, UNLESS OTHERWISE NOTED.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
8. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTE

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

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LOUIS R. LITZKO
STATE OF WASHINGTON
55681
REGISTERED
PROFESSIONAL ENGINEER
9-3-25

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01 - PARTIAL ONE-LINE DIAGRAM - REVISED (2 OF 2)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700

Building Number: 01

Drawing Number: 01EP612

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:32:35 AM

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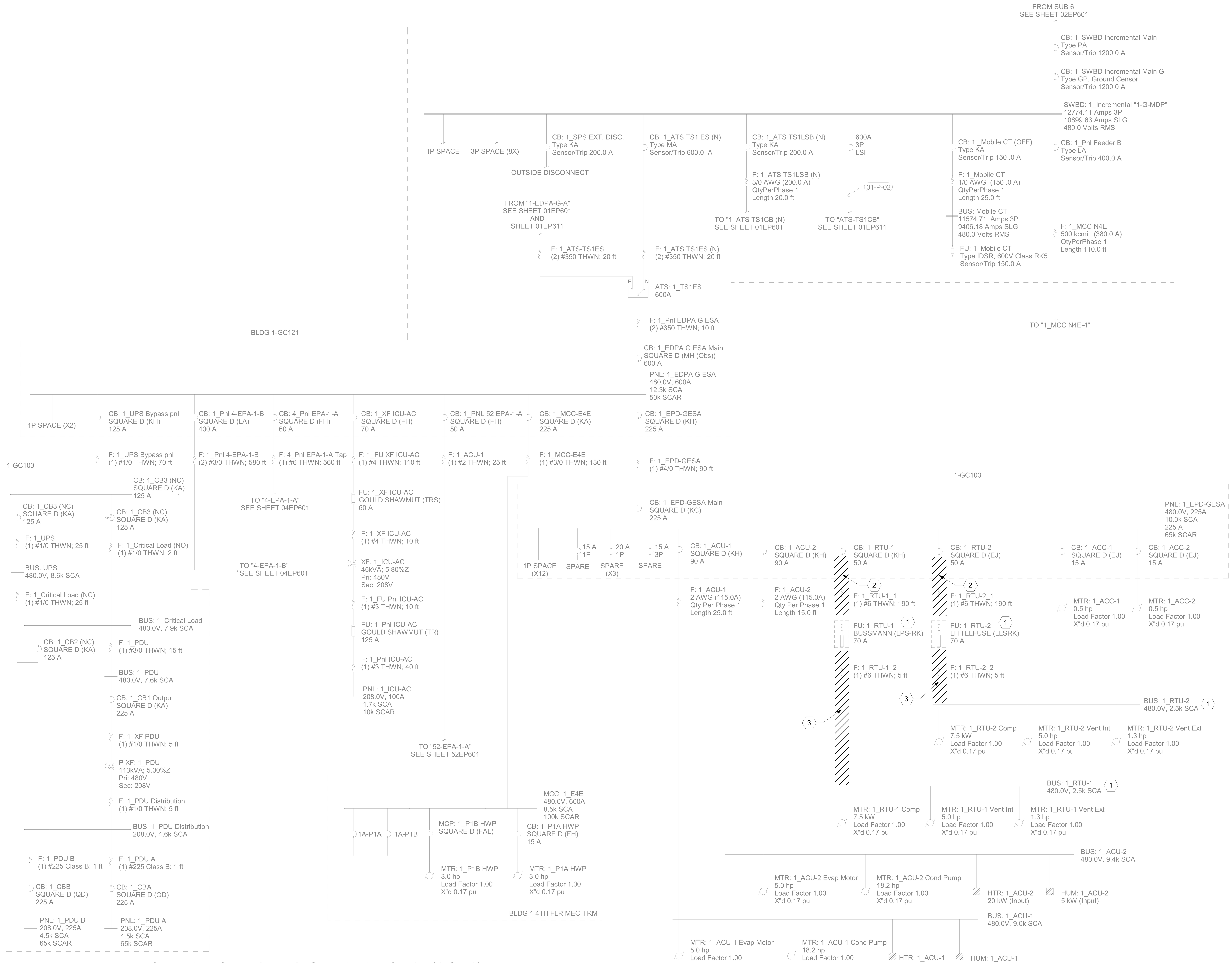

1 - DATA CENTER - ONE-LINE DIAGRAM - PHASE 1A (1 OF 2)

1/4" = 1'-0"

GENERAL NOTE

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30-DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL INTERIOR LOCATED EQUIPMENT SHALL BE NEMA 1 RATED AND ALL EXTERIOR LOCATED EQUIPMENT SHALL BE NEMA 3R RATED, UNLESS OTHERWISE NOTED.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTE

1. RELOCATED ROOFTOP MECHANICAL EQUIPMENT. EXTEND CONDUIT AND WIRE AS REQUIRED.
2. CONDUCTOR SHALL BE DEMOLISHED IN ITS ENTIRETY; RE-USE AND EXTEND EXISTING RACEWAYS AS PRACTICAL.
3. CONDUCTOR AND RACEWAY SHALL BE DEMOLISHED IN ITS ENTIRETY.

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

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SDB SPEESDESIGNBUILD

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LOUIS R. LITZKO, STATE OF WASHINGTON, 55681, REGISTERED PROFESSIONAL ENGINEER, 4-3-25

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: DATA CENTER - ONE-LINE DIAGRAM - PHASE 1A (1 OF 2)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700

Building Number: 01

Drawing Number: 01EP601-1A

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:32:58 AM

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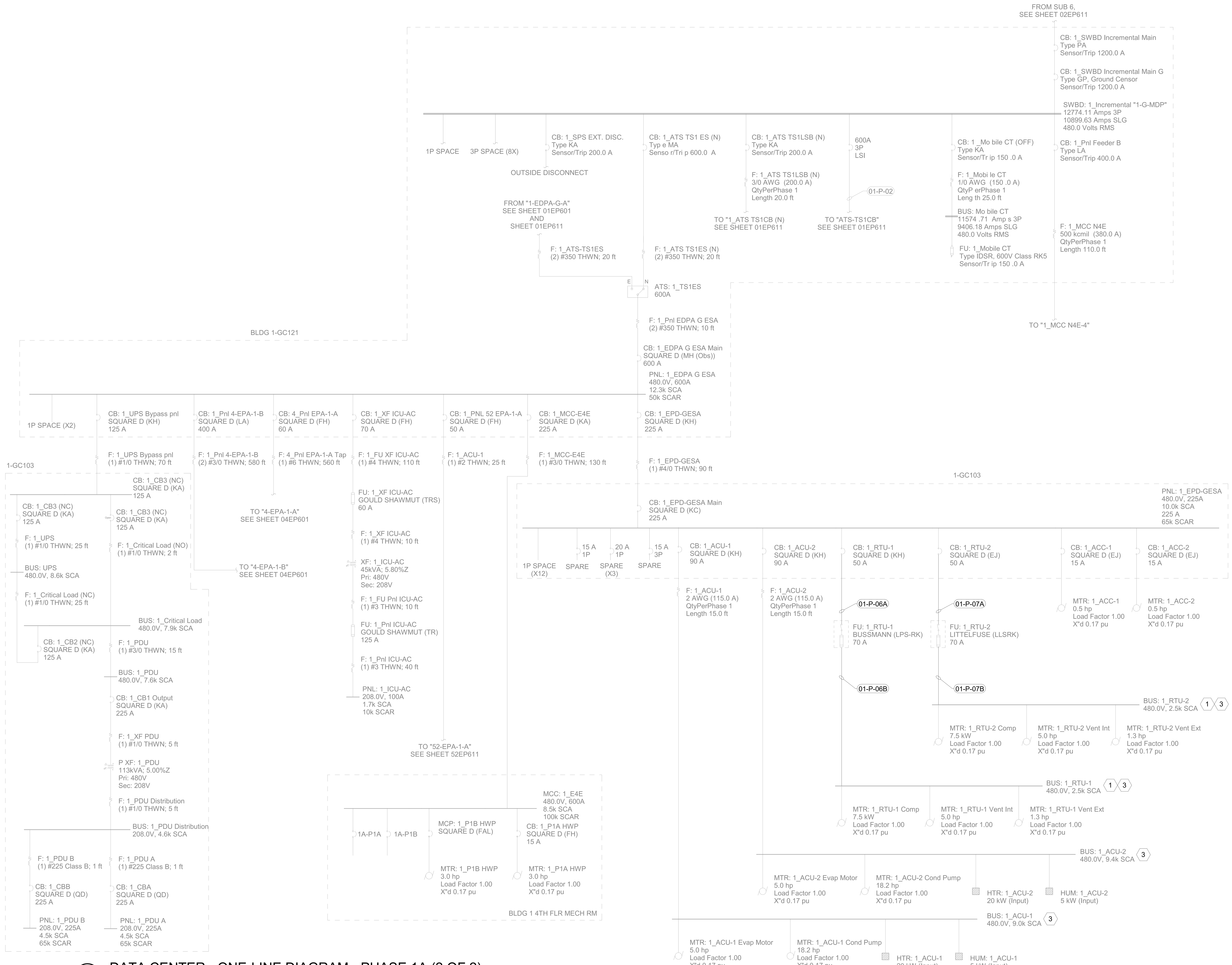

1 / -
DATA CENTER - ONE-LINE DIAGRAM - PHASE 1A (2 OF 2)
1/4" = 1'-0"

GENERAL NOTE

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30-DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL INTERIOR LOCATED EQUIPMENT SHALL BE NEMA 1 RATED AND ALL EXTERIOR LOCATED EQUIPMENT SHALL BE NEMA 3R RATED, UNLESS OTHERWISE NOTED.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
8. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTE

1. RELOCATED ROOFTOP MECHANICAL EQUIPMENT. EXTEND CONDUIT AND WIRE AS REQUIRED.
2. PROVIDE JUNCTION BOX AND INSULATED TAPS TO INTERCEPT AND EXTEND EXISTING CONDUIT AND WIRE TO RELOCATED EQUIPMENT. SIZE JUNCTION BOX PER NEC. CONTRACTOR SHALL FEILD COORDINATE JUNCTION BOX INSTALLATION LOCATION AS REQUIRED.
3. EQUIPMENT SHALL BE DEMOLISHED AND REPLACE IN PHASE 2B. REFERENCE ASSOCIATED DATA CENTER FLOOR PLANS FOR ADDITIONAL INFORMATION.

Revisions:	Date:

CONSULTANT
ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD
A/E:
SPEES DESIGN BUILD
23830 PACIFIC HIGHWAY S. SUITE 300
KENT, WA 98032
SPEESDESIGNBUILD

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LOUIS R. LITZKO · STATE OF WASHINGTON · 55681 · REGISTERED PROFESSIONAL ENGINEER · 9-3-25

Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
DATA CENTER - ONE-LINE DIAGRAM - PHASE 1A (2 OF 2)	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Building Number: 01
		Issue Date: 09/03/2025 · Checked: DCW/NVP · Drawn: RTS/JNV	Drawing Number: 01EP602-1A
			458 OF 693

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
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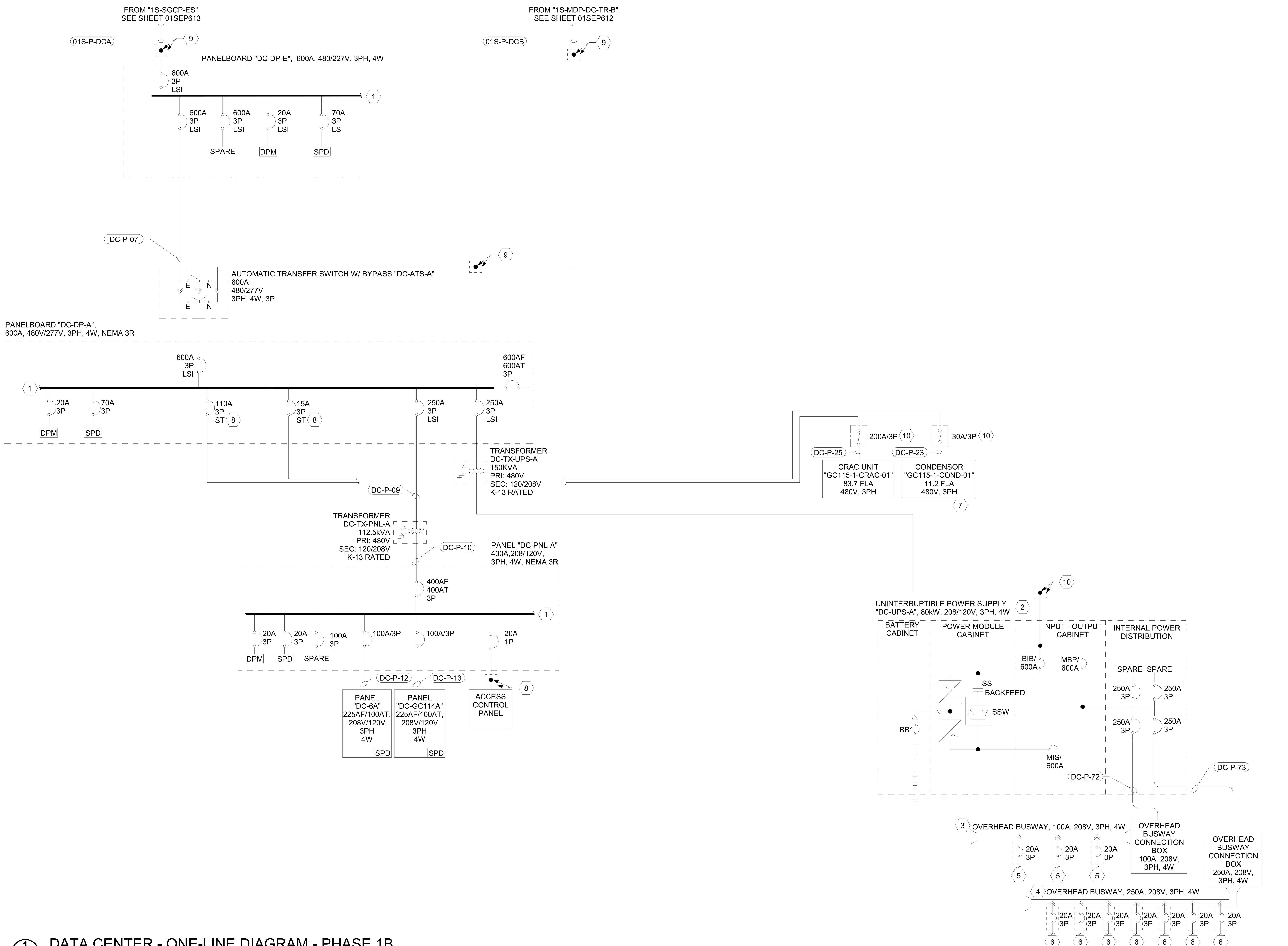

1 - DATA CENTER - ONE-LINE DIAGRAM - PHASE 1B
NTS

GENERAL NOTE

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30-DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL INTERIOR LOCATED EQUIPMENT SHALL BE NEMA 1 RATED AND ALL EXTERIOR LOCATED EQUIPMENT SHALL BE NEMA 3R RATED, UNLESS OTHERWISE NOTED.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
8. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTE

1. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. UNIT SHALL BE RATED FOR 80KW AND HAVE THE ABILITY TO BE EXPANDED TO 120KW. UNIT SHALL BE PROVIDED WITH 3-CB MAINTENANCE BYPASS SIDECAR. UNIT MINIMUM RUNTIME SHALL BY 10 MINUTES.
3. MDA ROW OVERHEAD BUSWAY, 100A, 208V, 3PH.
4. SERVER CABINET ROW OVERHEAD BUSWAY, 250A, 208V, 3PH.
5. CONTINUE TO MDA CABINET VIA PLUG/CORD SET AND CABINET RECEPTACLE. REFERENCE ELECTRICAL DETAILS FOR ADDITIONAL INFORMATION.
6. CONTINUE TO HDA OR SERVER CABINET VIA PLUG/CORD SET AND CABINET RECEPTACLE. REFERENCE ELECTRICAL DETAILS FOR ADDITIONAL INFORMATION.
7. MECHANICAL UNIT IS EQUIPPED WITH INTEGRAL STARTER. COORDINATE WITH MECHANICAL PLANS AND MANUFACTURER AS REQUIRED.
8. HVAC EQUIPMENT FEEDER BREAKERS SERVING DATA CENTER SHALL BE SHUNT TRIPPED UPON HVAC SHUTDOWN SWITCH ACTIVATION (HVAC SHUTDOWN SWITCHES ARE LOCATED IN MCR ROOM). REFERENCE SHEETS 01EF400Gc AND SHEET EF601 FOR ADDITIONAL INFORMATION.
9. PROVIDE JUNCTION BOX AND INSULATED TAPS FOR TEMPORARY AND PERMANENT POWER PROVISION. SIZE JUNCTION BOX PER NEC. REFERENCE SHEET 01ED400GI FOR JUNCTION BOX LOCATION. CONTRACTOR SHALL FIELD COORDINATE JUNCTION BOX INSTALLATION LOCATION AS REQUIRED.
10. PROVIDE EXTRA LUGS FOR TERMPORARY AND PERMANENT POWER PROVISIONS.

Revisions: | Date:

CONSULTANT
ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD
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STATE OF WASHINGTON
55681
REGISTERED
PROFESSIONAL ENGINEER
9-3-25

Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title: DATA CENTER - ONE-LINE DIAGRAM - PHASE 1B
Approved:

Phase: 100% CONSTRUCTION DOCUMENTS
FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location: VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date: 09/03/2025 | Checked: DCW/NVP | Drawn: RTS/JNV

Project Number: 581-22-700
Building Number: 01
Drawing Number: 01EP603-1B
459 OF 693

VA FORM 08 - 6231

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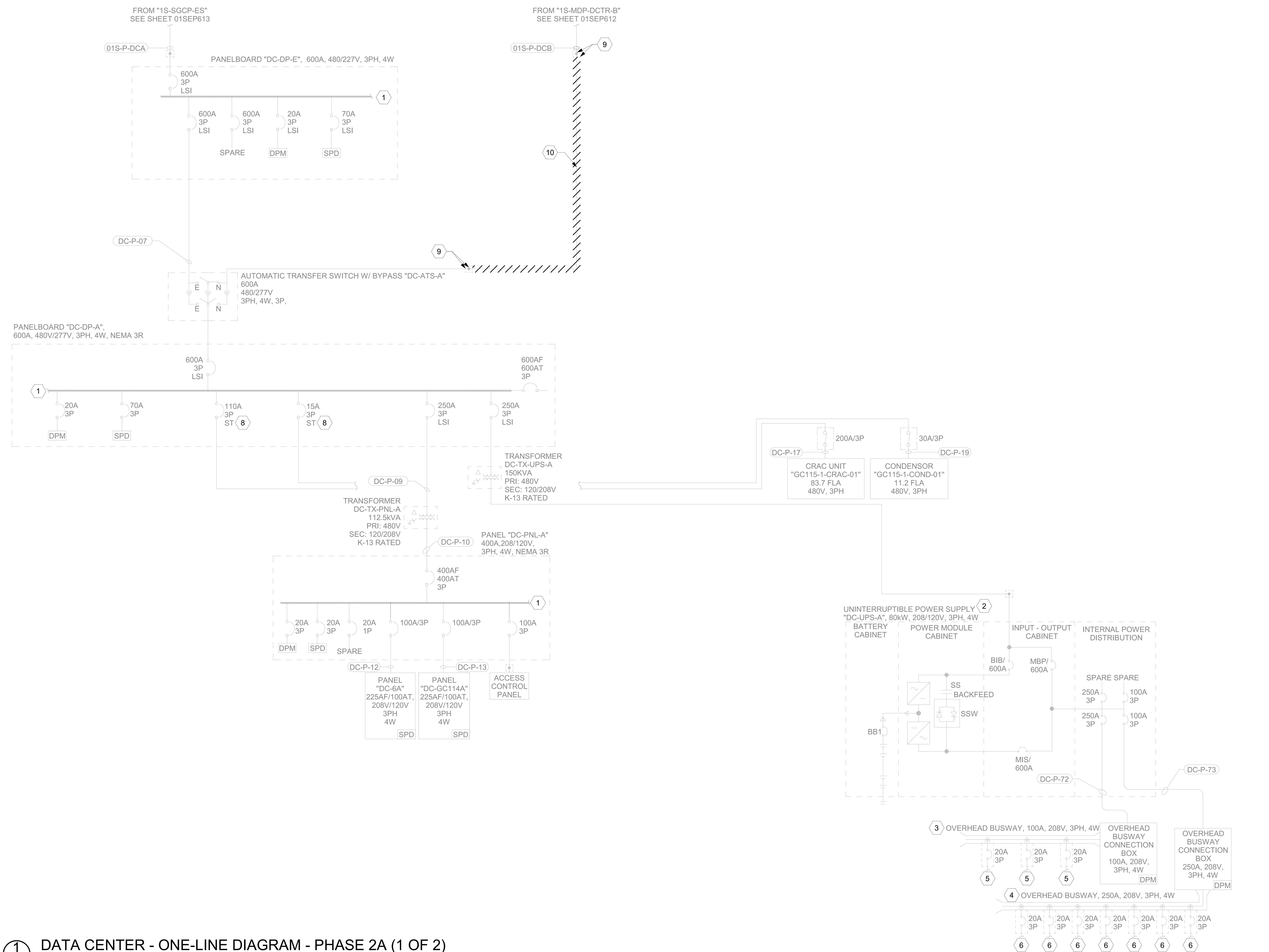

1 DATA CENTER - ONE-LINE DIAGRAM - PHASE 2A (1 OF 2)

NTS

GENERAL NOTE

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30-DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL INTERIOR LOCATED EQUIPMENT SHALL BE NEMA 1 RATED AND ALL EXTERIOR LOCATED EQUIPMENT SHALL BE NEMA 3R RATED, UNLESS OTHERWISE NOTED.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTE

1. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. UNIT SHALL BE RATED FOR 80KW AND HAVE THE ABILITY TO BE EXPANDED TO 120KW. UNIT SHALL BE PROVIDED WITH 3-CB MAINTENANCE BYPASS SIDECAR. UNIT MINIMUM RUNTIME SHALL BY 10 MINUTES.
3. MDA ROW OVERHEAD BUSWAY, 100A, 208V, 3PH.
4. SERVER CABINET ROW OVERHEAD BUSWAY, 250A, 208V, 3PH.
5. CONTINUE TO MDA CABINET VIA PLUG/CORD SET AND CABINET RECEPTACLE. REFERENCE ELECTRICAL DETAILS FOR ADDITIONAL INFORMATION.
6. CONTINUE TO HDA OR SERVER CABINET VIA PLUG/CORD SET AND CABINET RECEPTACLE. REFERENCE ELECTRICAL DETAILS FOR ADDITIONAL INFORMATION.
7. MECHANICAL UNIT IS EQUIPPED WITH INTEGRAL STARTER. COORDINATE WITH MECHANICAL PLANS AND MANUFACTURER AS REQUIRED.
8. HVAC EQUIPMENT FEEDER BREAKERS SERVING DATA CENTER SHALL BE SHUNT TRIPPED UPON HVAC SHUTDOWN SWITCH ACTIVATION (HVAC SHUTDOWN SWITCHES ARE LOCATED IN MCR ROOM). REFERENCE SHEETS 01EF400Gc AND SHEET EF601 FOR ADDITIONAL INFORMATION.
9. UTILIZE JUNCTION BOX AND INSULATED TAPS FOR TEMPORARY AND PERMANENT POWER PROVISION. SIZE JUNCTION BOX PER NEC. REFERENCE SHEET 01ED400GI FOR JUNCTION BOX LOCATION. CONTRACTOR SHALL FIELD COORDINATE JUNCTION BOX INSTALLATION LOCATION AS REQUIRED.
10. DEMOLISH TEMPORARY RACEWAY AND CONDUCTOR BETWEEN JUNCTION BOXES AS INDICATED.

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title	DATA CENTER - ONE-LINE DIAGRAM - PHASE 2A (1 OF 2)
Approved:	
Phase	100% CONSTRUCTION DOCUMENTS
	FULLY SPRINKLERED
Project Title	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location	VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date	09/03/2025
Checked	DCW/NVP
Drawn	RTS/JNV
Project Number	581-22-700
Building Number	01
Drawing Number	01EP604-2A
	460 OF 693

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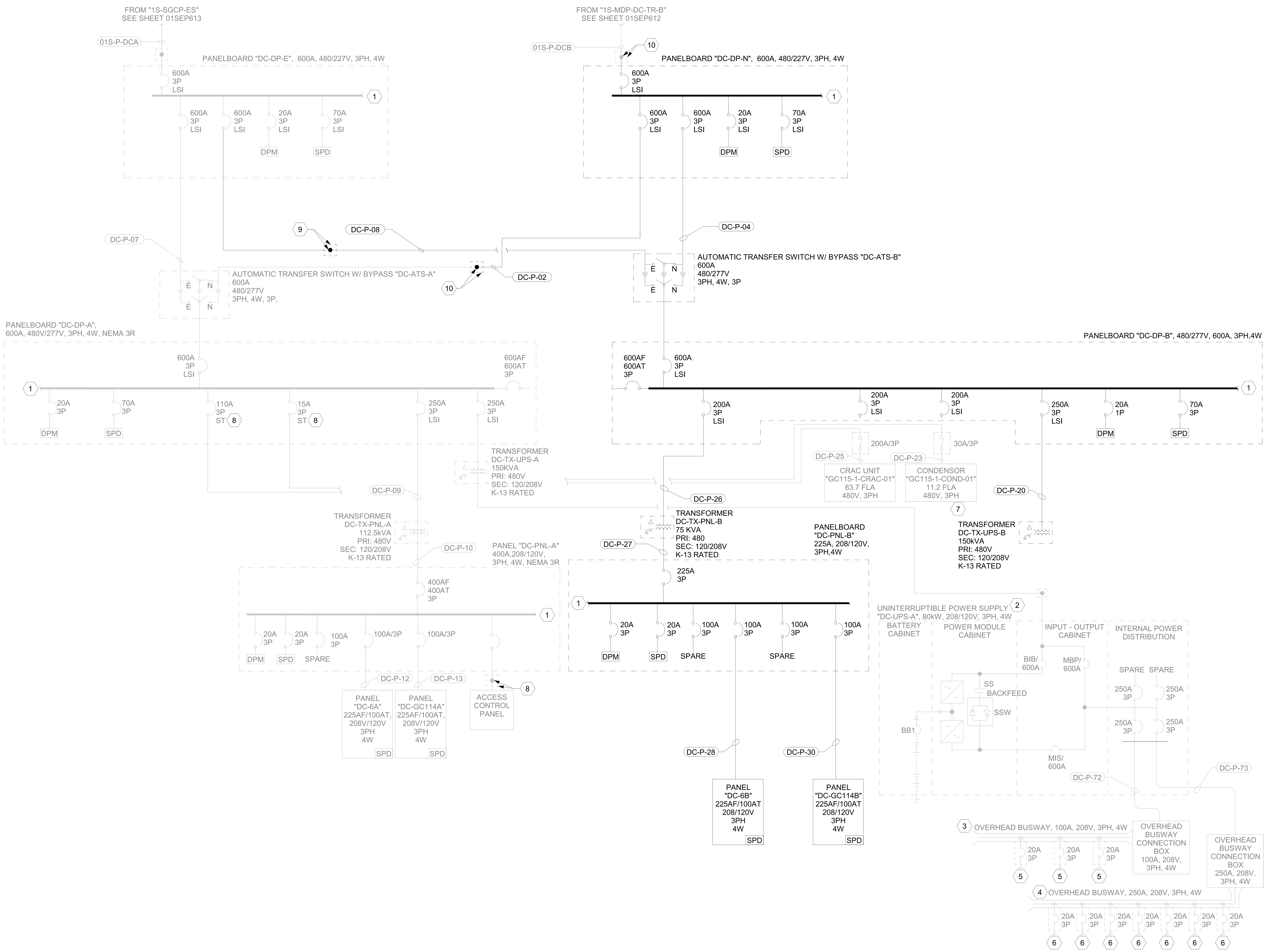
9/3/2025 9:32:24 AM



1 DATA CENTER - ONE-LINE DIAGRAM - PHASE 2A (2 OF 2)

NTS

GENERAL NOTE

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30-DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
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6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
8. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTE

1. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. UNIT SHALL BE RATED FOR 80KW AND HAVE THE ABILITY TO BE EXPANDED TO 120KW. UNIT SHALL BE PROVIDED WITH 3-CB MAINTENANCE BYPASS SIDECAR. UNIT MINIMUM RUNTIME SHALL BY 10 MINUTES.
3. MDA ROW OVERHEAD BUSWAY, 100A, 208V, 3PH.
4. SERVER CABINET ROW OVERHEAD BUSWAY, 250A, 208V, 3PH.
5. CONTINUE TO MDA CABINET VIA PLUG/CORD SET AND CABINET RECEPTACLE. REFERENCE ELECTRICAL DETAILS FOR ADDITIONAL INFORMATION.
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9. UTILIZE JUNCTION BOX AND INSULATED TAPS FOR TEMPORARY AND PERMANENT POWER PROVISION. SIZE JUNCTION BOX PER NEC. REFERENCE SHEET 01ED400GI FOR JUNCTION BOX LOCATION. CONTRACTOR SHALL FIELD COORDINATE JUNCTION BOX INSTALLATION LOCATION AS REQUIRED.
10. PROVIDE ENCLOSED CIRCUIT BREAKER "DC-ECB-TIE" BETWEEN PANELBOARD "DC-DP-A" AND PANELBOARD "DC-DP-B". CIRCUIT BREAKER SHALL ONLY BE ENERGIZED AFTER MAIN CIRCUIT BREAKER OF EITHER "DC-DP-A" OR "DC-DP-B" HAS BEEN DEENERGIZED. PROVIDE MECHANICAL KEYED INTERLOCKS AS NEEDED.

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

A/E:
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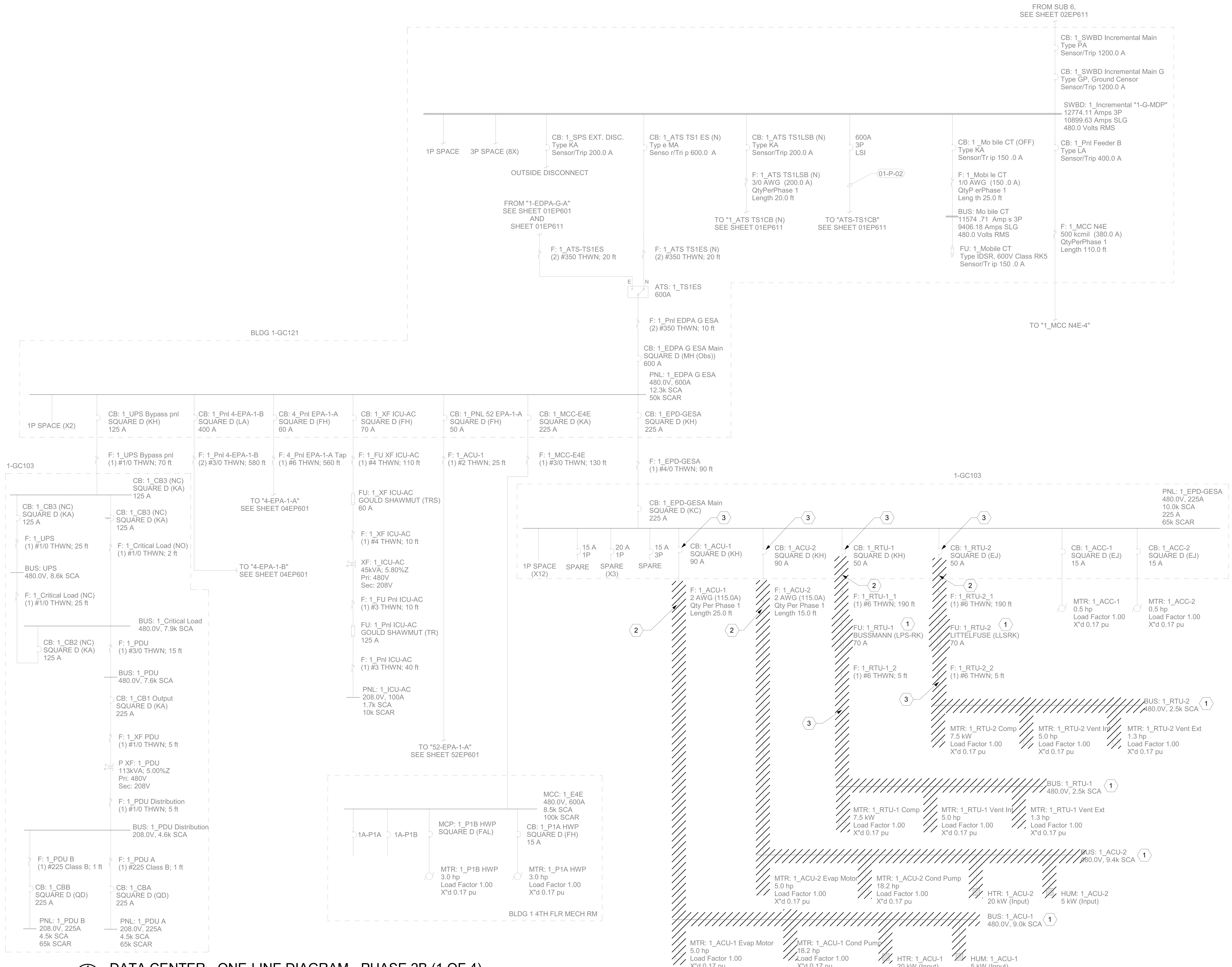
LOUIS R. LITZKO · STATE OF WASHINGTON · 55681 · REGISTERED PROFESSIONAL ENGINEER · 9-3-25

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: DATA CENTER - ONE-LINE DIAGRAM - PHASE 2A (2 OF 2)	Phase: 100% CONSTRUCTION DOCUMENTS	Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	Project Number: 581-22-700
			Building Number: 01
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Drawing Number: 01EP605-2A
		Issue Date: 09/03/2025 · Checked: DCW/NVP · Drawn: RTS/JNV	461 OF 693



1 - DATA CENTER - ONE-LINE DIAGRAM - PHASE 2B (1 OF 4)
NTS

GENERAL NOTE

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30-DAYS PRIOR TO IMPLEMENTATION.
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5. ALL INTERIOR LOCATED EQUIPMENT SHALL BE NEMA 1 RATED AND ALL EXTERIOR LOCATED EQUIPMENT SHALL BE NEMA 3R RATED, UNLESS OTHERWISE NOTED
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTE

1. EXISTING EQUIPMENT SHALL BE DEMOLISHED. ALL CONDUIT AND CONDUCTORS SHALL BE DEMOLISHED BACK TO SOURCE.
2. EXISTING CONDUIT AND CONDUCTORS SERVING EQUIPMENT BEING DEMOLISHED. ALL ROOFTOP PENETRATIONS SHALL BE PATCHED TO PREVENT DAMAGE DUE TO WEATHER.
3. EXISTING CIRCUIT BREAKERS SERVING DEMOLISHED LOADS SHALL BE PRESERVED. CIRCUIT DIRECTORY SHALL BE UPDATED AND CIRCUIT BREAKERS LABELED AS SPARES.

Revisions:	Date:

CONSULTANT
ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD
A/E:
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4-3-25

Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title: DATA CENTER - ONE-LINE DIAGRAM - PHASE 2B (1 OF 4)
Approved:

Phase: 100% CONSTRUCTION DOCUMENTS
FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700
Building Number: 01
Drawing Number: 01EP606-2B
462 OF 693

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:33:10 AM

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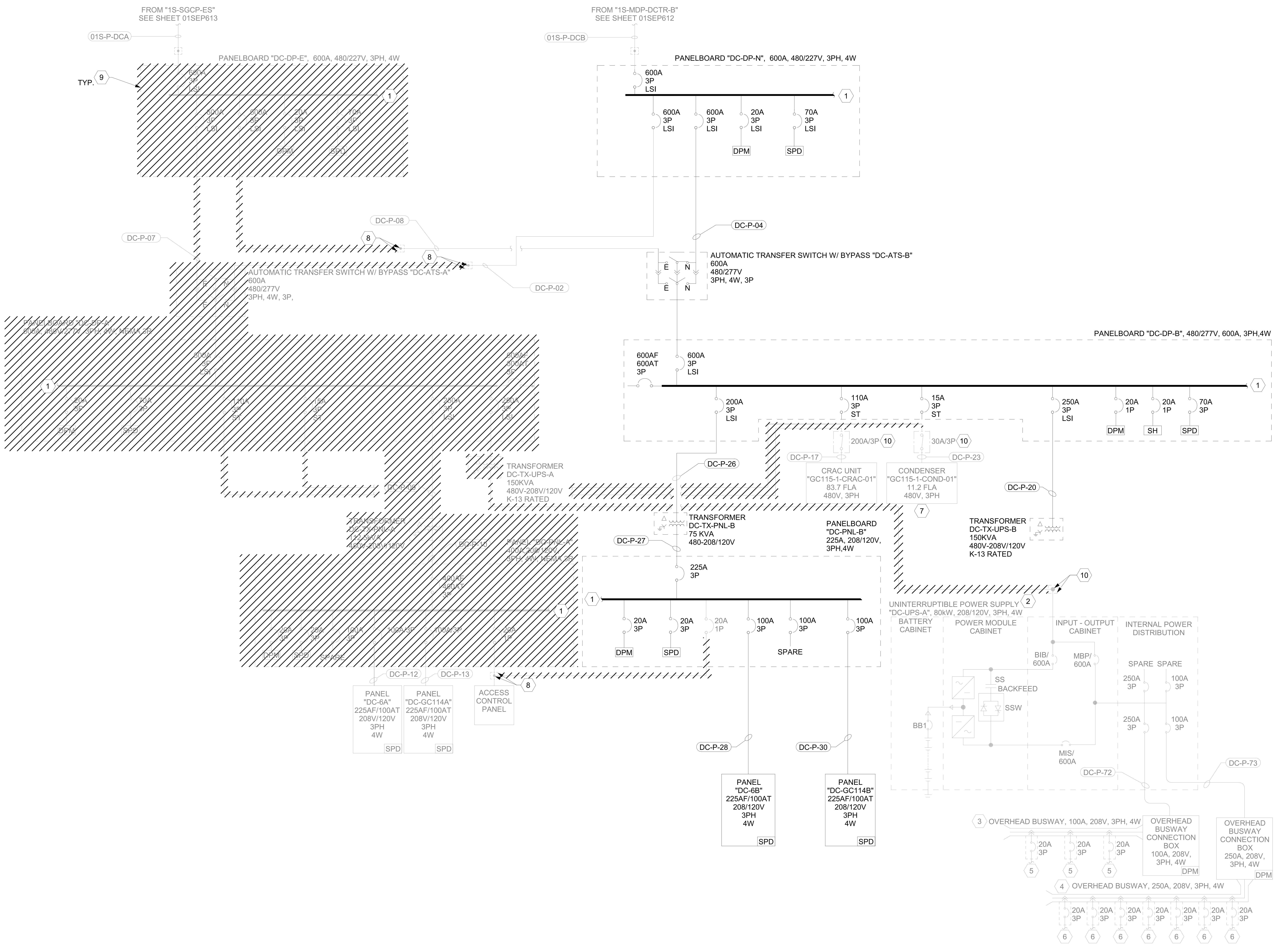

1 DATA CENTER - ONE-LINE DIAGRAM - PHASE 2B (2 OF 4)
NTS

GENERAL NOTE

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30-DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL INTERIOR LOCATED EQUIPMENT SHALL BE NEMA 1 RATED AND ALL EXTERIOR LOCATED EQUIPMENT SHALL BE NEMA 3R RATED, UNLESS OTHERWISE NOTED.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
7. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTE

1. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. UNIT SHALL BE RATED FOR 80KW AND HAVE THE ABILITY TO BE EXPANDED TO 120KW. UNIT SHALL BE PROVIDED WITH 3-CB MAINTENANCE BYPASS SIDECAR. UNIT MINIMUM RUNTIME SHALL BY 10 MINUTES.
3. MDA ROW OVERHEAD BUSWAY, 100A, 208V, 3PH.
4. SERVER CABINET ROW OVERHEAD BUSWAY, 250A, 208V, 3PH.
5. CONTINUE TO MDA CABINET VIA PLUG/CORD SET AND CABINET RECEPTACLE. REFERENCE ELECTRICAL DETAILS FOR ADDITIONAL INFORMATION.
6. CONTINUE TO HDA OR SERVER CABINET VIA PLUG/CORD SET AND CABINET RECEPTACLE. REFERENCE ELECTRICAL DETAILS FOR ADDITIONAL INFORMATION.
7. MECHANICAL UNIT IS EQUIPPED WITH INTEGRAL STARTER. COORDINATE WITH MECHANICAL PLANS AND MANUFACTURER AS REQUIRED.
8. UTILIZE JUNCTION BOX AND INSULATED TAPS FOR TEMPORARY AND PERMANENT POWER PROVISION. SIZE JUNCTION BOX PER NEC. REFERENCE SHEET 01ED400GI FOR JUNCTION BOX LOCATION. CONTRACTOR SHALL FIELD COORDINATE JUNCTION BOX INSTALLATION LOCATION AS REQUIRED.
9. DEMOLISH TEMPORARY EQUIPMENT, RACEWAY, AND CONDUCTOR AS INDICATED.
10. UTILIZE EXTRA LUGS FOR TEMPORARY AND PERMANENT POWER PROVISIONS. REFERENCE REVISED PHASE 2B ONE-LINE DIAGRAMS FOR ADDITIONAL INFORMATION.

Revisions:	Date:

CONSULTANT
ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD
A/E:
SPEES DESIGN BUILD
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STATE OF WASHINGTON
55681
REGISTERED PROFESSIONAL ENGINEER
4-3-25

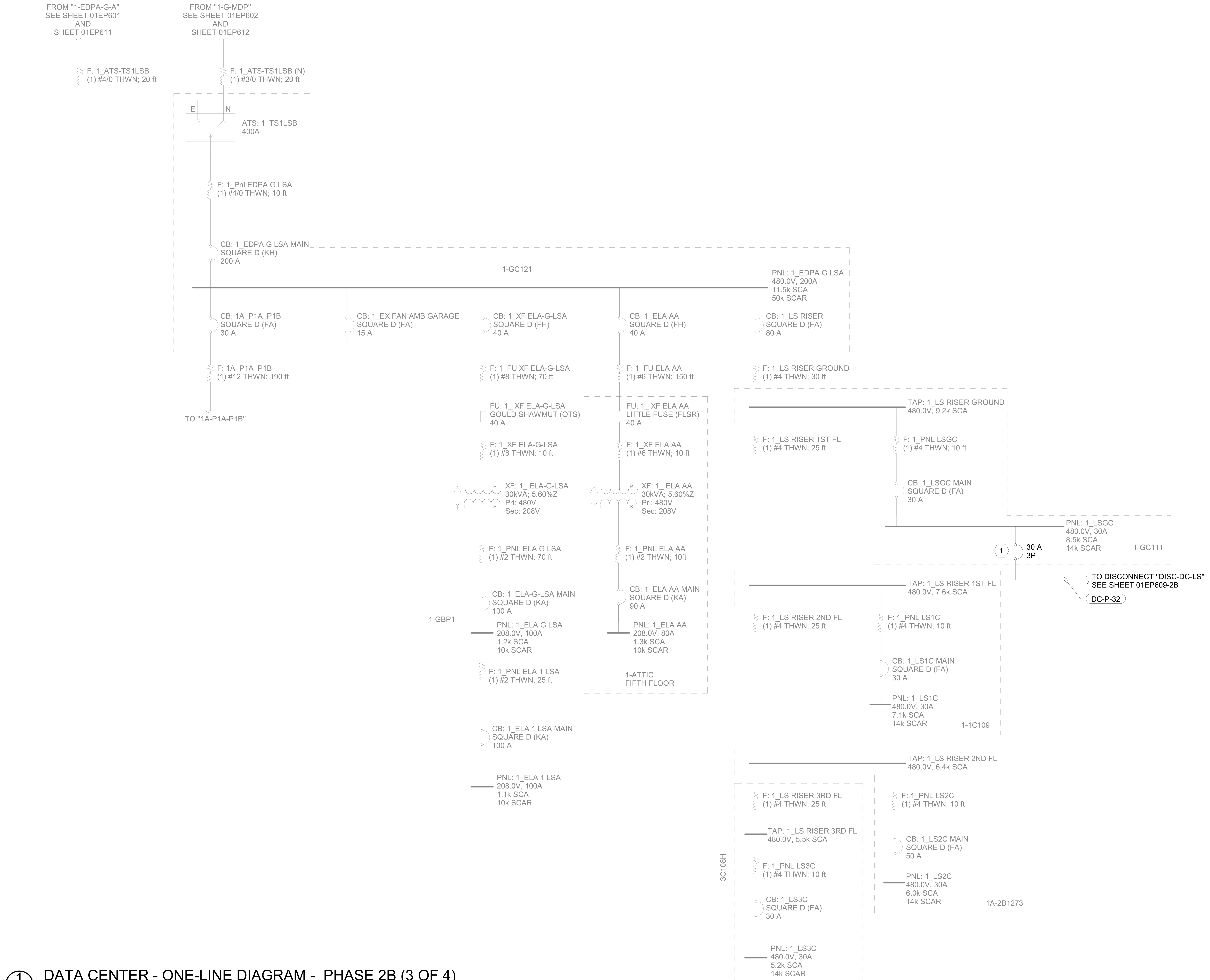
Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title: DATA CENTER - ONE-LINE DIAGRAM - PHASE 2B (2 OF 4)
Approved:

Phase: 100% CONSTRUCTION DOCUMENTS
FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date: 09/03/2025 | Checked: DCW/NVP | Drawn: RTS/JNV

Project Number: 581-22-700
Building Number: 01
Drawing Number: 01EP607-2B



1 / - DATA CENTER - ONE-LINE DIAGRAM - PHASE 2B (3 OF 4)

NTS

GENERAL NOTE

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30-DAYS PRIOR TO IMPLEMENTATION.
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5. ALL INTERIOR LOCATED EQUIPMENT SHALL BE NEMA 1 RATED AND ALL EXTERIOR LOCATED EQUIPMENT SHALL BE NEMA 3R RATED, UNLESS OTHERWISE NOTED.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
8. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTE

1 EXISTING DISTRIBUTION EQUIPMENT SHALL BE MODIFIED; PROVIDE OVERCURRENT PROTECTIVE DEVICE TO SUPPLY LOAD AS SHOWN.

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

A/E:
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SDB SPEESDESIGNBUILD

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LOUIS R. LITZKO · STATE OF WASHINGTON · 55681 · 9-3-25 · REGISTERED PROFESSIONAL ENGINEER

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: DATA CENTER - ONE-LINE DIAGRAM - PHASE 2B (3 OF 4)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700

Building Number: 01

Drawing Number: 01EP608-2B

464 OF 693

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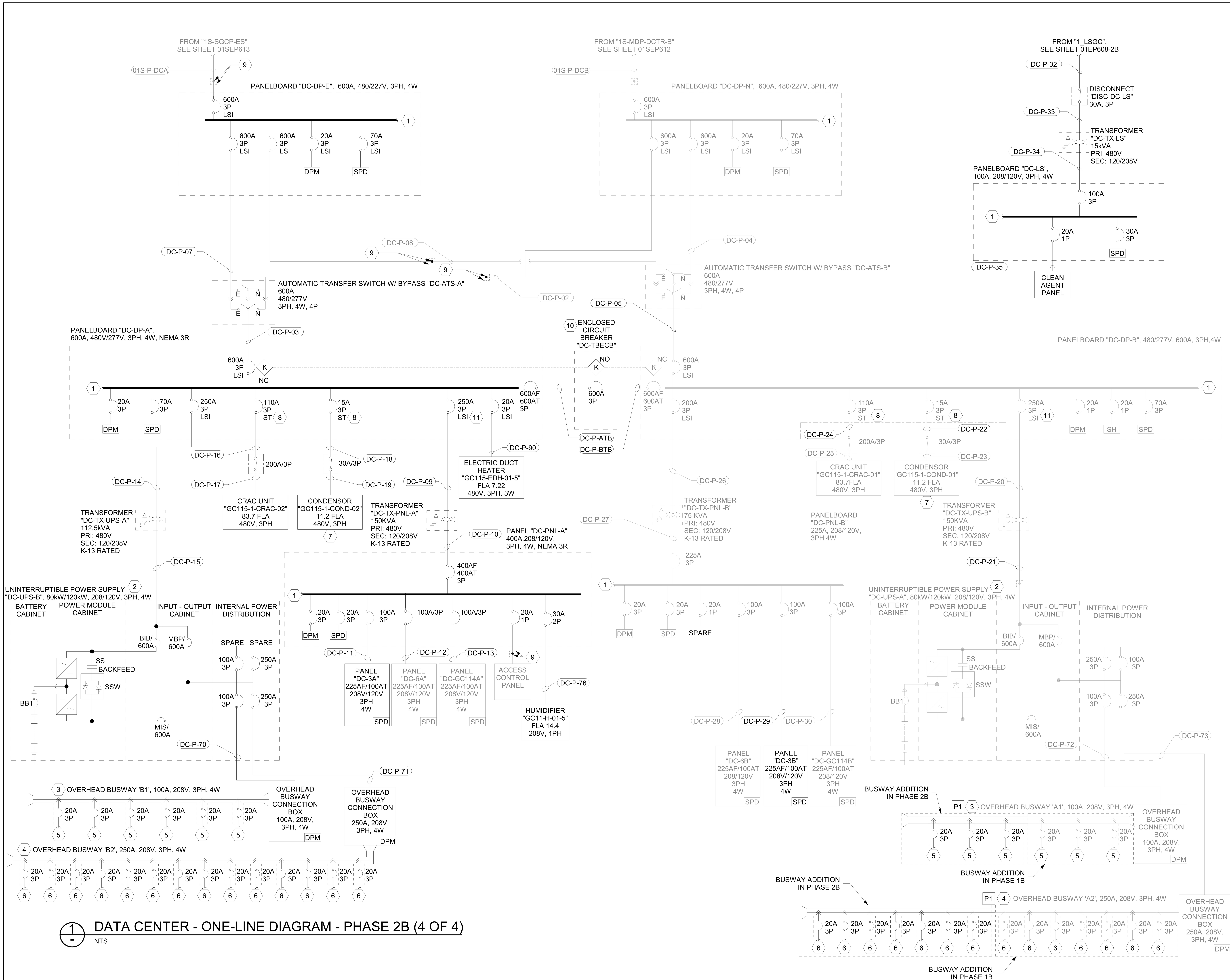

1 DATA CENTER - ONE-LINE DIAGRAM - PHASE 2B (4 OF 4)
NTS

GENERAL NOTE

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30-DAYS PRIOR TO IMPLEMENTATION.
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7. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTE

1. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. UNIT SHALL BE RATED FOR 80KW AND HAVE THE ABILITY TO BE EXPANDED TO 120KW. UNIT SHALL BE PROVIDED WITH AN INPUT-OUTPUT CABINET MAINTENANCE BYPASS SIDECAR, POWER MODULE SECTION, BATTERY CABINET SECTION, AND INTEGRAL POWER DISTRIBUTION SECTION AS INDICATED. UNIT MINIMUM RUNTIME SHALL BY 10 MINUTES. UNIT SHALL BE PROVIDED WITH EMERGENCY SHUTDOWN CAPABILITY; IT SHALL BE ABLE TO RECEIVE INPUT FROM REMOTE PUSHBUTTONS AND SIGNAL FROM FIRE ALARM SYSTEM TO DEACTIVATE POWER INPUT AND OUTPUT.
3. MDA ROW OVERHEAD BUSWAY, 100A, 208V, 3PH.
4. SERVER CABINET ROW OVERHEAD BUSWAY, 250A, 208V, 3PH.
5. CONTINUE TO MDA CABINET VIA PLUG/CORD SET AND CABINET RECEPTACLE. REFERENCE ELECTRICAL DETAILS FOR ADDITIONAL INFORMATION.
6. CONTINUE TO HDA OR SERVER CABINET VIA PLUG/CORD SET AND CABINET RECEPTACLE. REFERENCE ELECTRICAL DETAILS FOR ADDITIONAL INFORMATION.
7. MECHANICAL UNIT IS EQUIPPED WITH INTEGRAL STARTER. COORDINATE WITH MECHANICAL PLANS AND MANUFACTURER AS REQUIRED.
8. HVAC EQUIPMENT FEEDER BREAKERS SERVING DATA CENTER SHALL BE SHUNT TRIPPED UPON HVAC SHUTDOWN SWITCH ACTIVATION (HVAC SHUTDOWN SWITCHES ARE LOCATED IN MCR ROOM). REFERENCE SHEETS 01EF400Gc AND SHEET EF601 FOR ADDITIONAL INFORMATION.
9. UTILIZE JUNCTION BOX AND INSULATED TAPS FOR TEMPORARY AND PERMANENT POWER PROVISION. SIZE JUNCTION BOX PER NEC. REFERENCE SHEET 01ED400GI FOR JUNCTION BOX LOCATION. CONTRACTOR SHALL FIELD COORDINATE JUNCTION BOX INSTALLATION LOCATION AS REQUIRED.
10. PROVIDE ENCLOSED CIRCUIT BREAKER "DC-ECB-TIE" BETWEEN PANELBOARD "DC-DP-A"AND PANELBOARD "DC-DP-B". CIRCUIT BREAKER SHALL ONLY BE ENERGIZED AFTER MAIN CIRCUIT BREAKER OF EITHER "DC-DP-A" OR "DC-DP-B" HAS BEEN DEENERGIZED. PROVIDE MECHANICAL KEYED INTERLOCKS AS NEEDED.
11. MCR POWER EQUIPMENT FEEDER BREAKERS SERVING DATA CENTER SHALL BE SHUNT TRIPPED UPON EPO SHUTDOWN SWITCH ACTIVATION (EPO SWITCHES ARE LOCATED IN MCR ROOM). REFERENCE SHEETS 01EF400Gf-2B AND EF601 FOR ADDITIONAL INFORMATION. PROVIDE CONDUIT AND WIRE AS REQUIRED.

PHASING NOTES

PHASING GENERAL NOTE: ALL PHASING SHALL BE COORDINATED WITH VA PRIOR TO ELECTRICAL INSTALLATION. PHASING NOTES SHOW ONE WAY OF PROVIDING WORK WHILE POSSIBLY MINIMIZING OUTAGES AND ARE INFORMATIONAL ONLY; OTHER METHODS MAY BE ACCEPTABLE. PHASING NOTES ARE NOTED IN SEQUENCE OF WORK.

P1 A-SIDE BUSWAY SHALL BE DE-ENERGIZED FOR INSTALLATION OF BUS EXTENSION. B-SIDE BUSWAY SHALL BE INSTALLED PRIOR AND REMAIN ENERGIZED TO ENSURE NO EQUIPMENT OUTAGES.

Revisions: | Date:

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LOUIS R. LITZKO — STATE OF WASHINGTON — 55681 — REGISTERED PROFESSIONAL ENGINEER — 9-3-25

Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
DATA CENTER - ONE-LINE DIAGRAM - PHASE 2B (4 OF 4)	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Building Number: 01
		Issue Date: 09/03/2025 · Checked: DCW/NVP · Drawn: RTS/JNV	Drawing Number: 01EP609-2B
			465 OF 693

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:32:12 AM

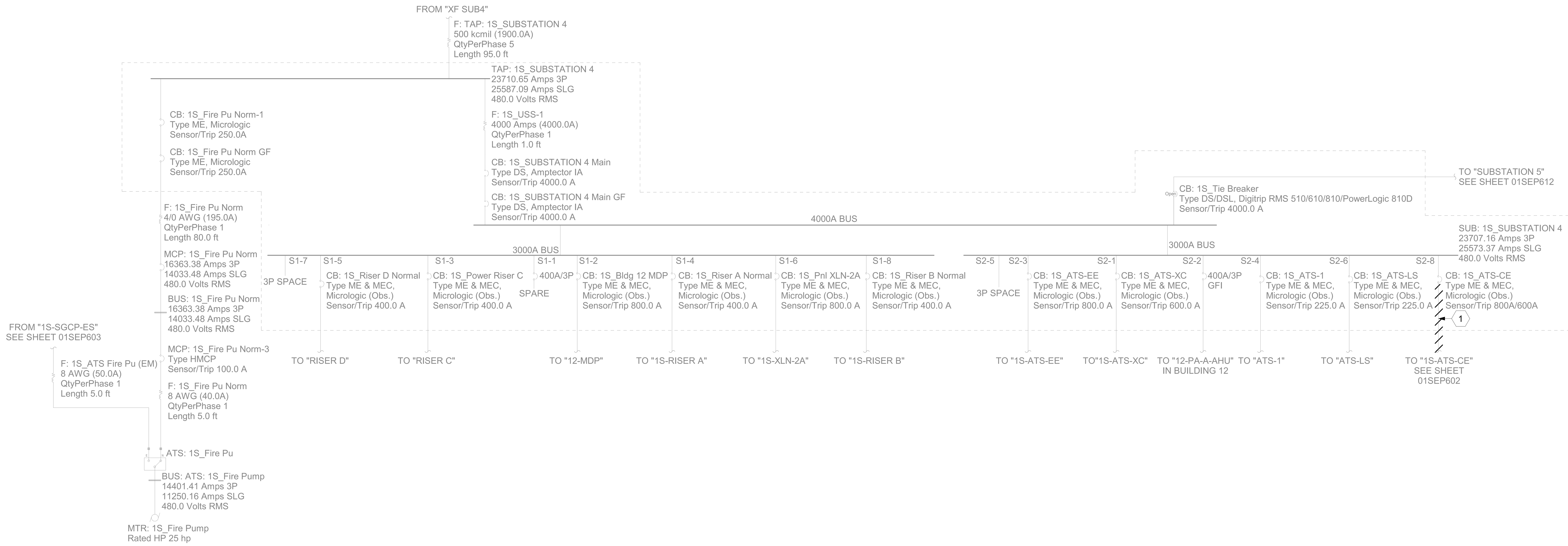
VA FORM 08 - 6231

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTES

1. EXISTING CONDUCTORS SHALL BE DEMOLISHED. EXISTING CONDUIT SHALL REMAIN.



1 BLDG 01S - PARTIAL ONE-LINE DIAGRAM - EXISTING (1 OF 3)

NTS

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01S - PARTIAL ONE-LINE DIAGRAM - EXISTING (1 OF 3)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700

Building Number: 01S

Drawing Number: 01SEP601

VA FORM 08 - 6231

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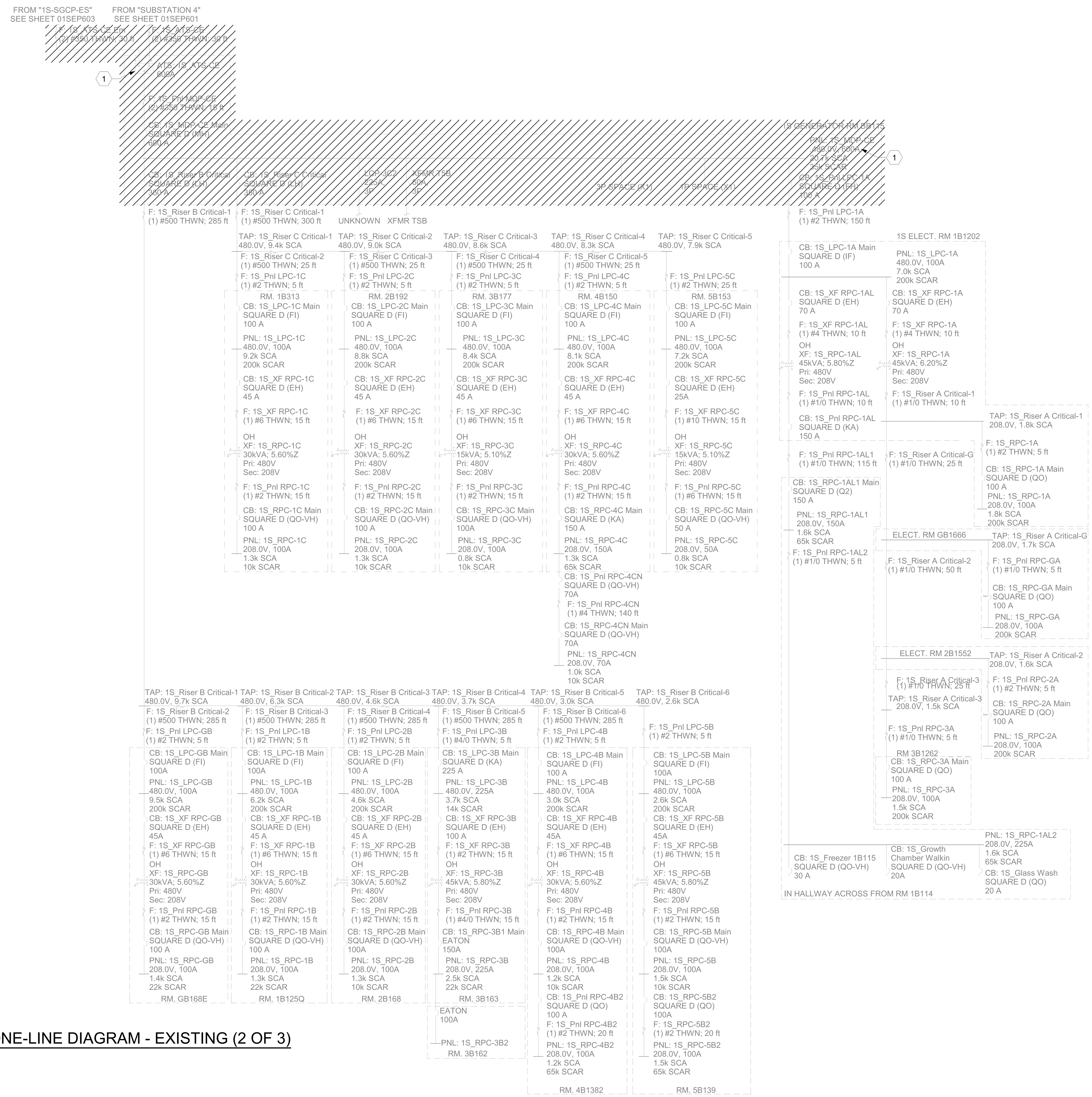

1 BLDG 01S - PARTIAL ONE-LINE DIAGRAM - EXISTING (2 OF 3)

1/8" = 1'-0"

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTES

1 EXISTING EQUIPMENT SHALL BE DEMOLISHED AND REPLACED. PRESERVE CIRCUITS FOR RETERMINATION. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01S' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.

Revisions:	Date:

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VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01S - PARTIAL ONE-LINE DIAGRAM - EXISTING (2 OF 3)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700

Building Number: 01S

Drawing Number: 01SEP602

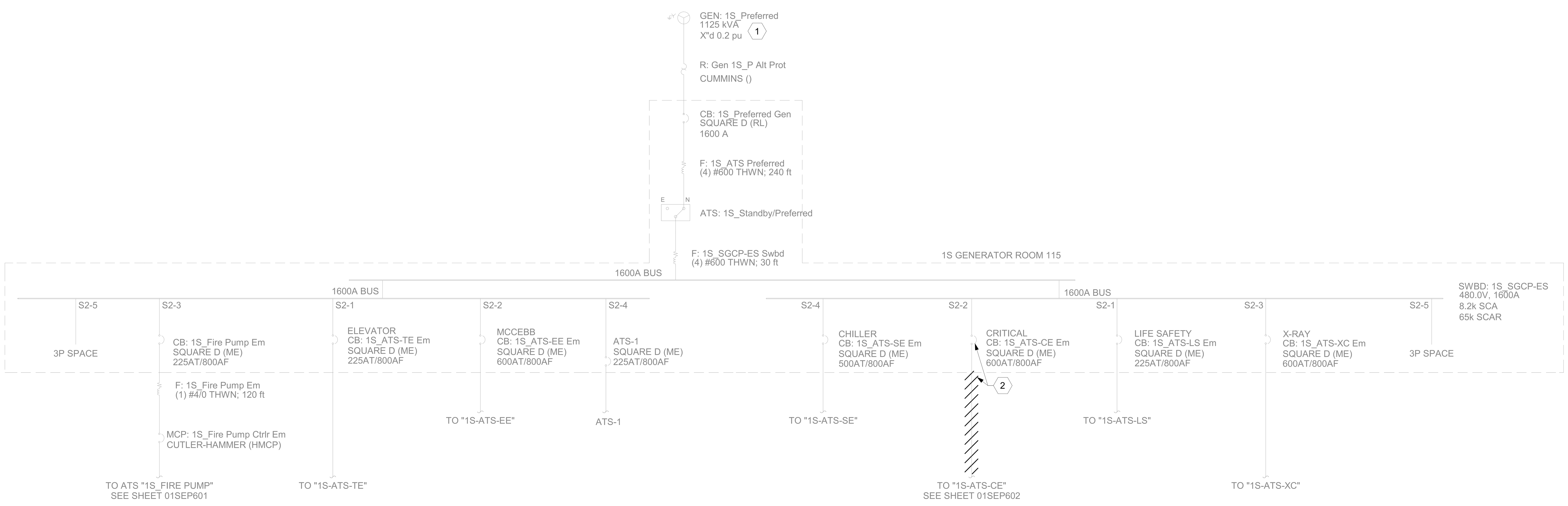
GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTES

1 REFERENCE ASSOCIATED GENERATOR CAPACITY AND LOAD INFORMATION SCHEDULE ON ELECTRICAL SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.

2 DEMOLISH EXISTING FEEDER FROM SWITCHBOARD 1S-SGCP-ES TO 1S-ATS-CE. ADJUST EXISTING BREAKER TO 800A TRIP.



1 - BLDG 01S - PARTIAL ONE-LINE DIAGRAM - EXISTING (3 OF 3)

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01S - PARTIAL ONE-LINE DIAGRAM - EXISTING (3 OF 3)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

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Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date: 09/03/2025 | Checked: DCW/NVP | Drawn: RTS/JNV

Project Number: 581-22-700

Building Number: 01S

Drawing Number: 01SEP603

VA FORM 08 - 6231

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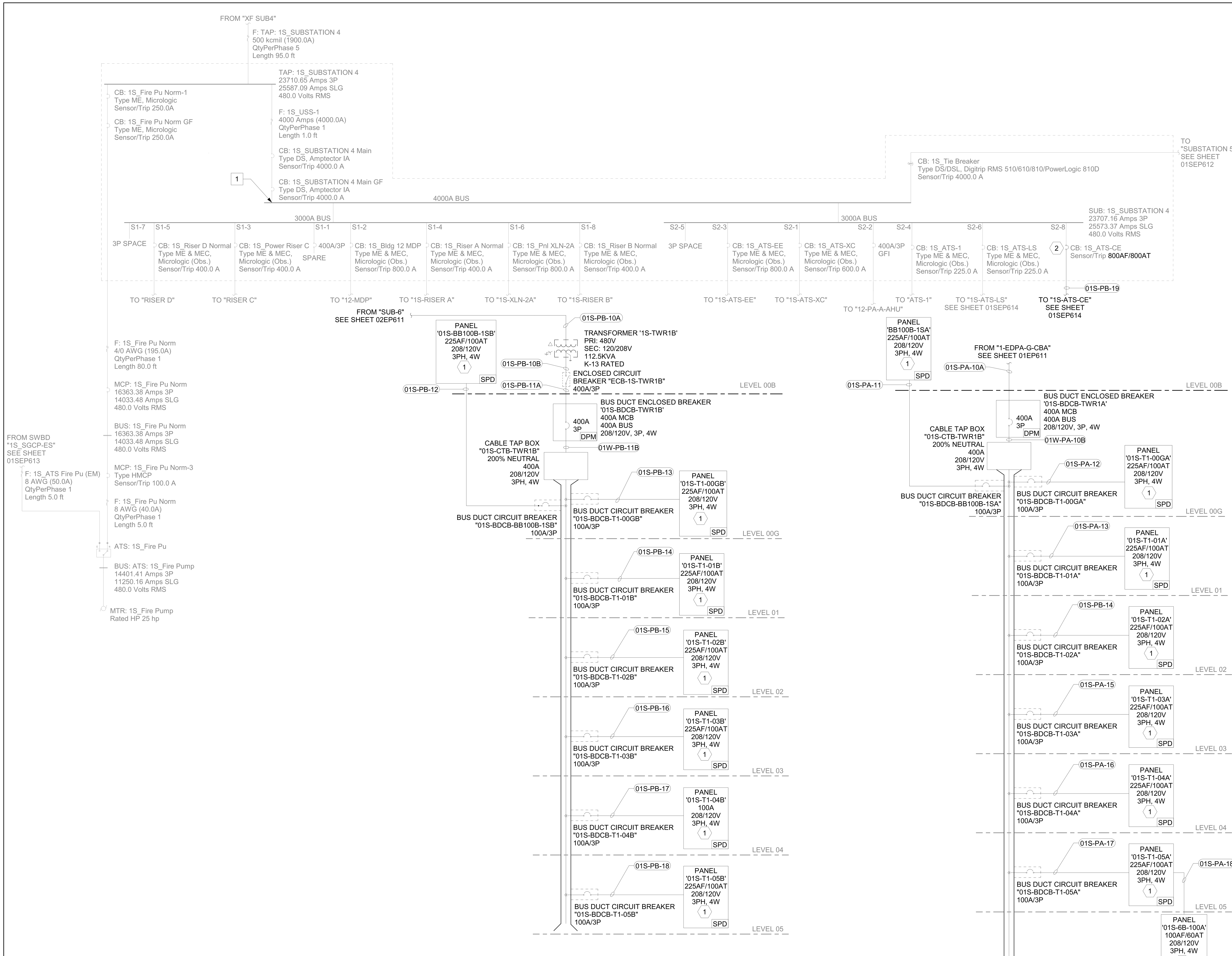

1 - BLDG 01S - ONE-LINE DIAGRAM - REVISED (1 OF 4)
NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
7. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01S' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. CIRCUIT BREAKER TRIP SHALL BE ADJUSTED TO 800A.

LOAD DATA POINT NOTES

1. LOAD DATA POINT: EXISTING SWITCHBOARD "1S_SUBSTATION 4" MAIN. REFERENCE "METERING AND LOAD INFORMATION SCHEDULE" ON SHEET 01SPEP704 FOR ADDITIONAL INFORMATION.

Revisions:	Date:

CONSULTANT
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Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01S - ONE-LINE DIAGRAM - REVISED (1 OF 4)
Approved:
Phase: 100% CONSTRUCTION DOCUMENTS
FULLY SPRINKLERED
Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date: 09/03/2025
Checked: DCW/NVP
Drawn: RTS/JNV
Project Number: 581-22-700
Building Number: 01S
Drawing Number: 01SEP611

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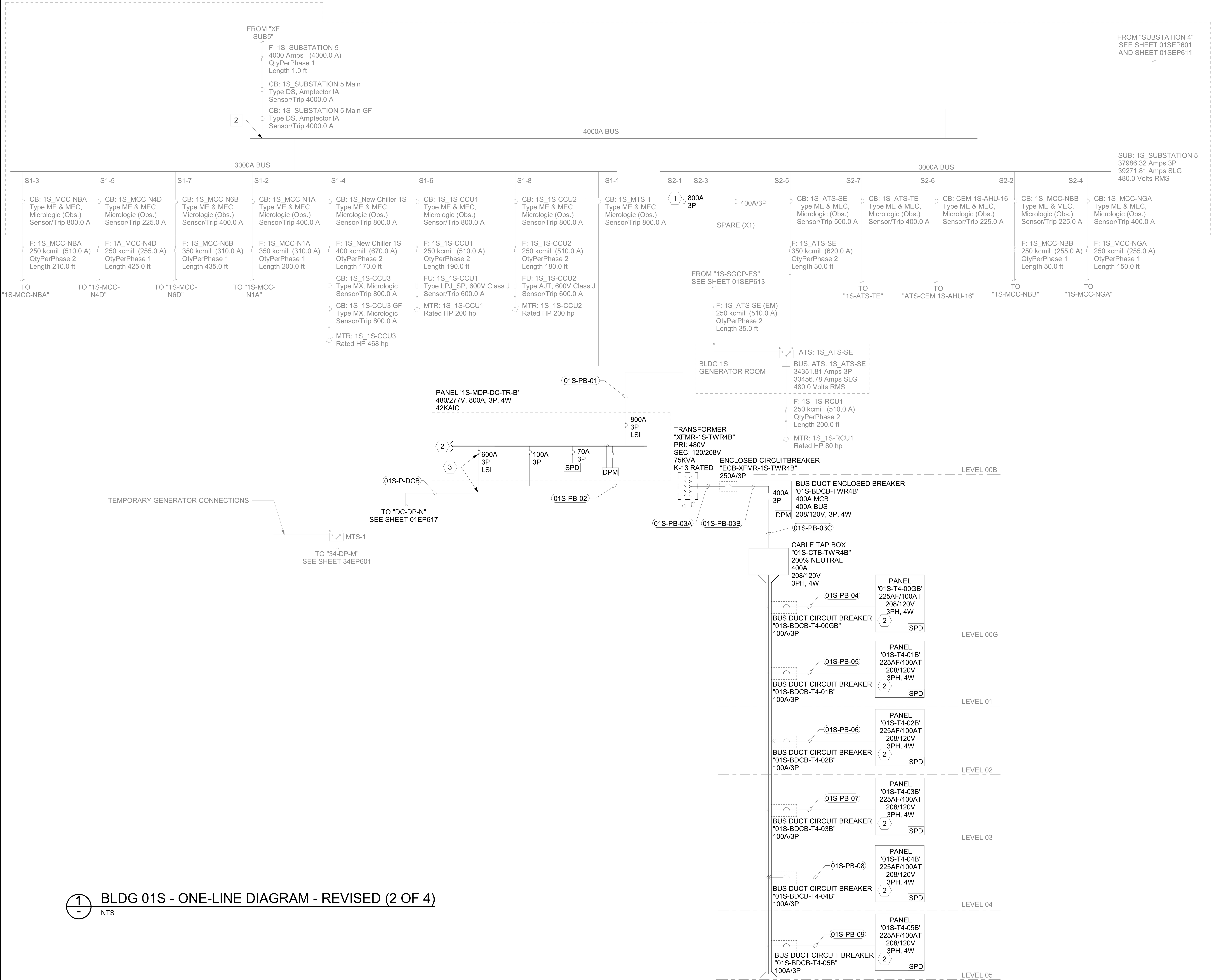

1 - BLDG 01S - ONE-LINE DIAGRAM - REVISED (2 OF 4)
NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
7. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1. SALVAGE EXISTING 400A SPARE TO THE VA AND PROVIDE CIRCUIT BREAKER AS SHOWN. REFERENCE REVISED PANEL SCHEDULES FOR ADDITIONAL INFORMATION.
2. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01S' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
3. BREAKER, CONDUIT AND CONDUCTORS SHALL BE INSTALLED DURING DATA CENTER PHASE "1B". REFERENCE DATA CENTER ONE-LINE DIAGRAMS FOR ADDITIONAL INFORMATION.

LOAD DATA POINT NOTES

2. LOAD DATA POINT: EXISTING SWITCHBOARD "1S_SUBSTATION 5" MAIN. REFERENCE "METERING AND LOAD INFORMATION SCHEDULE" ON SHEET 01SEP703 FOR ADDITIONAL INFORMATION.

BIM 360://581-22-700 VAHuntington EHRM/VA HUNTINGTON - Legend, General Notes, Diagrams and Details.rvt
9/3/2025 9:30:41 AM

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CONSULTANT
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Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01S - ONE-LINE DIAGRAM - REVISED (2 OF 4)
Approved:
Phase: 100% CONSTRUCTION DOCUMENTS
FULLY SPRINKLERED
Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date: 09/03/2025
Checked: DCW/NVP
Drawn: RTS/JNV
Project Number: 581-22-700
Building Number: 01S
Drawing Number: 01SEP612
470 OF 693
VA FORM 08 - 6231

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
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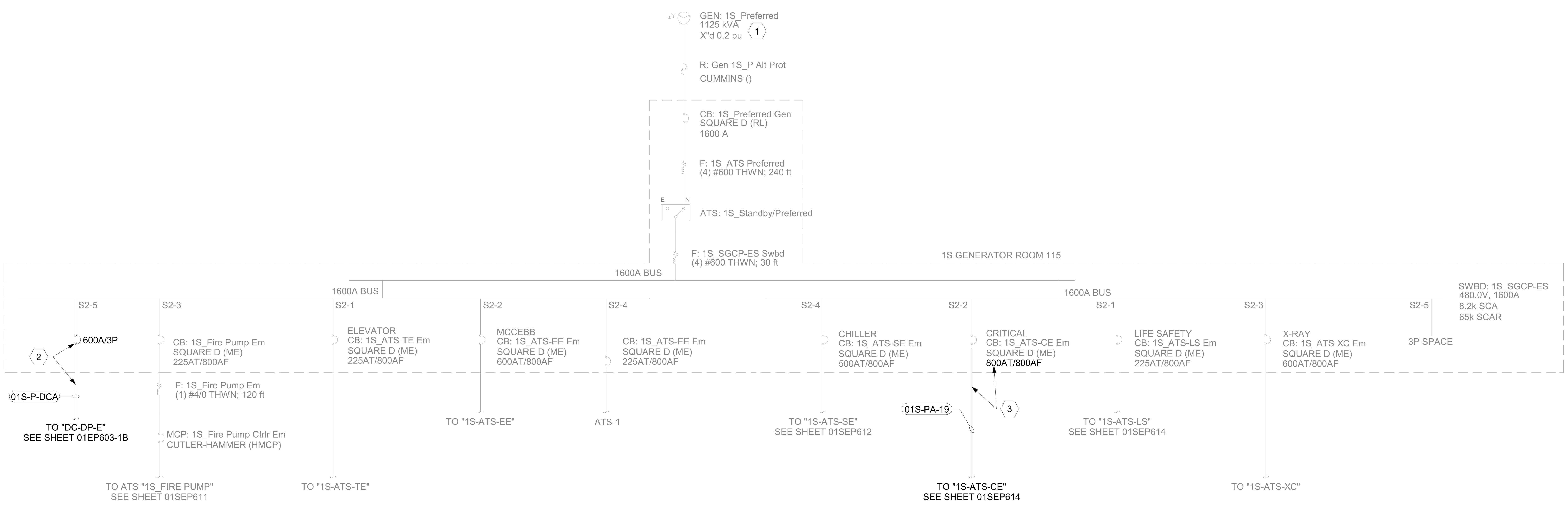
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
7. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1. REFERENCE ASSOCIATED GENERATOR CAPACITY AND LOAD INFORMATION SCHEDULE ON ELECTRICAL SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. BREAKER, CONDUIT AND CONDUCTORS SHALL BE INSTALLED DURING DATA CENTER PHASE "1B". REFERENCE DATA CENTER ONE-LINE DIAGRAMS FOR ADDITIONAL INFORMATION.
3. PROVIDE FEEDER, RACEWAY, AND CONDUCTORS FROM EXISTING CIRCUIT BREAKER TO SERVE AUTOMATIC TRANSFER SWITCH '1S-ATS-CE'. EXISTING CIRCUIT BREAKER ADJUSTED TO 800A TRIP. REFERENCE SHEET 01SEP614 FOR ADDITIONAL INFORMATION.



1 - BLDG 01S - ONE-LINE DIAGRAM - REVISED (3 OF 4)

NTS

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

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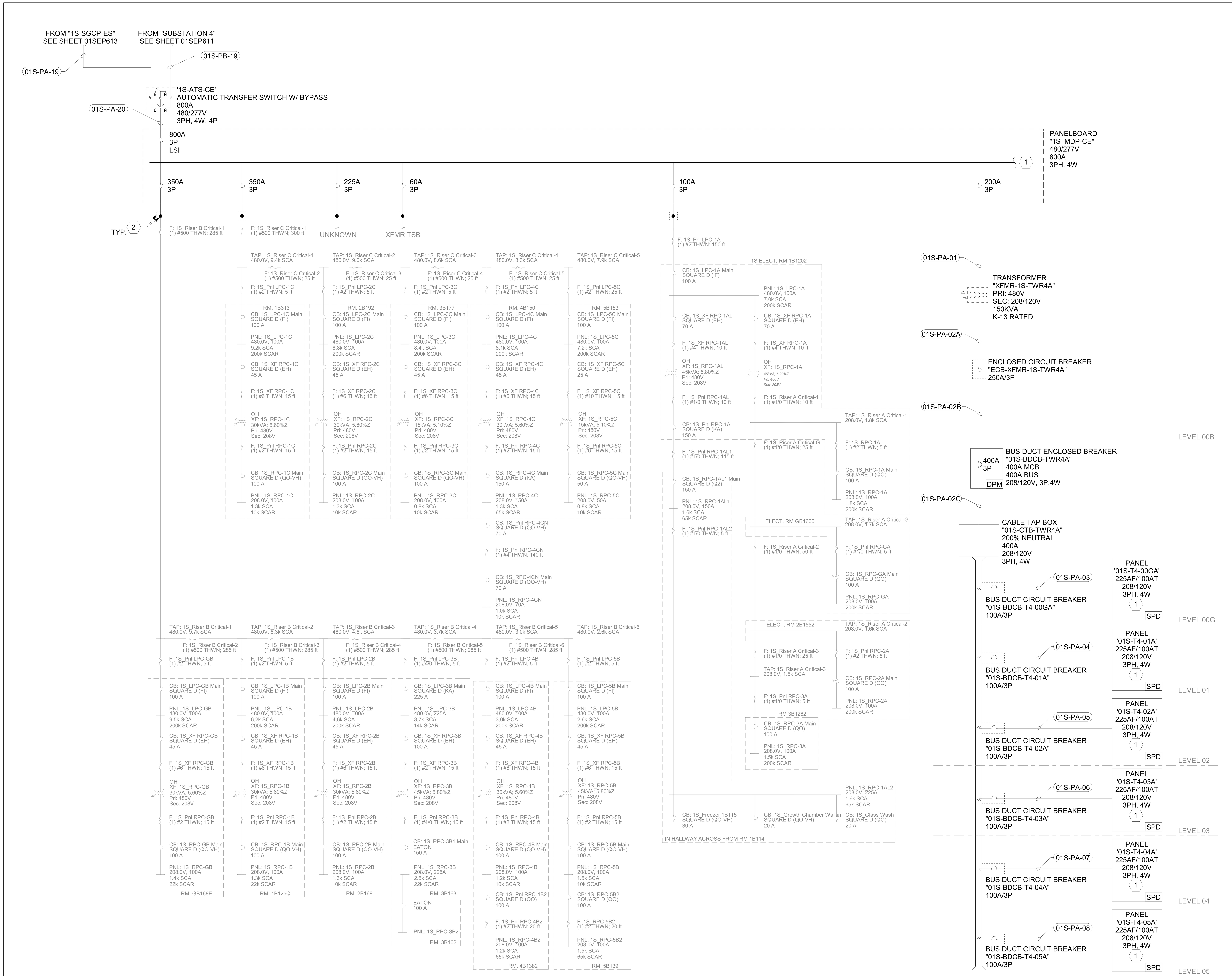
Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
BLDG 01S - ONE-LINE DIAGRAM - REVISED (3 OF 4)	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Building Number: 01S
		Issue Date: 09/03/2025; Checked: DCW/NVP; Drawn: RTS/JNV	Drawing Number: 01SEP613



1 BLDG 01S - ONE-LINE DIAGRAM - REVISED (4 OF 4)

NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. REFERENCE SHEET 01SEP635 FOR REVISED ONE-LINE DIAGRAM.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
8. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01S' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. PROVIDE JUNCTION BOX AND INSULATED TAPS TO EXTEND EXISTING CONDUIT AND WIRE TO PANELBOARD AS REQUIRED. SIZE JUNCTION BOX PER NEC. CONTRACTOR SHALL FIELD COORDINATE JUNCTION BOX INSTALLATION LOCATION AS REQUIRED.

Revisions:	Date:

CONSULTANT

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9-3-25

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
BLDG 01S - ONE-LINE DIAGRAM - REVISED (4 OF 4)	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700
			Building Number: 01S
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Drawing Number: 01SEP614
		Issue Date: 09/03/2025; Checked: DCW/NVP; Drawn: RTS/JNV	472 OF 693

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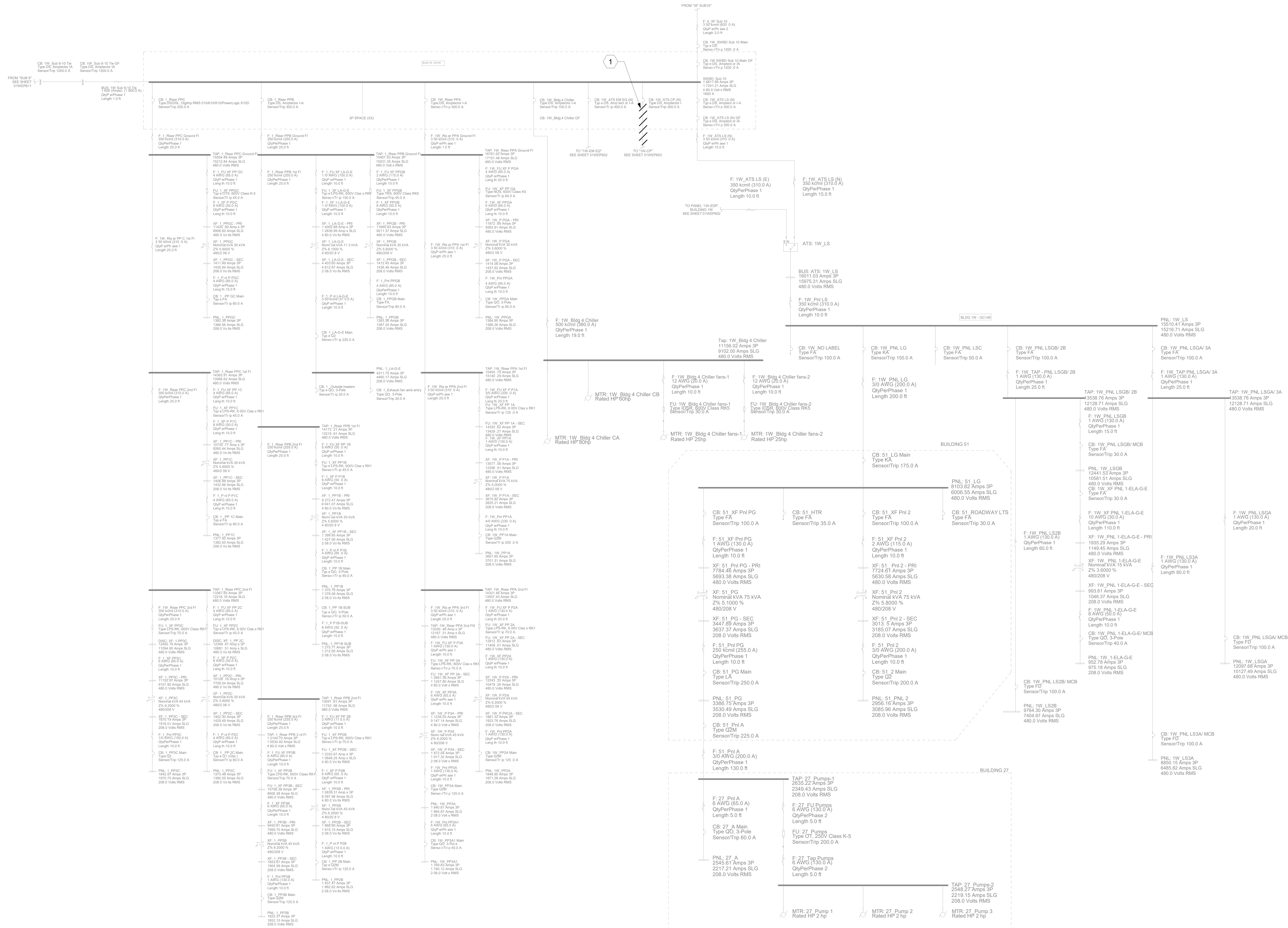

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BLDG 01W - ONE-LINE DIAGRAM - EXISTING (1 OF 3)
NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTES

1 DEMOLISH EXISTING FEEDER TO AUTOMATIC TRANSFER SWITCH"1W-CP".

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

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STATE OF WASHINGTON
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REGISTERED PROFESSIONAL ENGINEER

Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
BLDG 01W - ONE-LINE DIAGRAM - EXISTING (1 OF 3)	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Building Number: 01W
		Issue Date: 09/03/2025; Checked: DCW/NVP; Drawn: RTS/JNV	Drawing Number: 01WEP601
			473 OF 693

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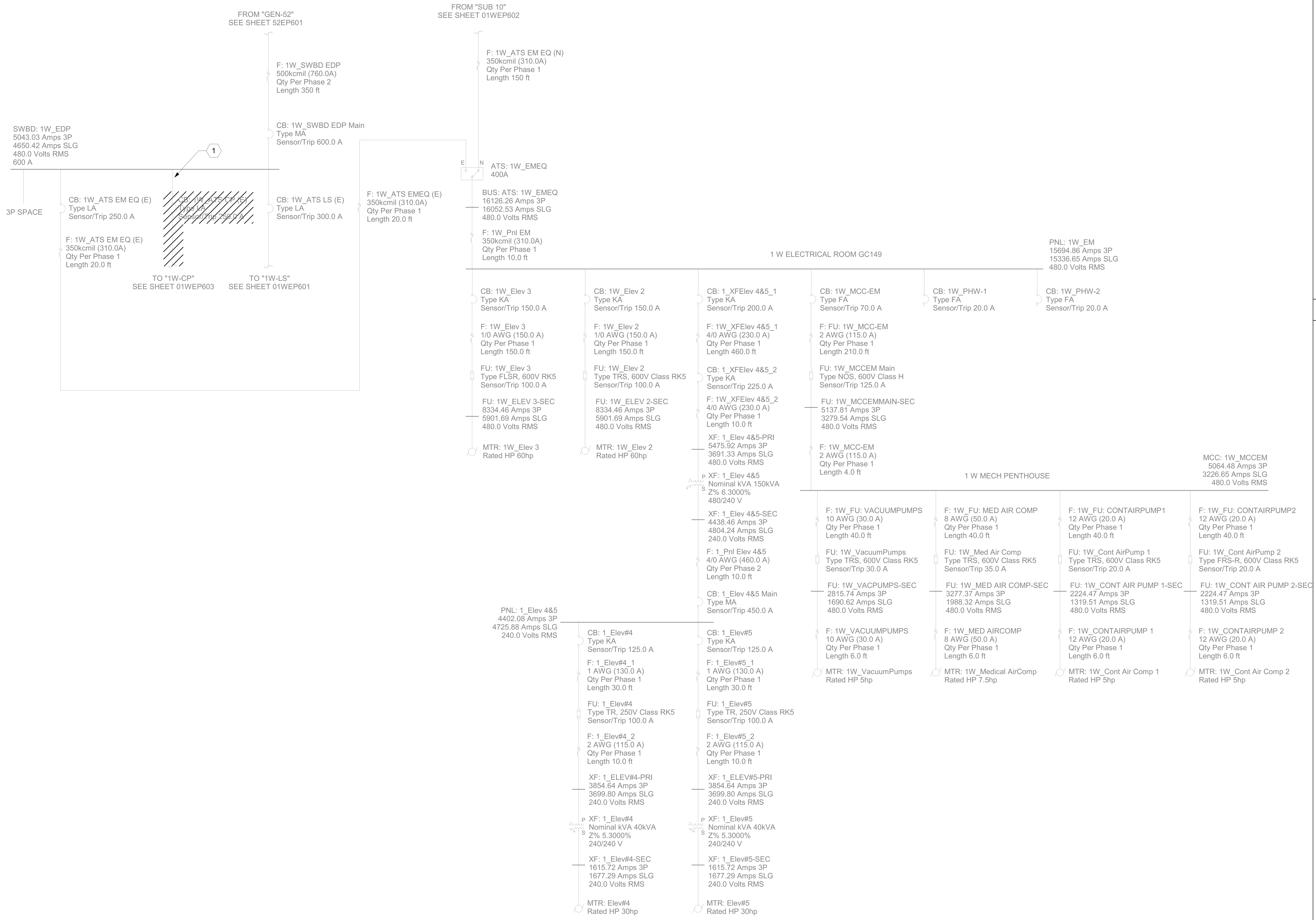

1 BLDG 01W - ONE-LINE DIAGRAM - EXISTING (2 OF 3)

NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. REFERENCE SHEET 01WEP611 FOR REVISED ONE-LINE DIAGRAM.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTES

1. DEMOLISH EXISTING CIRCUIT BREAKER AND FEEDER TO AUTOMATIC TRANSFER SWITCH "1W-CP".

Revisions:	Date:

CONSULTANT

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VA U.S. Department of Veterans Affairs

Drawing Title	BLDG 01W - ONE-LINE DIAGRAM - EXISTING (2 OF 3)
Approved:	
Phase	100% CONSTRUCTION DOCUMENTS
	FULLY SPRINKLERED
Project Title	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location	VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date	09/03/2025
Checked	DCW/NVP
Drawn	RTS/JNV
Project Number	581-22-700
Building Number	01W
Drawing Number	01WEP602
	474 OF 693

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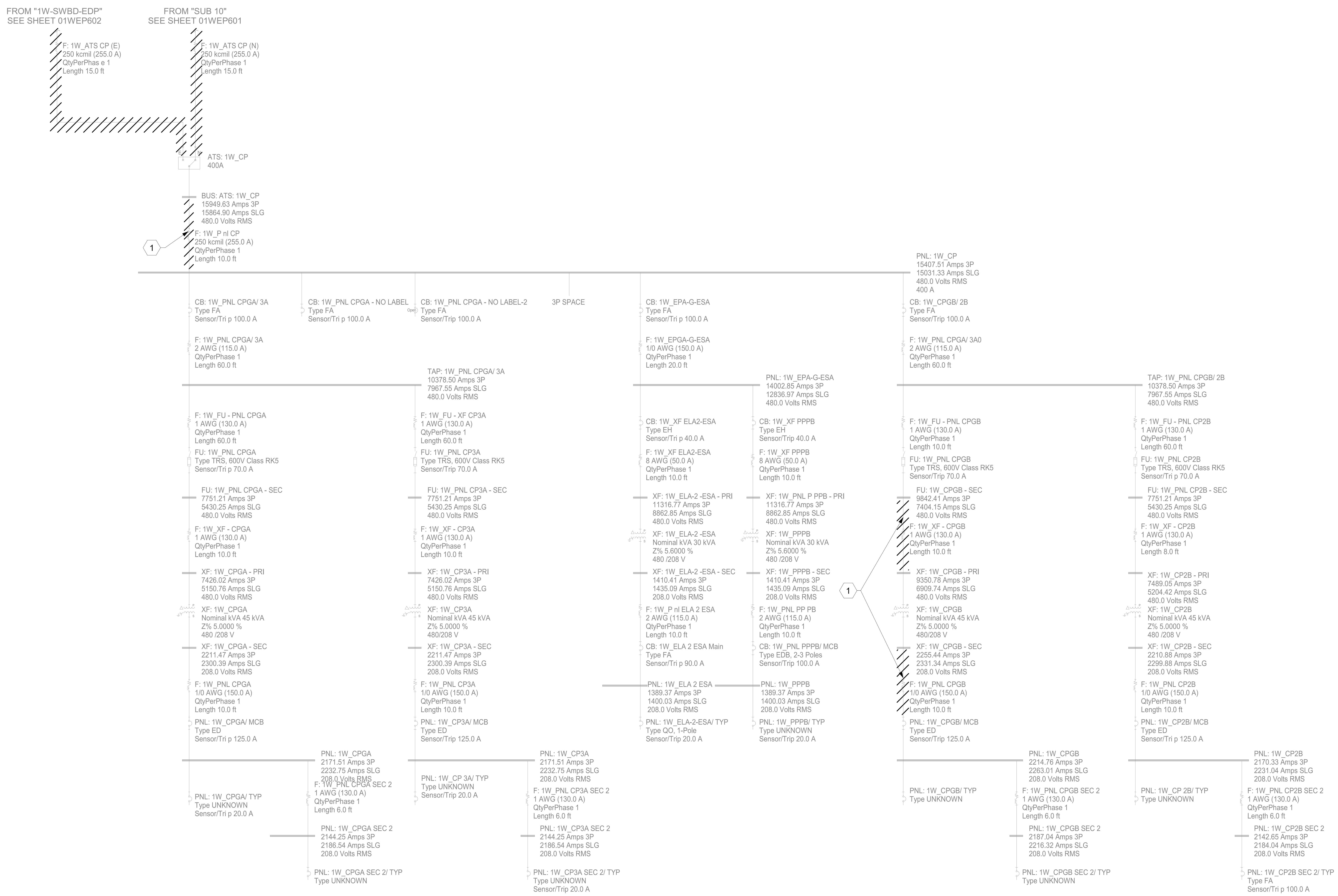

1 - BLDG 01W - ONE-LINE DIAGRAM - EXISTING (3 OF 3)

NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. REFERENCE SHEET 01WEP611 FOR REVISED ONE-LINE DIAGRAM.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTES

1. EXISTING FEEDER, RACEWAY, AND CONDUCTORS SHALL BE DEMOLISHED.

Revisions:	Date:

CONSULTANT

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VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01W - ONE-LINE DIAGRAM - EXISTING (3 OF 3)	Phase: 100% CONSTRUCTION DOCUMENTS	Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	Project Number: 581-22-700
			Building Number: 01W
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Drawing Number: 01WEP603
		Issue Date: 09/03/2025; Checked: DCW/NVP; Drawn: RTS/JNV	475 OF 693

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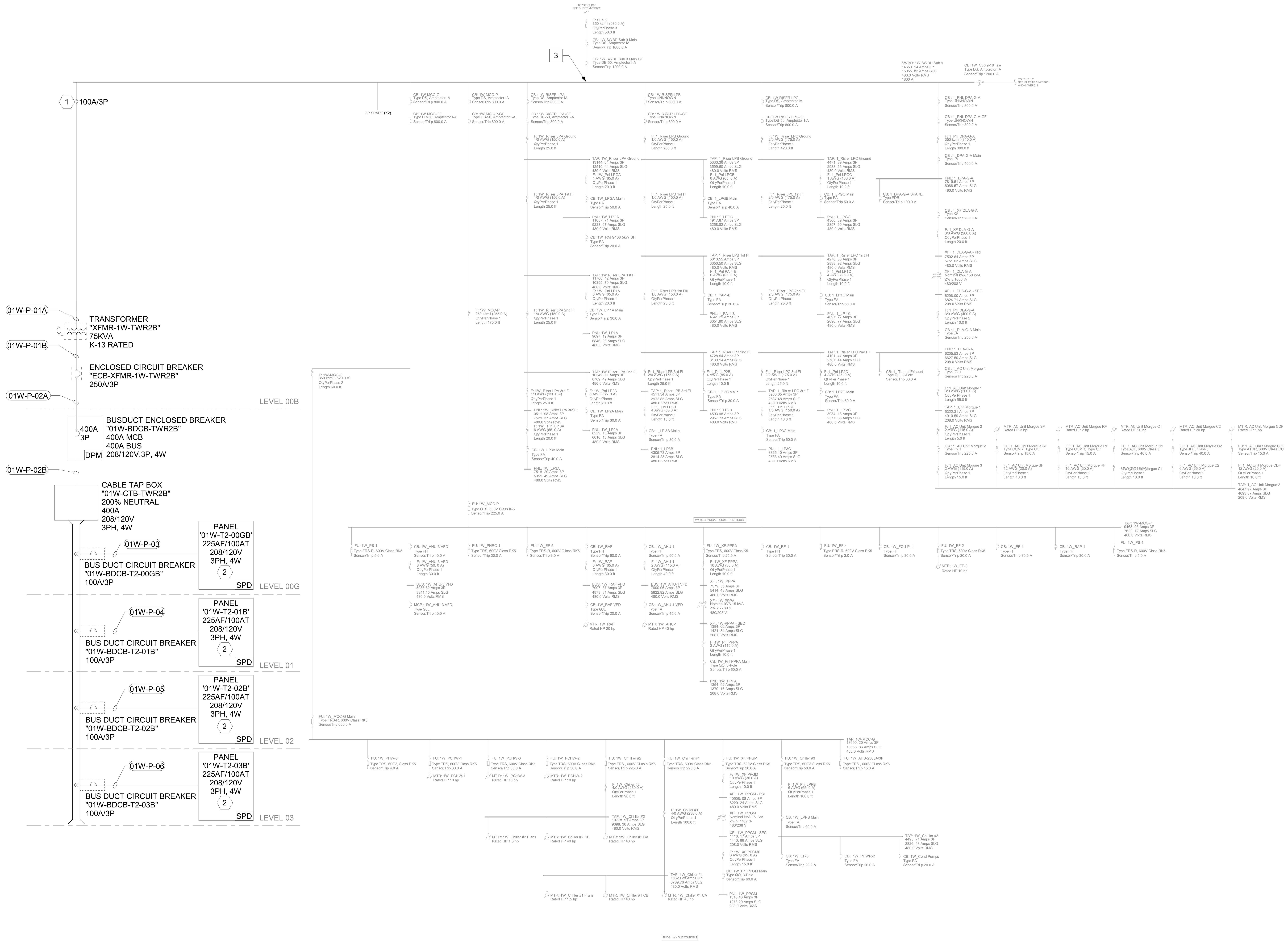

1 - BLDG 01W - ONE-LINE DIAGRAM - REVISED (1 OF 4)

NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
7. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1 UTILIZE EXISTING SPACE TO PROVIDE CIRCUIT BREAKER AS SHOWN. REFERENCE CORRESPONDING REVISED PANEL SCHEDULES FOR ADDITIONAL INFORMATION.

2 REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01W' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.

LOAD DATA POINT NOTES

3 LOAD DATA POINT: EXISTING SWITCHBOARD "1W SWBD Sub 9". REFERENCE "METERING AND LOAD INFORMATION SCHEDULE" ON SHEET 01WEP701 FOR ADDITIONAL INFORMATION.

Revisions:	Date:

CONSULTANT

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LOUIS R. LITZKO — STATE OF WASHINGTON — 55681 — REGISTERED PROFESSIONAL ENGINEER — 9-3-25

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
BLDG 01W - ONE-LINE DIAGRAM - REVISED (1 OF 4)	100% CONSTRUCTION DOCUMENTS	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	581-22-700
Approved:	FULLY SPRINKLERED	Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Building Number: 01W
		Issue Date: 09/03/2025; Checked: DCW/NVP; Drawn: RTS/JNV	Drawing Number: 01WEP611
			476 OF 693

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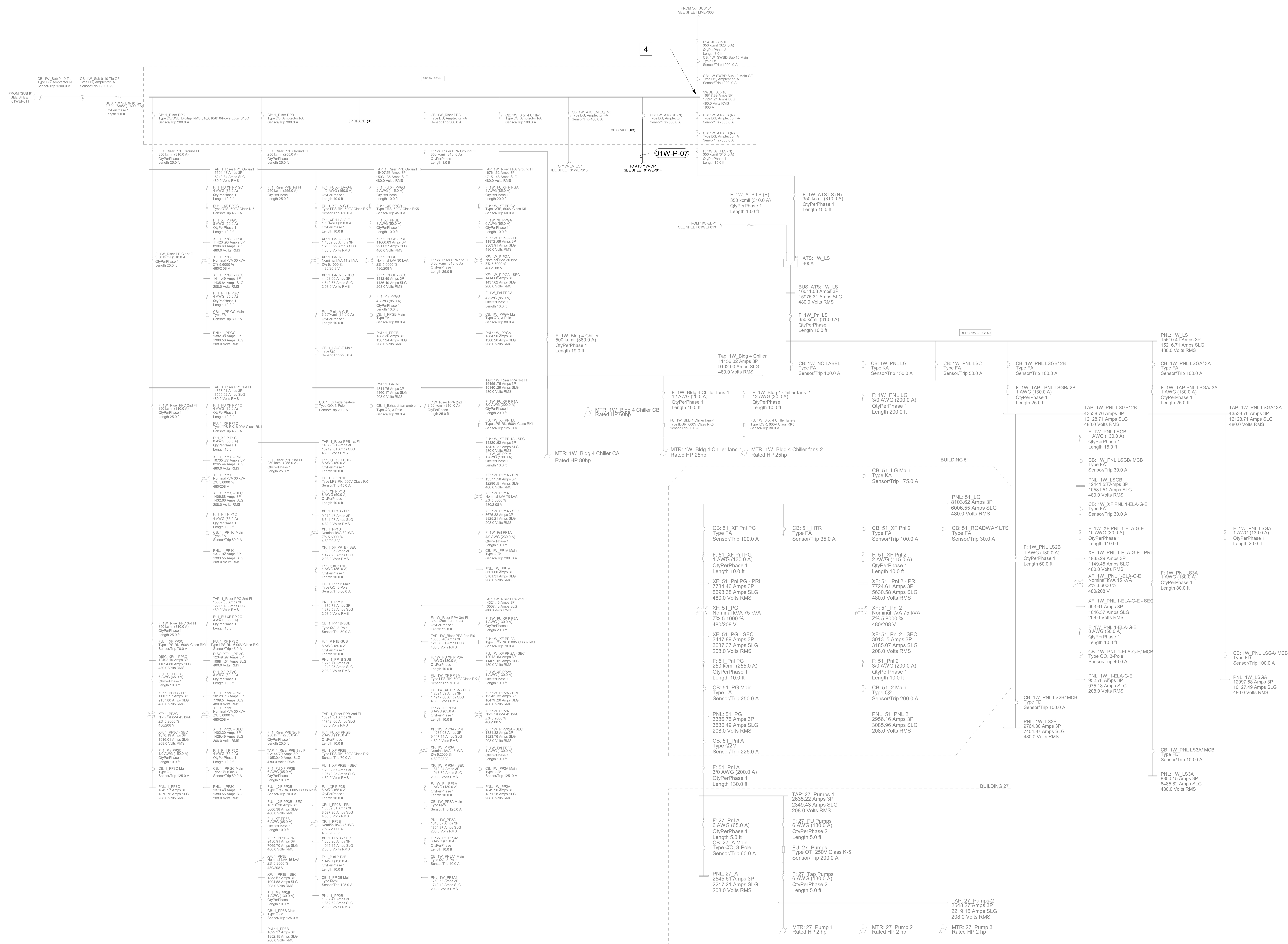

1 - BLDG 01W - ONE-LINE DIAGRAM - REVISED (2 OF 4)

NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
7. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

LOAD DATA POINT NOTES

4 LOAD DATA POINT: EXISTING SWITCHBOARD "Sub 10". REFERENCE "METERING AND LOAD INFORMATION SCHEDULE" ON SHEET 01WEP702 FOR ADDITIONAL INFORMATION.

Revisions:	Date:

CONSULTANT

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01W - ONE-LINE DIAGRAM - REVISED (2 OF 4)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700

Building Number: 01W

Drawing Number: 01WEP612

477 OF 693

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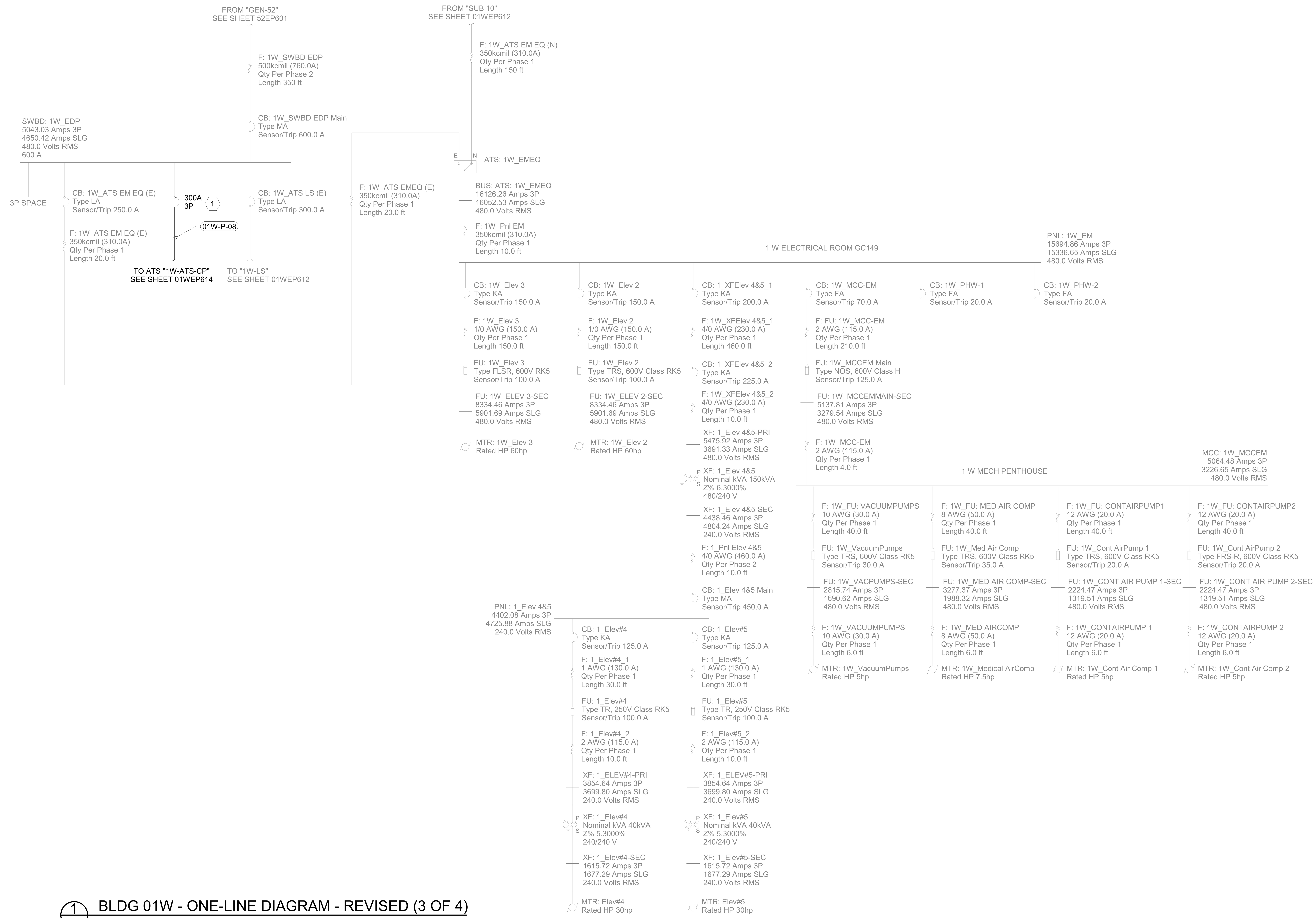

1 / -
BLDG 01W - ONE-LINE DIAGRAM - REVISED (3 OF 4)
NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. REFERENCE SHEET 01WEP611 FOR REVISED ONE-LINE DIAGRAM.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
8. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1. UTILIZE EXISTING SPACE TO PROVIDE CIRCUIT BREAKER AS SHOWN.

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01W - ONE-LINE DIAGRAM - REVISED (3 OF 4)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

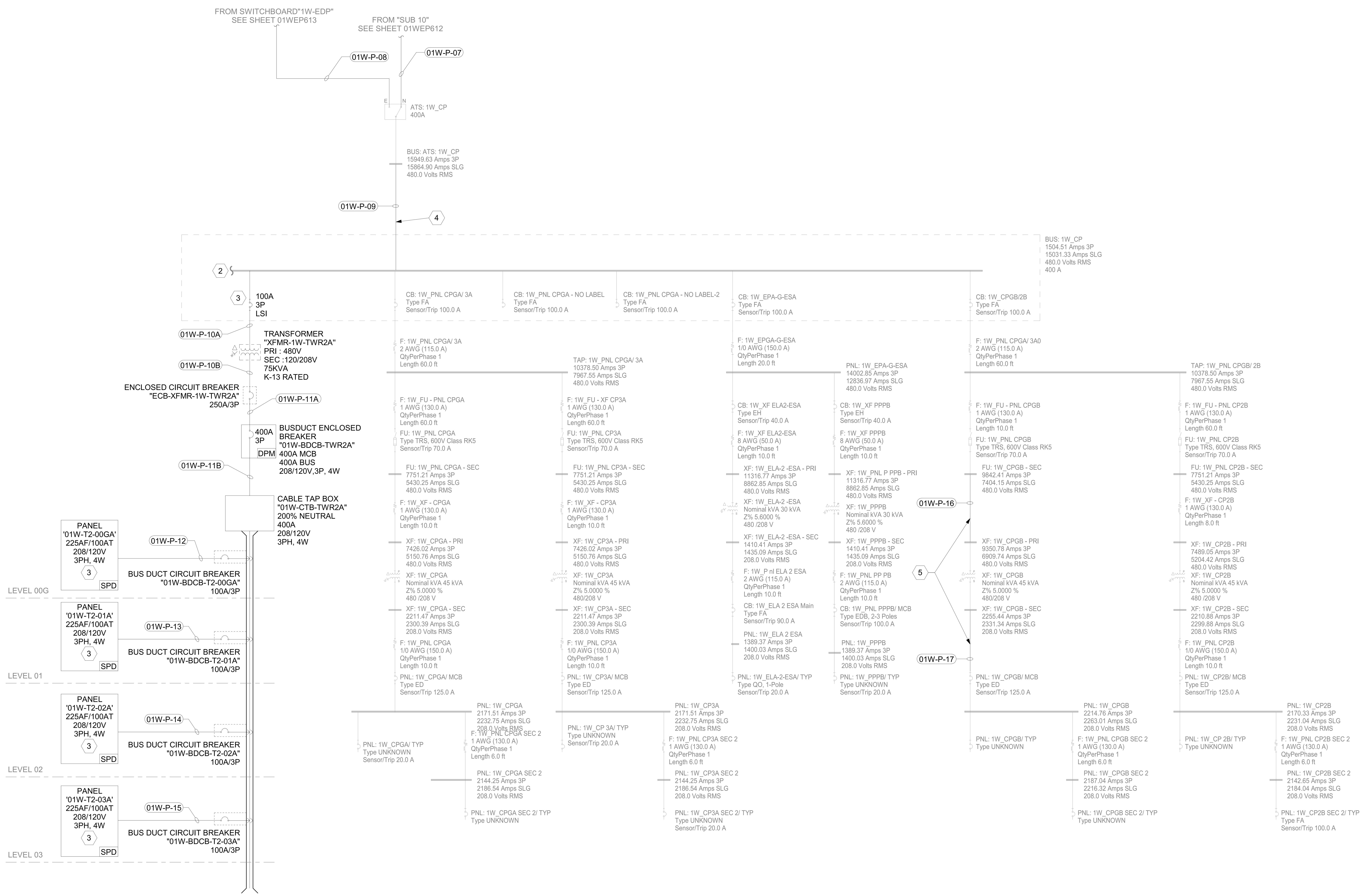
Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700

Building Number: 01W

Drawing Number: 01WEP613



1 - BLDG 01W - ONE-LINE DIAGRAM - REVISED (4 OF 4)

NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. REFERENCE SHEET 01WEP611 FOR REVISED ONE-LINE DIAGRAM.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
8. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1. MODIFIED EQUIPMENT. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '02' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '01W' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
3. UTILIZE EXISTING SPACE TO PROVIDE CIRCUIT BREAKER AS SHOWN.
4. PROVIDE OVERCURRENT PROTECTION FEEDER AS INDICATED FOR ADDED EHRM LOAD.
5. PROVIDE FEEDER, RACEWAY, AND CONDUCTORS TO REFEED EXISTING EQUIPMENT.

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

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9-3-25

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 01W - ONE-LINE DIAGRAM - REVISED (4 OF 4)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700

Building Number: 01W

Drawing Number: 01WEP614

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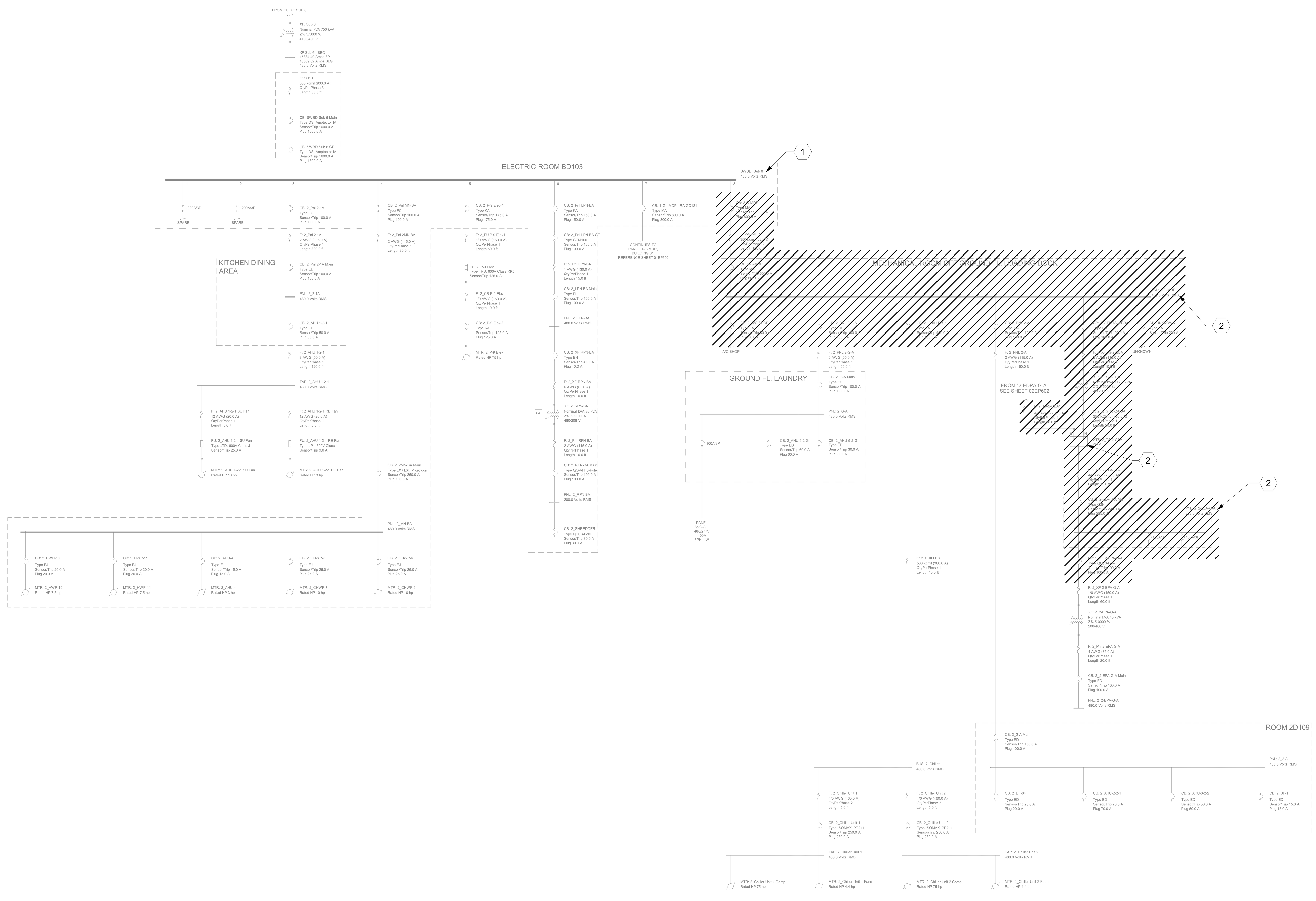

1 - BLDG 02 - PARTIAL ONE-LINE DIAGRAM - EXISTING (1 OF 2)

NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. REFERENCE SHEET 02EP611 FOR REVISED ONE-LINE DIAGRAM.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTES

1. EXISTING EQUIPMENT SHALL BE MODIFIED. REFERENCE ASSOCIATED REVISED ONE-LINE DIAGRAM FOR ADDITIONAL INFORMATION.
2. EXISTING EQUIPMENT SHALL BE DEMOLISHED AND REPLACED. PRESERVE CIRCUITS FOR RETERMINATION. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '02' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

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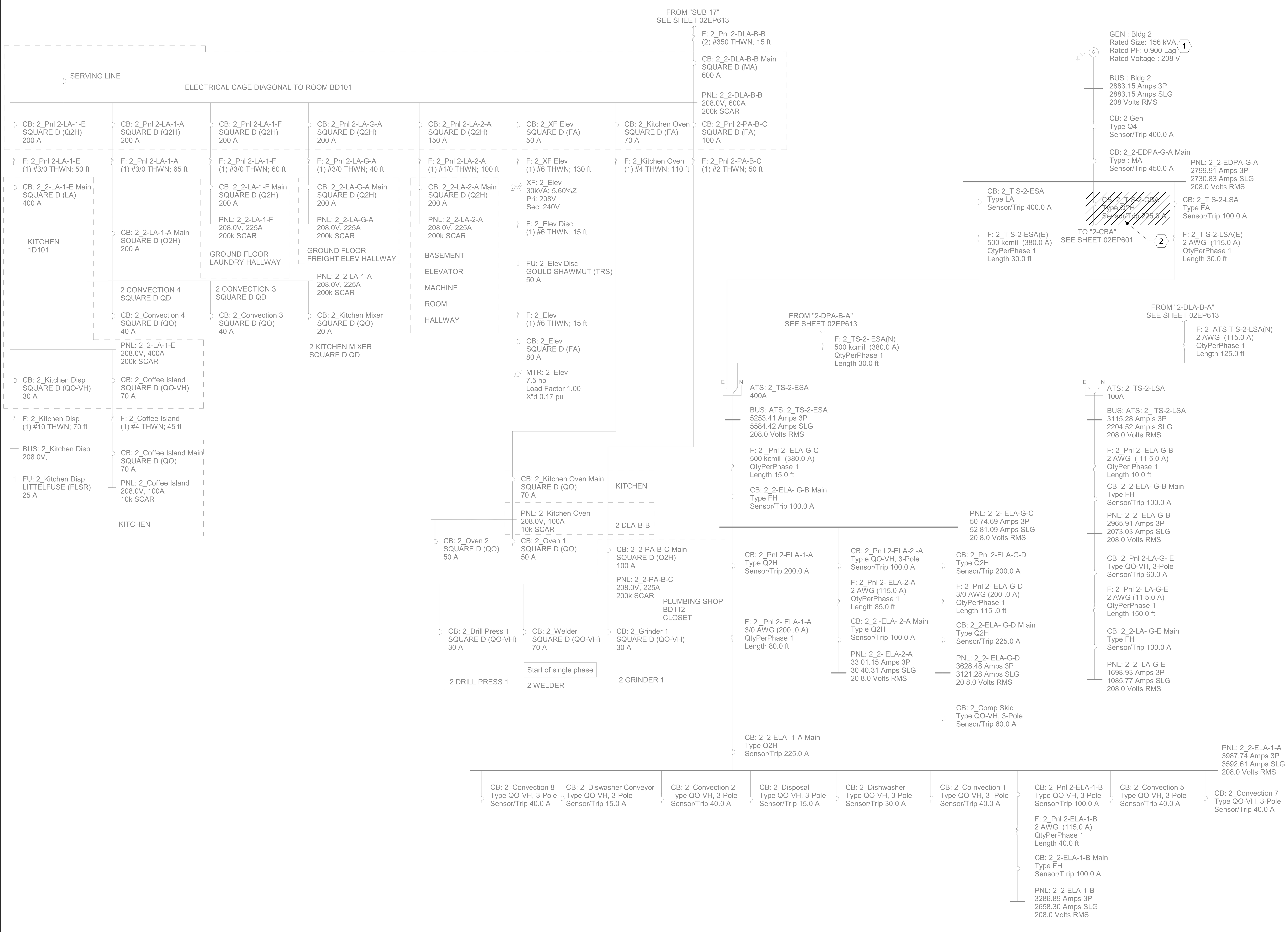
VA U.S. Department of Veterans Affairs

Drawing Title	BLDG 02 - PARTIAL ONE-LINE DIAGRAM - EXISTING (1 OF 2)
Approved:	
Phase	100% CONSTRUCTION DOCUMENTS
	FULLY SPRINKLERED
Project Title	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location	VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date	09/03/2025
Checked	DCW/NVP
Drawn	RTS/JNV
Project Number	581-22-700
Building Number	02
Drawing Number	02EP601



1 - BLDG 02 - PARTIAL ONE-LINE DIAGRAM - EXISTING (2 OF 2)
NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. REFERENCE SHEET 02EP611 FOR REVISED ONE-LINE DIAGRAM.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTES

1. REFERENCE ASSOCIATED GENERATOR CAPACITY AND LOAD INFORMATION SCHEDULE ON ELECTRICAL SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. EXISTING FEEDER AND BREAKER SHALL BE DEMOLISHED.

Revisions: | Date:

CONSULTANT
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LOUIS R. LITZKO STATE OF WASHINGTON 55681 REGISTERED PROFESSIONAL ENGINEER 4-3-25

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VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 02 - PARTIAL ONE-LINE DIAGRAM - EXISTING (2 OF 2)
Approved:
Phase: 100% CONSTRUCTION DOCUMENTS
FULLY SPRINKLERED
Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location: VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date: 09/03/2025
Checked: DCW/NVP
Drawn: RTS/JNV
Project Number: 581-22-700
Building Number: 02
Drawing Number: 02EP602
481 OF 693

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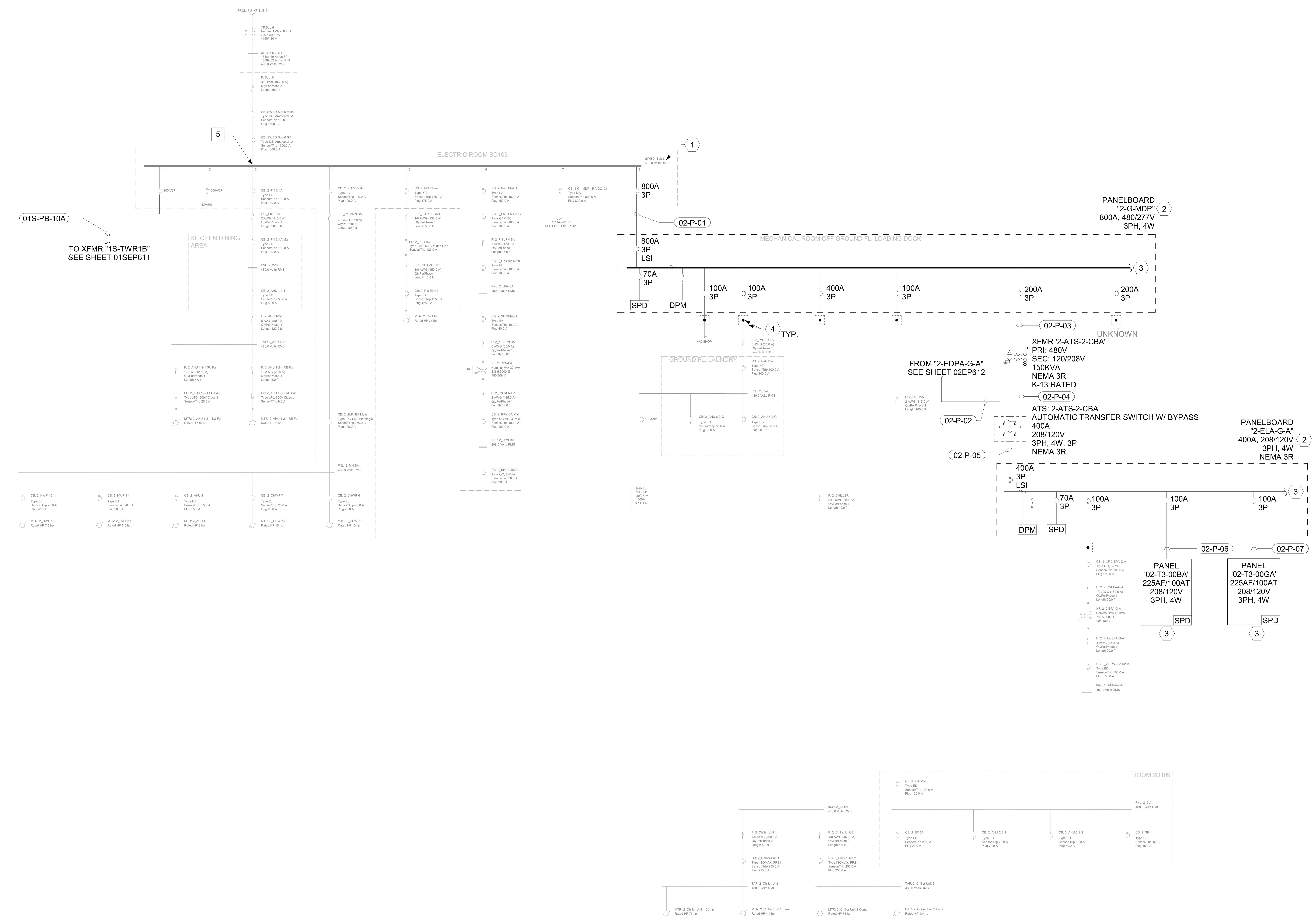

1 - BLDG 02 - ONE-LINE DIAGRAM - REVISED (1 OF 3)

NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. REFERENCE SHEET 02EP611 FOR REVISED ONE-LINE DIAGRAM.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
8. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1 MODIFIED EQUIPMENT. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '02' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.

2 EXISTING EQUIPMENT SHALL BE DEMOLISHED AND REPLACED. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '02' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.

3 REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '02' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.

4 PROVIDE JUNCTION BOX AND INSULATED TAPS TO EXTEND EXISTING CONDUIT AND WIRE TO PANELBOARD AS REQUIRED. SIZE JUNCTION BOX PER NEC. CONTRACTOR SHALL FIELD COORDINATE JUNCTION BOX INSTALLATION LOCATION AS REQUIRED.

LOAD DATA POINT NOTES

5 LOAD DATA POINT: EXISTING SWITCHBOARD "Sub 6". REFERENCE "METERING AND LOAD INFORMATION SCHEDULE" ON SHEET 02EP701 FOR ADDITIONAL INFORMATION.

Revisions:	Date:

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 02 - ONE-LINE DIAGRAM - REVISED (1 OF 3)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

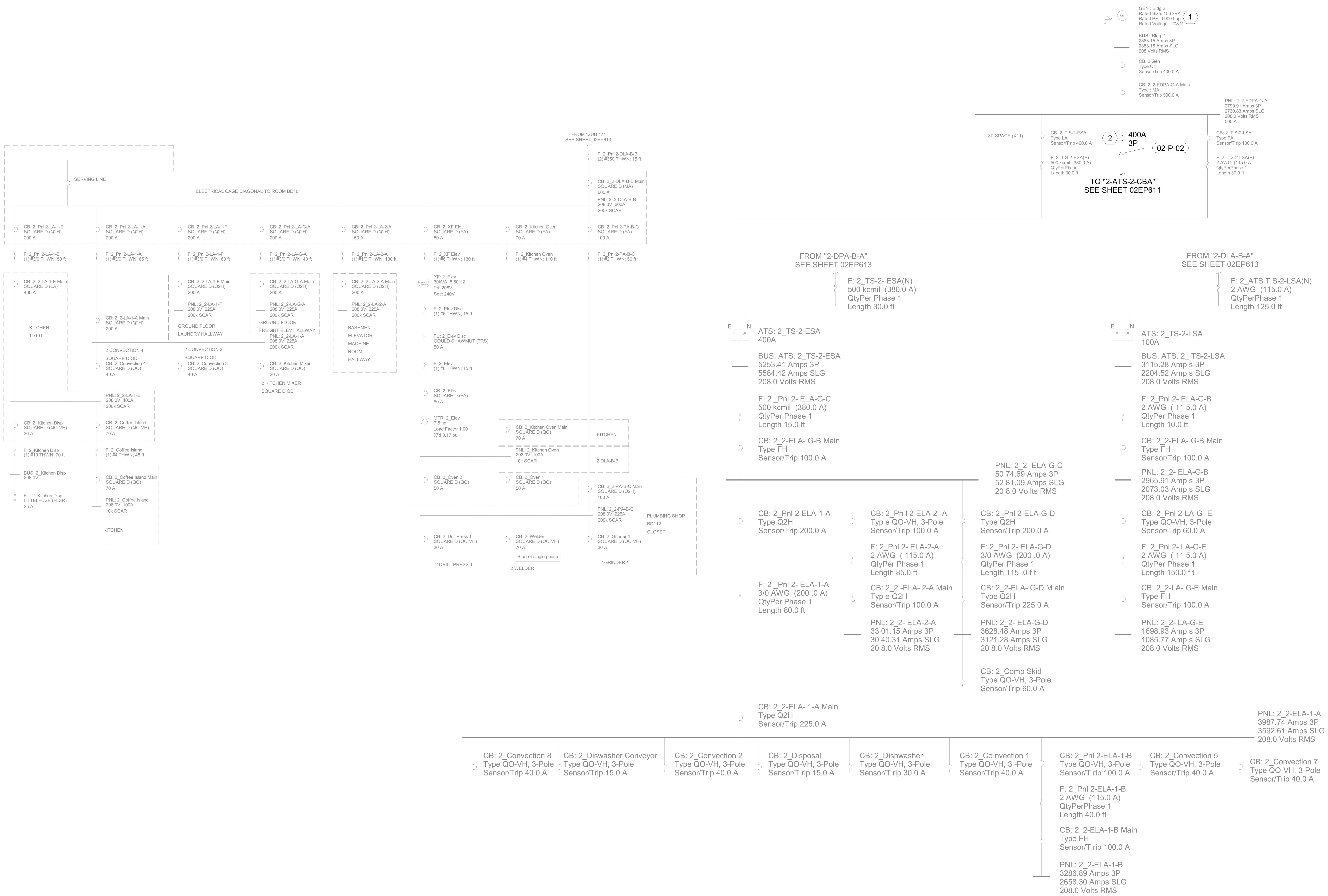
Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date	Checked	Drawn
09/03/2025	DCW/NVP	RTS/JNV

Project Number: 581-22-700

Building Number: 02

Drawing Number: 02EP611



1 BLDG 02 - ONE-LINE DIAGRAM - REVISED (2 OF 3)
NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
7. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1. REFERENCE ASSOCIATED GENERATOR CAPACITY AND LOAD INFORMATION SCHEDULE ON ELECTRICAL SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. UTILIZE EXISTING SPACE TO PROVIDE CIRCUIT BREAKER AS SHOWN.

Revisions:	Date:

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Drawing Title: BLDG 02 - ONE-LINE DIAGRAM - REVISED (2 OF 3)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

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Building Number: 02

Drawing Number: 02EP612

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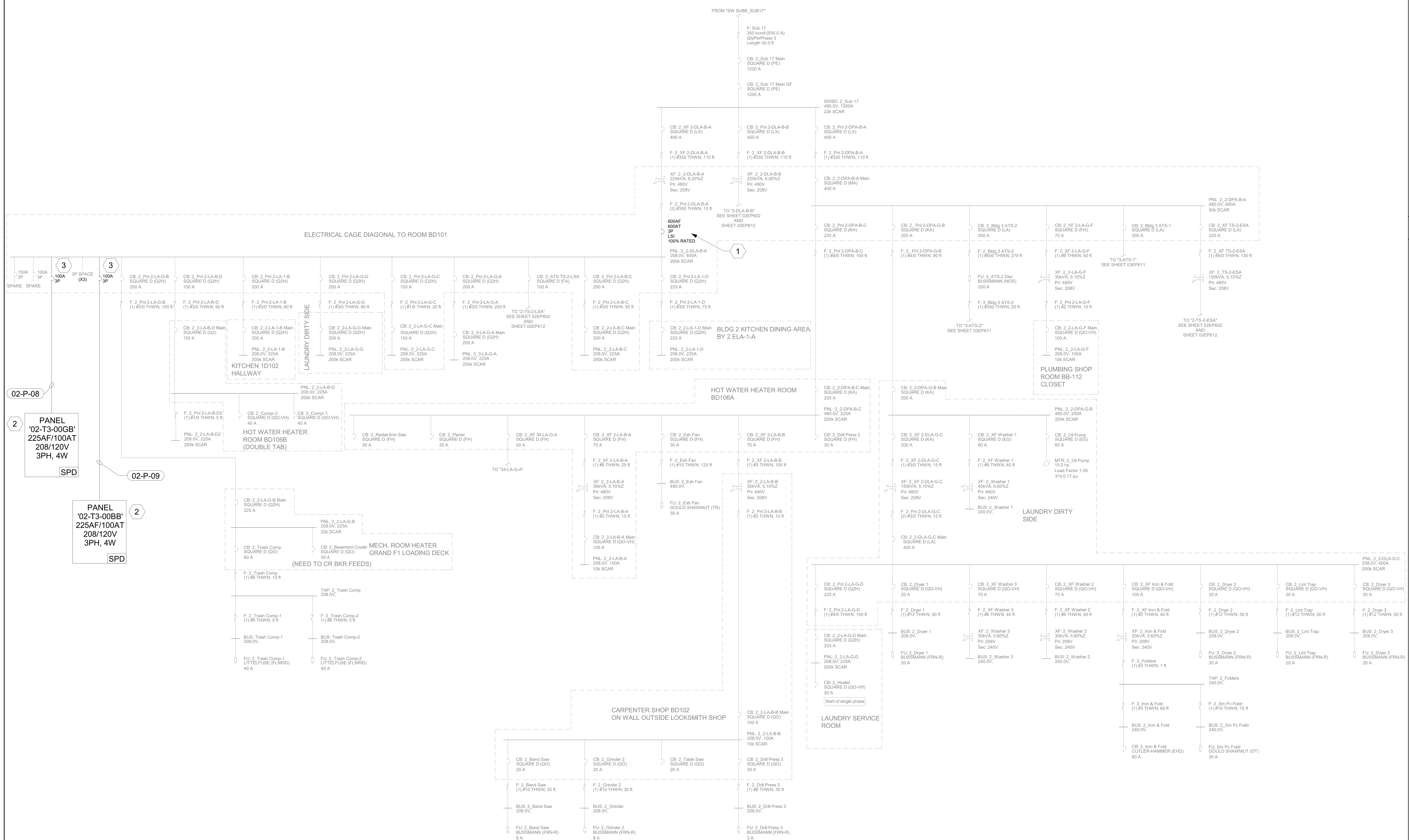

1 BLDG 02 - ONE-LINE DIAGRAM - REVISED (3 OF 3)

NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
6. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.
7. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1. REPLACE MAIN BREAKER WITH 600A 100% RATED AS INDICATED.
2. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '02' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
3. UTILIZE EXISTING SPACE TO PROVIDE CIRCUIT BREAKER AS SHOWN.

Revisions: Date:

CONSULTANT

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: BLDG 02 - ONE-LINE DIAGRAM - REVISED (3 OF 3)

Approved:

Phase: 100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title: EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location: VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

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Building Number: 02

Drawing Number: 02EP613

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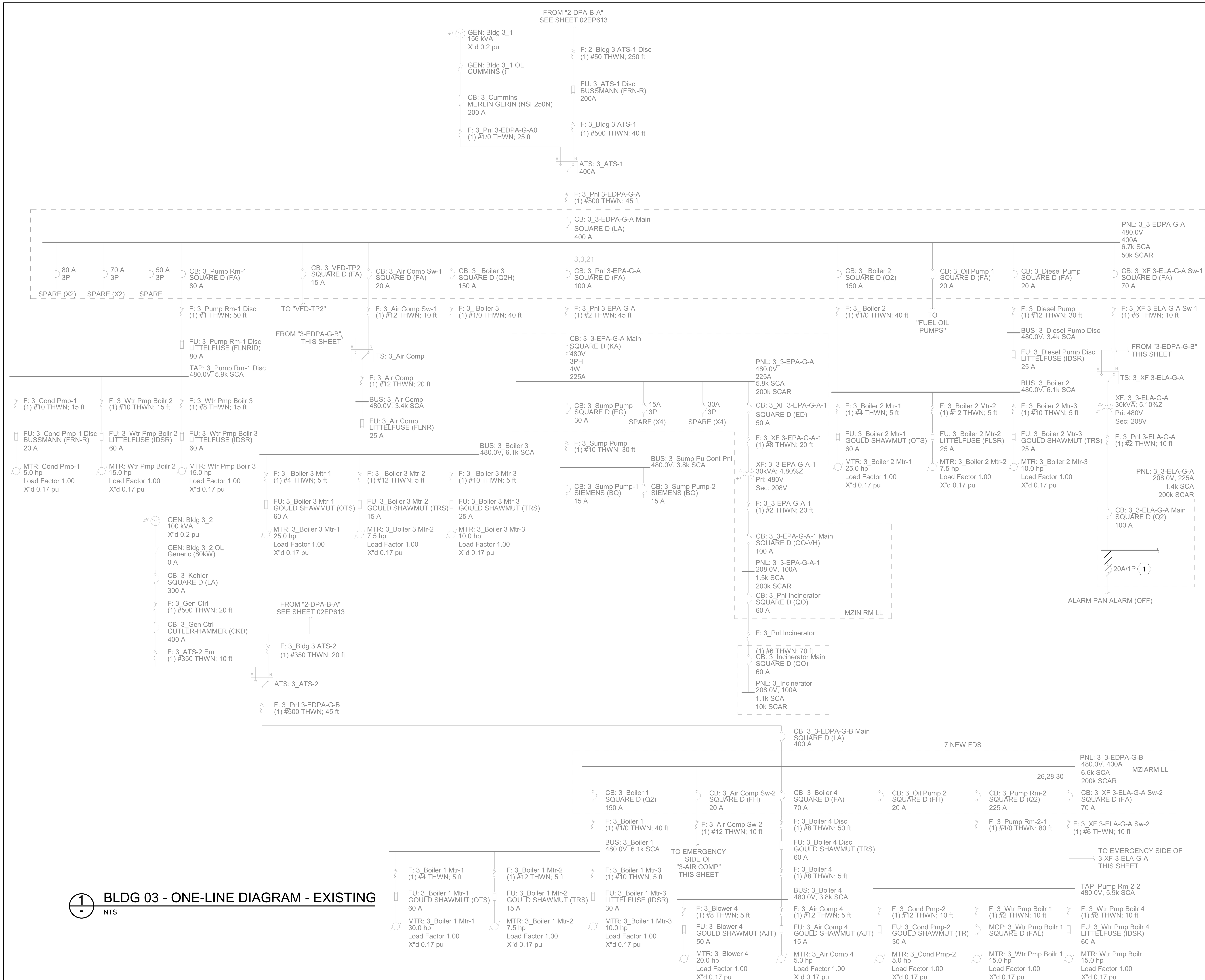

1 BLDG 03 - ONE-LINE DIAGRAM - EXISTING
NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
2. CONTRACTOR SHALL PROVIDE ALL SHUTDOWN PLANS, TEMPORARY POWER PLANS, AND PHASING PLANS FOR VA AND ENGINEER REVIEW AND APPROVAL PRIOR TO COMMENCING WORK. ALL PLANS SHALL BE SUBMITTED FOR REVIEW 30 DAYS PRIOR TO IMPLEMENTATION.
3. SHORT CIRCUIT CALCULATIONS BASED ON VA PROVIDED AS-BUILTS.
4. NOT ALL ONE-LINE DIAGRAM DRAWING SHEETS MAY HAVE WORK SHOWN; ALL ONE-LINE DRAWINGS HAVE BEEN INCLUDED IN THIS DRAWING SET FOR REFERENCE PURPOSES.
5. REFERENCE SHEET 03EP611 FOR REVISED ONE-LINE DIAGRAM.
6. ALL PROVIDED DIGITAL POWER METERS (DPM) SHALL BE INTEGRATED INTO THE EXISTING FACILITY BMS SYSTEM; PROVIDE ALL REQUIRED RACEWAYS, CABLING, AND INTEGRATION TO PROVIDE A COMPLETE AND FULLY FUNCTIONAL SYSTEM.
7. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS WITH AIC RATING EQUAL TO OR GREATER THAN HIGHEST FAULT CURRENT RATING INDICATED ON ASSOCIATED PANELBOARD SCHEDULE AND ONE-LINE DIAGRAM.

KEY NOTES

1. (1) EXISTING 20A, 1P CIRCUIT BREAKERS SHALL BE DEMOLISHED. REFERENCE REVISED ONE-LINE DIAGRAMS AND ASSOCIATED PANEL SCHEDULES FOR ADDITIONAL INFORMATION.

Revisions:	Date:

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VA U.S. Department of Veterans Affairs

Drawing Title	BLDG 03 - ONE-LINE DIAGRAM - EXISTING
Approved:	
Phase	100% CONSTRUCTION DOCUMENTS
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Project Title	EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON
Location	VA HUNTINGTON, 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
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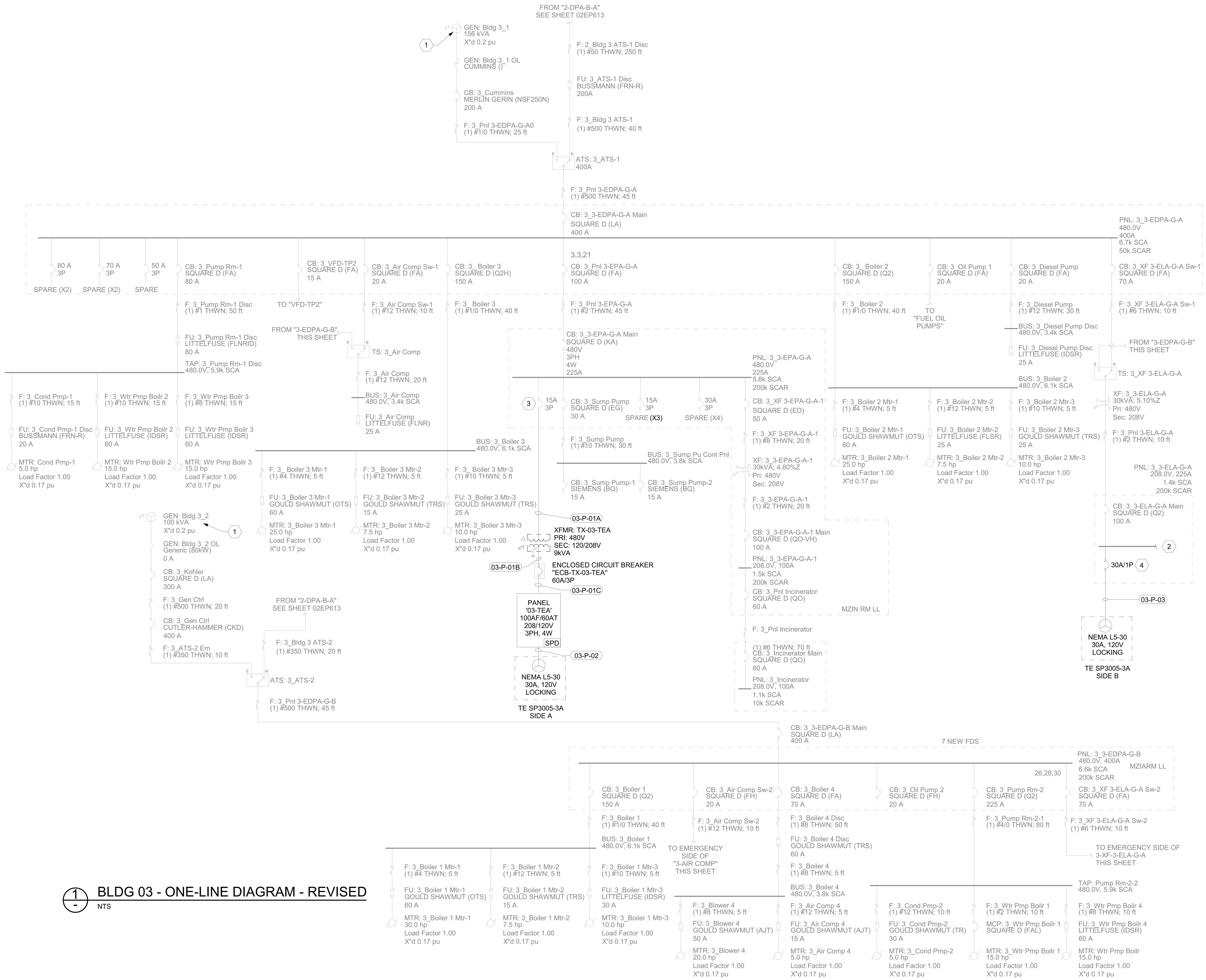
Project Number	581-22-700
Building Number	03
Drawing Number	03EP601



1 - BLDG 03 - ONE-LINE DIAGRAM - REVISED
NTS

GENERAL NOTES

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS. ANY DISCREPANCIES FOUND SHALL BE REPORTED TO THE ENGINEER.
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8. CIRCUIT BREAKERS SERVING TRANSFORMERS SHALL BE LOCKABLE.

KEY NOTES

1. REFERENCE ASSOCIATED GENERATOR CAPACITY AND LOAD INFORMATION SCHEDULE ON ELECTRICAL SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
2. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '03' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
3. UTILIZE EXISTING SPARE AS INDICATED. REFERENCE ASSOCIATED PANEL SCHEDULE ON BUILDING '03' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.
4. EXISTING 20A CIRCUIT BREAKER DEMOLISHED AND REPLACED WITH CONTRACTOR PROVIDED 30A CIRCUIT BREAKER. REFERENCE ASSOCIATED PANEL SCHEDULES ON BUILDING '03' SCHEDULE SHEETS FOR ADDITIONAL INFORMATION.

Revisions:	Date:

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Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title	BLDG 03 - ONE-LINE DIAGRAM - REVISED
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Issue Date	09/03/2025
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Drawing Number	03EP611
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